

Data Exchange Framework v12.5 Fundamentals

Global Education Services



Course Overview

Data Exchange Framework v12.5
Fundamentals

Create a
NiCE..
world 

Training Expectations



During your session

Do these...

Setup Out of Office



Be on Time



Ask Questions

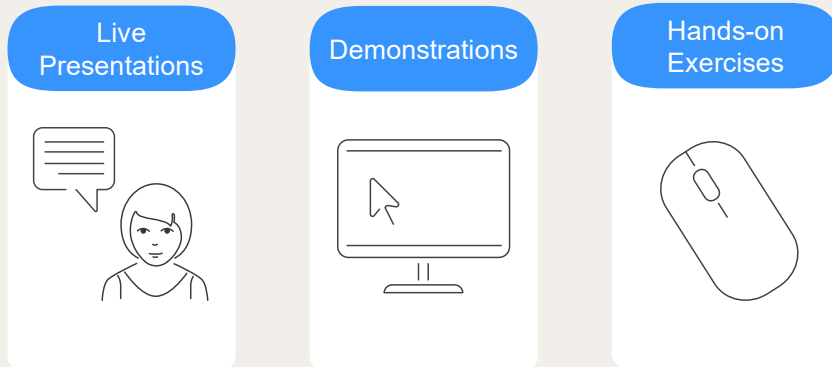


NiCE

Even though this is a virtual training environment, it should be treated like a classroom environment. Setup your email out of office and update your voicemail that you will be unavailable during training. Be punctual when returning from the breaks and lunches. Active participation is appreciated and contributes to the learning process.



Instructor-led Training



This training is delivered over the web as 'Virtual Classroom Training' which combines live presentations and demonstrations. Every day during this training course, we will open a Teams session which will remain open throughout the day. This is your platform to communicate with the trainer, ask questions and comment. It will also be used for the live presentations and demos.

NICE Training Environment

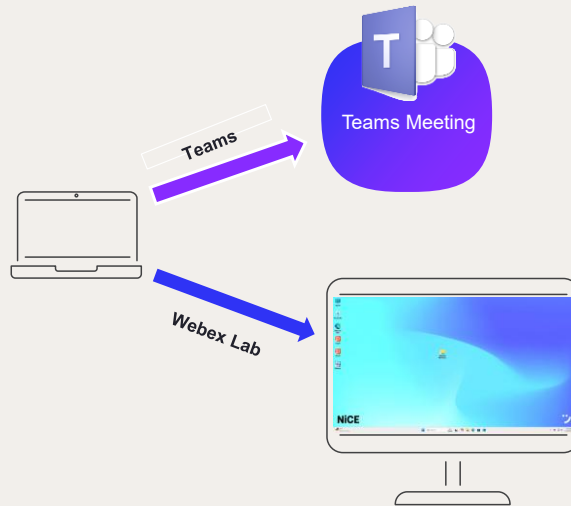
Training Environment consists of 2 sessions (Teams + Webex)

Teams Meeting for Trainer Presentations, Demos

Webex Lab Session for Hands-on Exercises



Two monitors are required so both sessions can be visible at the same time



- This is a diagram of the virtual environment. A Teams session will be used for the audio conference. A second Webex connection will be made to computer.

Navigating the Teams Session

01:34

Take control

Pop out

Chat

People

Raise

React

View

Rooms

Apps

More

Camera

Mic

Share

Leave

Pop-out for Full Screen

Participants

React

Mute / Unmute

NiCE Product Intro and Application Design

NiCE Training

Create a NiCE world

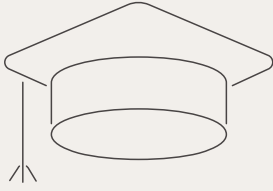
Participant 1

Participant 2

NiCE

A stylized logo icon consisting of two blue dots above a black curved line, resembling a smile or a checkmark.

Certification



Need 80%
or better



Practical



- To receive a certification, you must have a scores above 80% on the practical and theoretical exams.



Course Materials



Course Book



Exercise Book

Let's take a look!



agenda

Day 1

- DEF Installation
- Introduction to Process
- Build and Edit Workflow – Data Folder

Day 2

- Build and Edit Workflow - DB
- Data for NIA
- Call Scenarios

Day 3

- Hierarchy Import
- Encryption Process

Day 4

- Upgrades
- Troubleshooting
- Practice

Day 5

- Practical Exam





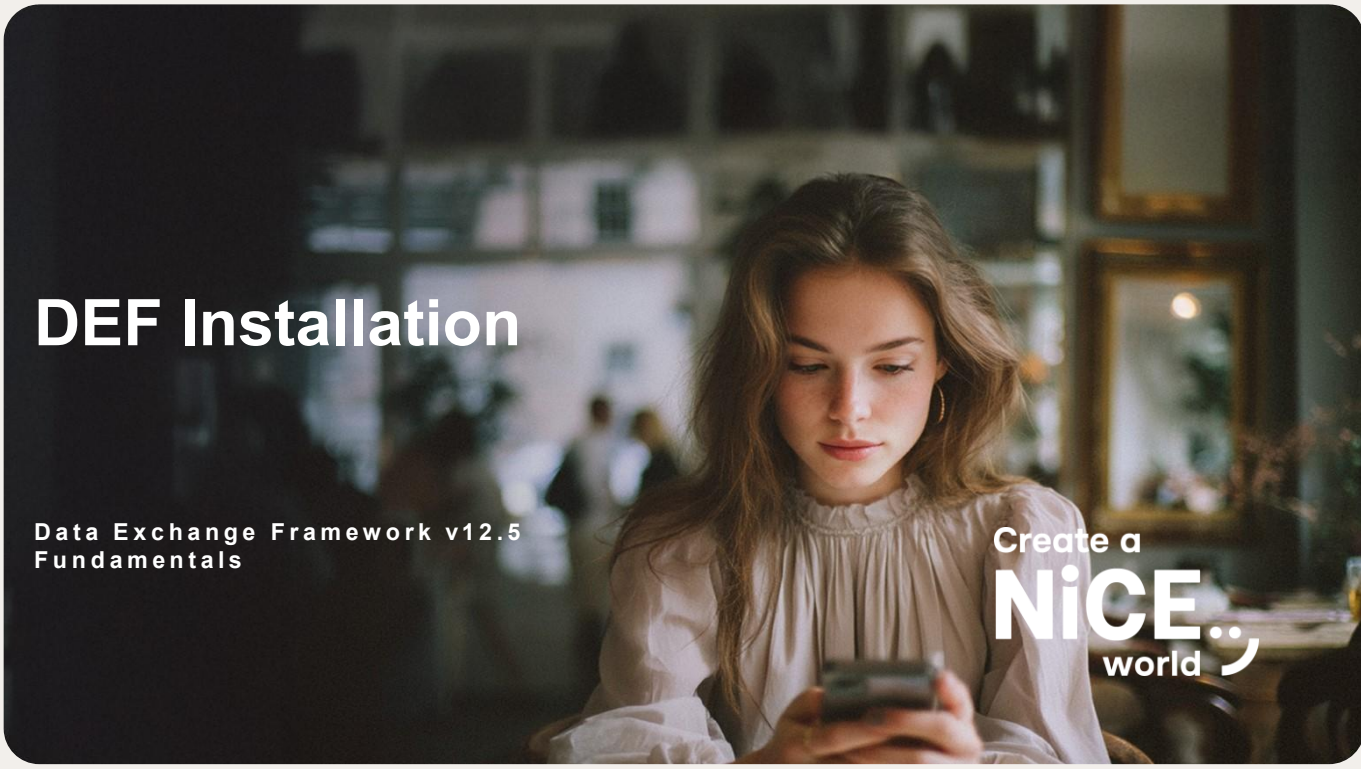
Before we begin...

- What is your experience with NiCE Solutions?
- Where are you located?
- What do you want to learn about in this course?





Thank You



DEF Installation

Data Exchange Framework v12.5
Fundamentals

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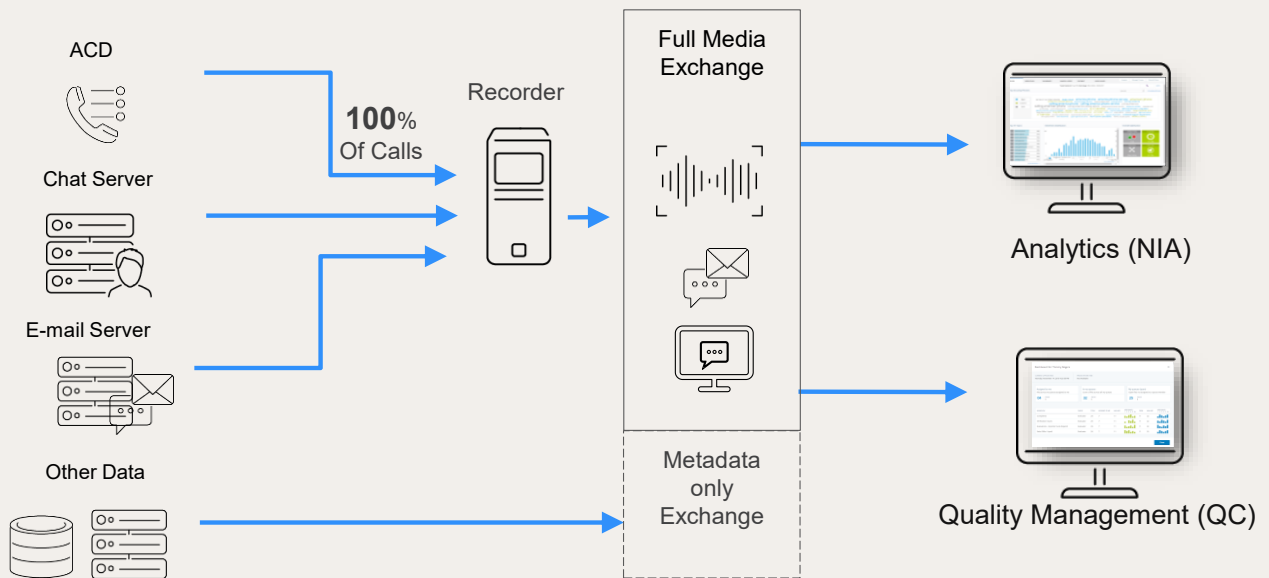
objectives

By the end of this lesson, you will be able to:

- Understand the pre-requisites to install DEF
- Install the NEAT Server
- Install the NEST Server



How Does It Work



Interaction Analytics is not a recording platform; it is an interaction analytics platform; to analyze the call, we first need to import them from the recording system. For that purpose, Interaction Analytics has the extractor tool, which extracts the *audio files, text interactions, and related metadata* from the recording system and copies them into Interaction Analytics servers.

DEF securely extracts information, such as audio files, text interactions, and related metadata, from diverse and geographically distributed data sources.

Extract the data, format, and process the data according to a data structure expected by Interaction Analytics 's application server such as QC & IA, and finally transfer the data to the destination location.



Prerequisites



Installation Files

-  CumulativeUpdateTool 12.5.0.2
-  CustomDEFActions 12.5.0.2
-  DataExchangeFramework Install 12.5.0.2



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The Installation files are in the software zip file, and they are also available on the SDC



System Requirements

Operating System

- Microsoft Windows Server 2016
- Microsoft Windows Server 2019

Browser

- Chrome 84 or later
- Microsoft Edge 84 or later

SQL Server

- SQL Server 2017
- SQL Server 2019
- External Databases (for resources) must accept TLS 1.2 connections



System Requirements

Service Account

- R/W privileges for all file resources
- DB owner privileges for database
- Logon as service right

DEF runs under service account, this account must have read and write privileges for all network, and local files resources that DEF will use. The service account must have DB owner privileges for the DEF database.



DEF Installation – NEAT Server



DEF Components



Both components must be installed, even if they are installed on the same server

The Data Exchange Framework consists of 2 components:

1. The Administration component which stores the global configuration is where you define the settings and build the process (the action you want to perform: import user, extract interaction metadata, etc.)
2. The Processing component executes the process, connect to the customer source system, extract the data, format, and process the data according to a data structure expected by Interaction Analytics 's application\server such as QC & IA, and finally, transfer the data to the destination location.

These components can be deployed in a single-server deployment model or a distributed-server deployment model.

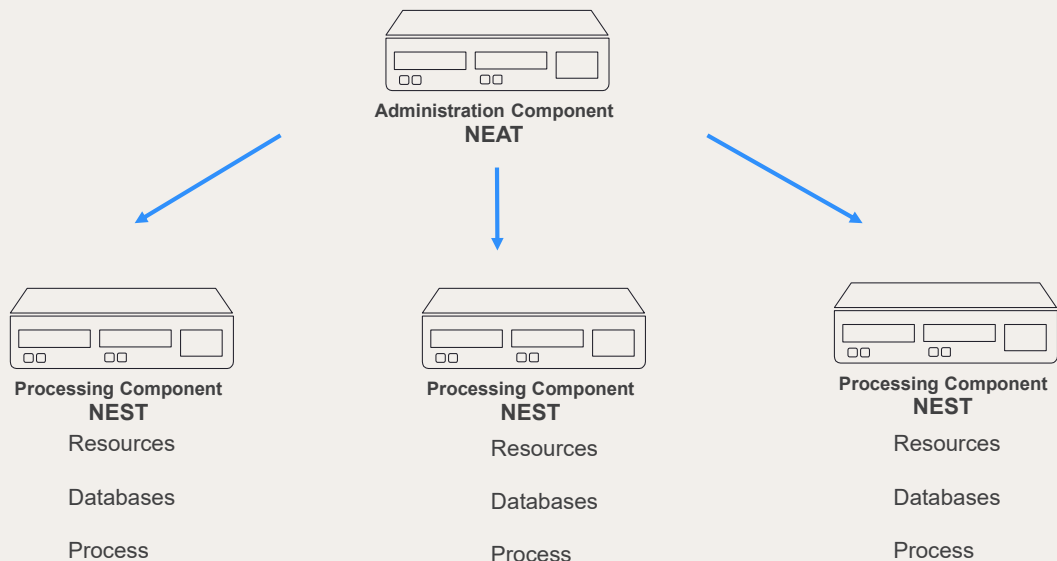


You will typically have one administration server with many associated processing servers. The processing servers report state and status information to the administration server via a “heartbeat” message.

NOTE: As of DEF v5.4, all installations are considered to multi-server. This means that a NEAT and NEST component must be installed. Both components can be installed together on the same server to create a single-server installation. Additional NEST servers can be added later if needed. This was done to allow for an easy expansion from a single-server to multi-server deployment.



Multi Server Installation



On a site with a multi-server setup, we have one NEAT server (administrator component – DEF UI) and multiple NEST servers. The NEST server executes the process we build\define on the NEAT server.

Each NEST (processing) server has its own process, resources, and database.

When we create a process and resources, we create them beneath a server hierarchy, so a process is associated with a NEST(processing) server. This process will only run on that NEST server.

Both the NEAT (administrator) and the NEST (processing) server have an SQL database installed. The NEST (processing) server has the database installed since it does not communicate directly back to the NEAT (administrator) server. It is only communicating back to the NEAT server via the heartbeat messaging.

In the meantime, for the NEST (processing) server to do its job, it needs a “copy” of the database with all the working tables that the NEAT(administrator) server has.

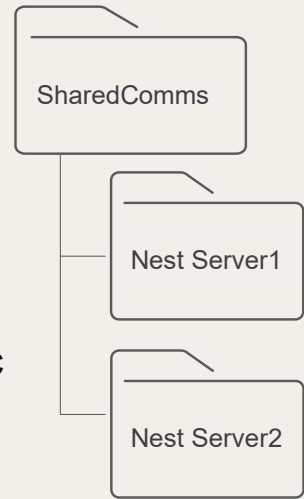
The 2 DBs are synched by the messaging between the NEAT and the NEST server.

Each NEST server has its own recourses, database and process



Communication Folder

- Build a directory to be used for heartbeat communication between DEF Servers
Example - \\NEAT1\\SharedComm
- Sub-folders will be created for each NEST server (Created automatically by DEF)
- When installing NEST, you will be prompted to enter the UNC path to the sub-folder



Create a shared folder that can be used for heartbeat communication between the DEF servers.

In training we will create this folder on NEAT1 (D:\SharedComm)

Give access to niceservices account

NEST Server Configuration

Server - NEST (Decryption Server)

Copy Deactivate Delete Process Analysis Disposition Report Wizard Refresh

Properties Edit Recent Activity Compute Nodes

ID	2
State	ONLINE
Status	ONLINE
Name	NEST (Decryption Server)
Communication Directory	/nest
Encrypt communications	true
Threads per action	2
Process debug level	WARN

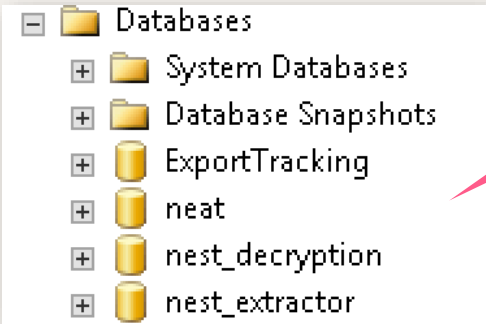
DEF will create this sub-folder automatically

When the NEST server is defined in NEAT, enter the path to the NEST sub-folder used for heartbeat communication with NEAT. Use the format /<folder name>.

NOTE: DEF will create this folder for you. Make note of the UNC path to this folder, you will need to enter it when installing the NEST software.



DEF Databases



DBs can be created manually or automatically by the installer



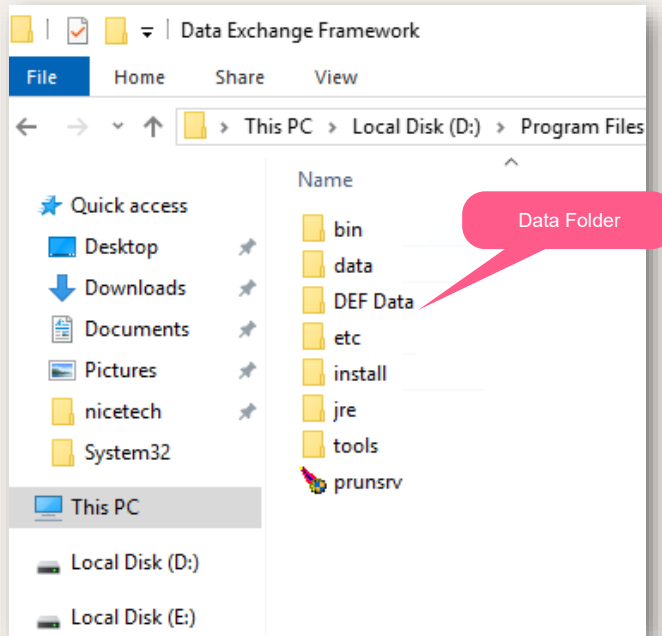
If you need to specify DB data and log file locations, then create the DBs manually in SQL

As of DEF 5.4, you no longer need to manually create the databases in SQL. The installer can create it automatically. However, if you need to specify the database data and log file locations, then the database should be created manually in SQL.

To create the database manually, login to SQL and create a new DB. Verify that you have created the database in the SQL Server instance you are using. The database you create must be owned by the DEF service account user. Follow the same steps to create a database for NEST.

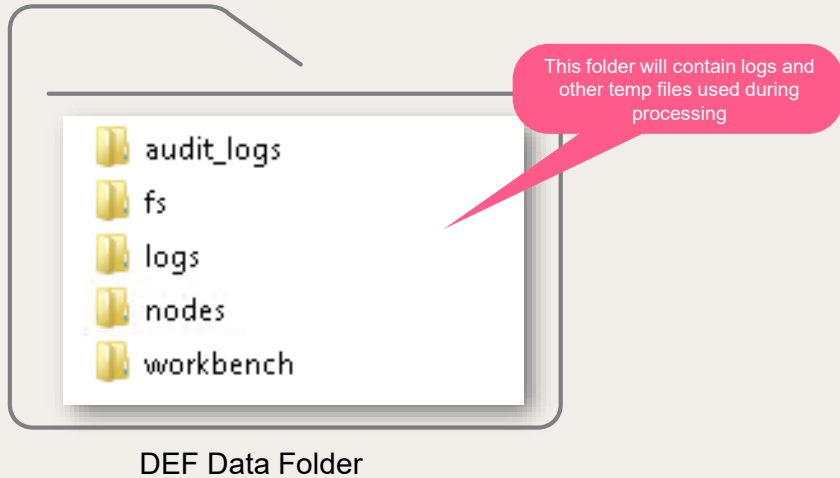


Create Data Folder



After the installation files are extracted, create a folder called “DEF Data”. You will be prompted to browse to this folder during the install.

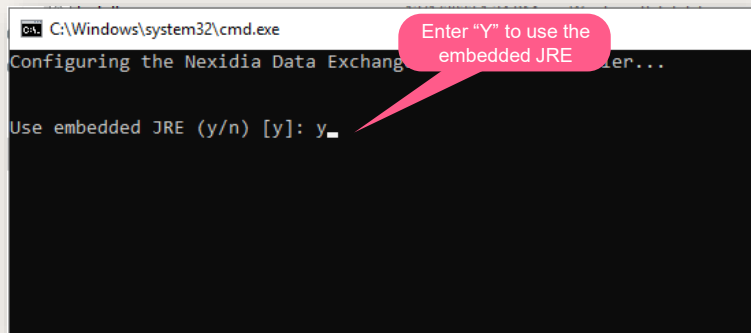
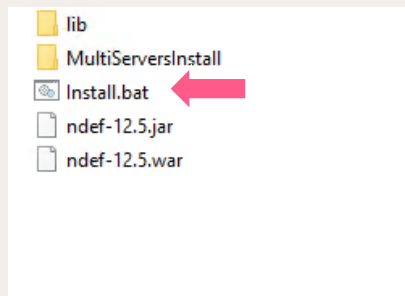
Data Folder



The DEF Data folder is where DEF will create log files and temporary directories used during processing.

The recommended path is a data directory under the install directory.

DEF Installation - NEAT



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The installation process will be the same for both NEAT and NEST. If they are being installed on the same server, you will run the install twice, once to install NEAT and again to install NEST.

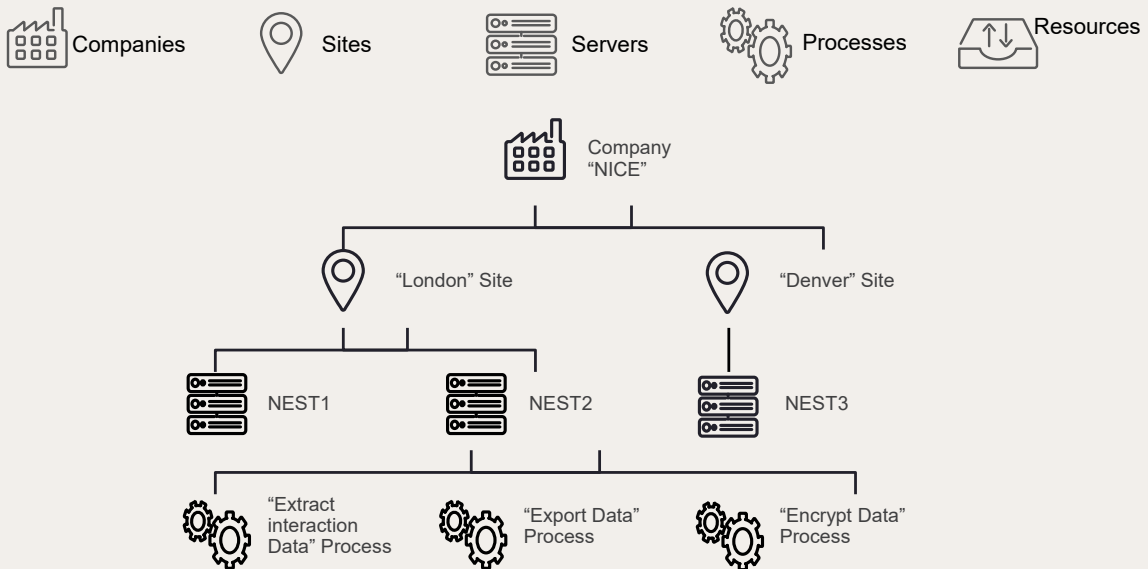
Navigate to the DEF installation directory: Installation Drive:\Program Files\Data Exchange Framework\install

Execute Install.bat as an administrator. You are prompted to specify if you want to use the embedded JRE or not.

Note: If you don't want to use the embedded JRE, then you will need to have JAVA_HOME installed in your environment. Refer to the system requirements guide to determine the version of Java that is required.



The Basic Entities



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We have five basic entities:

- Companies - use to define the organization
- Sites within the Company – use to define different regions\branches\offices within the company
- Servers are the locations of the DEF processing component (NEST)
- Process and Resources within the Servers – an example of the process is to copy files for both interactions and metadata from the recorder

This is one too many relationships: 1 company - many sites, one site – many servers, one server – many processes and resources.

The Company, Site, and Server views allow you to organize the processes. The Company view allows you to identify the owner of a process, the Site view provides location information, and the Server view contains the computing environment hosting the processes. You must define these before you can define a process.



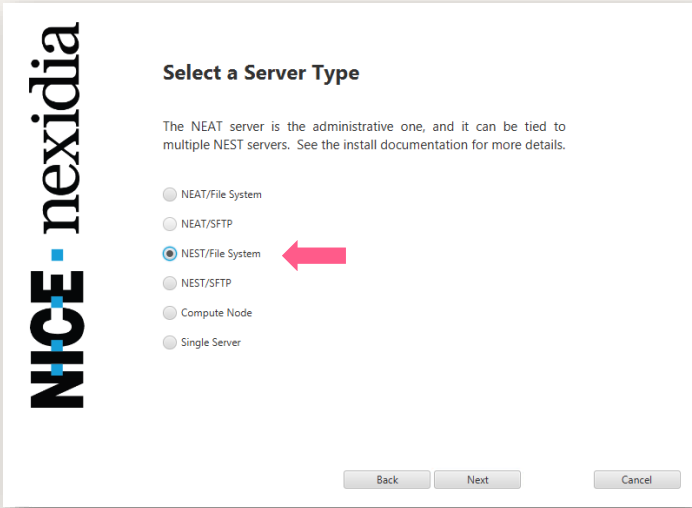
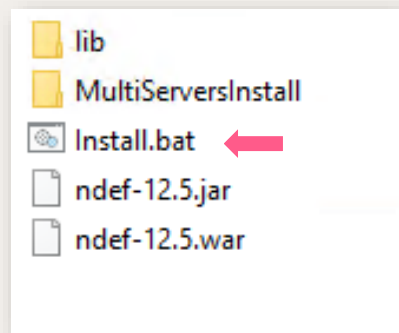
Do it Yourself

- Refer to Exercises 1 – Install the NEAT Server

DEF Installation –
NEST Server



NEST Installation



The installation process will be the same for both NEAT and NEST. If they are being installed on the same server, you will run the install twice, once to install NEAT and again to install NEST. Go to the NEST1 server and extract the installation files just as you did for the NEAT server. Be sure to create a DEF Data folder as well.

Select the NEST install option, based on the usage of File System or SFTP for communication to NEAT.



Do it Yourself

- Refer to Exercises 2 – Install the NEST Server

summary

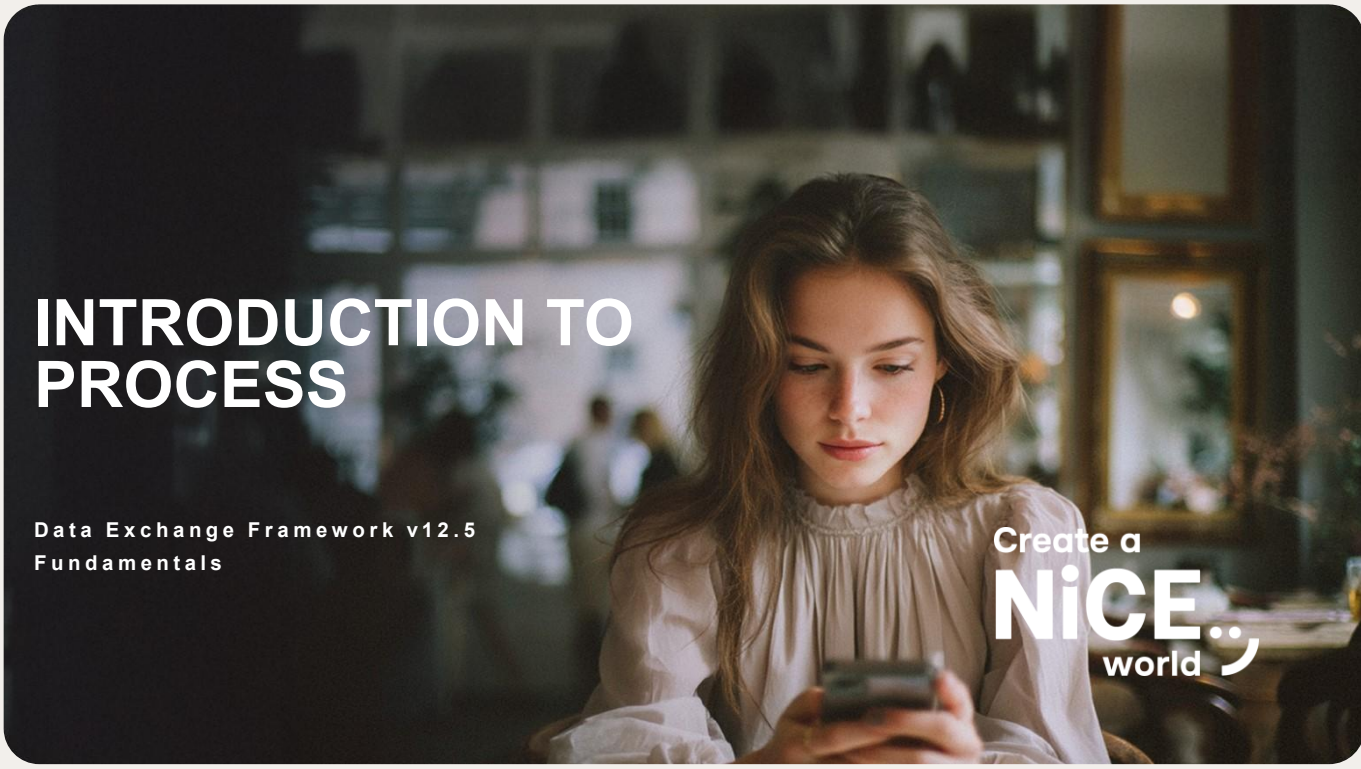
In this lesson we covered:

- DEF installation pre-requisites
- Installing the NEAT Server
- Installing the NEST Server





Thank You



INTRODUCTION TO PROCESS

Data Exchange Framework v12.5
Fundamentals

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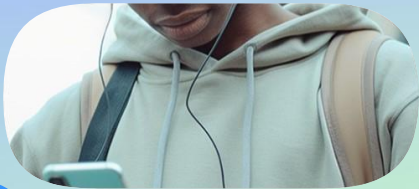
objectives

By the end of this lesson, you will be able to:

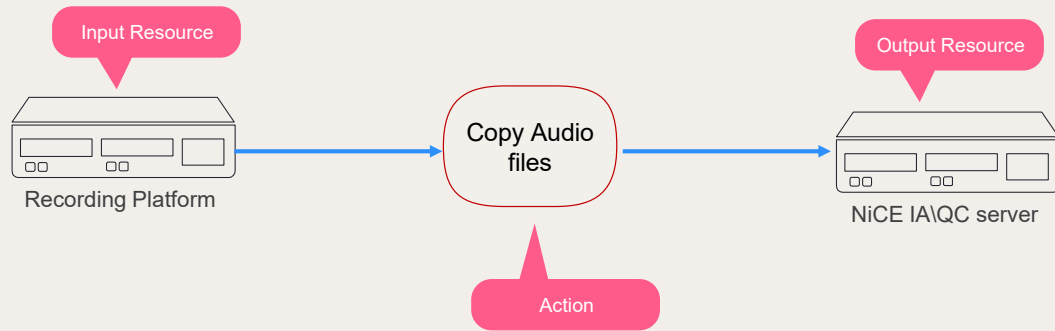
- Understand how to create a process workflow using actions and resources
- Understand the purpose of a working table



Process



What Is a Process



As we have seen before, DEF is an Extract Load and Transfer utility. In the UI, we will define all the tasks we want the DEF to perform. Those tasks are called processes.

An example of a process is an Extraction process that copy files (audio files\interaction) from location A (recording platform) to location B (IA\QC server).

Each process is combined from resources and actions, in our example :

Location A and B are the resources that specify the source and destination directories, and the action is copy file.

Process Resources

- A Resource is a container of data in the form of a database or as files in a folder
- There are four types of data sources
 - Folder type of data source
 - SFTP type data source
 - Database type of data source
 - Amazon S3 – Cloud resource using Amazon AWS

A Resource is a container of data in the form of a database or as files in a folder

There are four types of data sources

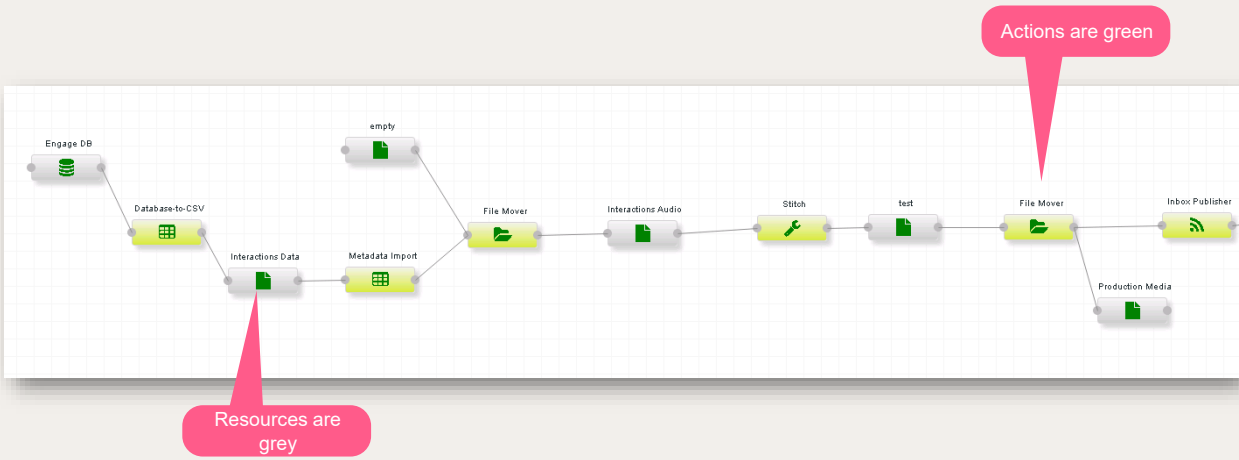
- Folder type of data source
- SFTP type data source
- Database type of data source
- Amazon S3 – Cloud resource using Amazon AWS

Database type of data source

We will select the relevant data source type according to where data is located.



Process



This is an example of how process looks like



Creating a Process

The screenshot displays the NICE Processes management interface. On the left, a navigation tree shows a hierarchy: Companies > NICE > Sites > Training > Servers > NEST > **Processes**. A red arrow points to the 'Processes' item in this tree. A pink callout bubble points to the 'Processes' item with the text: "Import a process template or create your own".

The main area shows the 'Processes' section with a toolbar containing a plus icon, a refresh icon, a download icon, and a share icon. A red arrow points to the plus icon. Below the toolbar is the 'Add a Process' dialog. The dialog has tabs for 'View Quantum' and 'View Schedule'. It contains a 'Process Template' section with a text area for the template and a 'Select a Process Template (*.json)' button. Below this are input fields for 'Name', 'Description', 'Start Time' (set to 04/26/2022 00:00), and 'Interval' (set to 1). A 'Submit' button is at the bottom right. A 'Show Advanced' link is at the bottom left. A note at the bottom right states 'Fields in bold are required'.

Add a new process by right-clicking on the process icon from the hierarchy tree or click on action buttons in the toolbar. You can import a process template or build your own.

 Add a Process ✕

 View Quantum

 View Schedule

Name

Description

Start Time

Interval

(min: 1; max: 60)

Interval Type

Blackout Start Time

Blackout End Time

Quantum (minutes: 0 = none)

Submit

[Show Advanced](#) Fields in bold are required

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The Interval is the next scheduled run time calculated from the start date + interval.

Interval Type is time measurement value: Second, Minute, Hour, or Day.

Blackout Start and End Time windows are the time-frames in which the process cannot be scheduled to start/stop. If there is an existing process that started 2 min before the blackout window, it will not stop it, but the process won't start again.

Enter the process name in the Name field.

Blackout Time – during the blackout time, the process will not run, for example, a process can be scheduled to run every hour but only after the working hours

Creating a Process

Add a Process ×

View Quantum

View Schedule

Quantum Upper Date Bound

12/31/2100 00:00

?

Quantum Timezone Offset (minutes)

0

?

(min: -1440; max: 1440)

Quantum Start Time Shift

+00:00:00

?

Quantum Stop Time Shift

+00:00:00

?

Process Group

?

Notification email on completion

false

▼

?

Pct variance notification threshold

0

?

(max: 100)

Process Quantum will be covered later in the course

Submit

Show Advanced

Fields in bold are required

Working Table Name

Process Group - set a name to the Process Group; this parameter is optional. This action will create a new table in the DEF DB with the same name. All relevant data to this process will be inserted into this table.

Every process inside this process group will work against the same working table, and everything will be tracked against it, so it is recommended to separate them logically by giving each a different name.

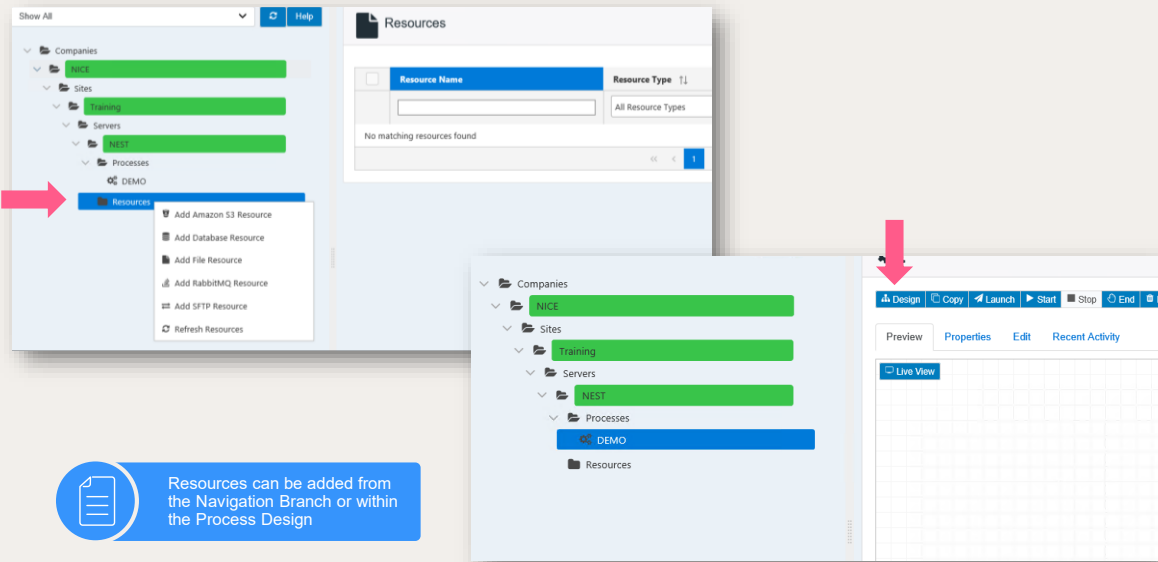
If you don't set a value in this field, it will use the default "working table," and it will be hard to identify your process.

Notification email - select if you want to receive an email if the process completes successfully.

Note: you will always receive an email if the process fails



Creating a Resource



Resources can be added from the hierarchy tree, or in the process designer window

Creating a File Resource

Add a Resource

Name

Demo input File

Path

D:\Demo input

?

User Name

?

User Password

?

Submit

Fields in bold are required

There is no need to pre-define the folder, the path provided will be created automatically by DEF

Enter the name to of the file source and the path to the file

Note: username and password is required for windows CIFS authentication only



Available actions

ACTION	INPUT RESOURCE	OUTPUT RESOURCE	DESCRIPTION
Alternate Media ID	File System or SFTP (required only when identifying using a file system test).	File system (used only to resolve/capture the full path to the alternate media).	Identifies alternate media by looking either for specific metadata field values, or by testing against the file system (either by input resource scanning or processing from metadata). The HasAlternateMedia, AlternateMediaSourcePath, and AlternateMediaExpirationDateUTC metadata fields are populated when the application finds alternate media.
Chat CSV to JSON	File System or SFTP	File System or SFTP	Transforms web chats into an expected format so that Nexidia Analytics and Nexidia Text Grid can process web chats.

 Refer to the user guide to view all available actions.

In the Interaction Analytics Data Exchange Framework User Guide there is a table that provides a list of all available actions and the required resources for inputs and output



Available actions

WELCOME, NIN ? ⓘ ↩

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Save Save & Close Publish

Available Actions - Profile 1 - Microsoft Edge

localhost:8080/NEAT/webhelp/content/_def_onlinehelp/topics/actions_information/available%20actions.ht...

GO TO ACTIONS

Available Actions

Agent Hierarchy Action

Agent/User Import V2 Action

Alternate Media ID Action

AppLink Metadata Retriever

AppLink SAP Business Data Upload

AppLink SAP Upload

AppLink SAP User Upload

Calabrio Media Retriever Action

Calabrio Metadata Retriever Action

Call Summarization Request

Call Summarization Response

CSV Reformatter Action

CSV Bulk-load Action

CSV Text Extractor Action

CSV-to-Database Action

CXone Agents Retriever V2 Action

Process Workflows > Actions > Available Actions

SHARE THIS PAGE GIVE FEEDBACK

ON THIS PAGE Available Actions

Available Actions

This table provides a list and description of the available actions and their required input and output resources. Click on the links to display the generic configuration options. The required fields for each action are displayed in bold font in the user interface. You can show or hide the advanced configuration options as needed.

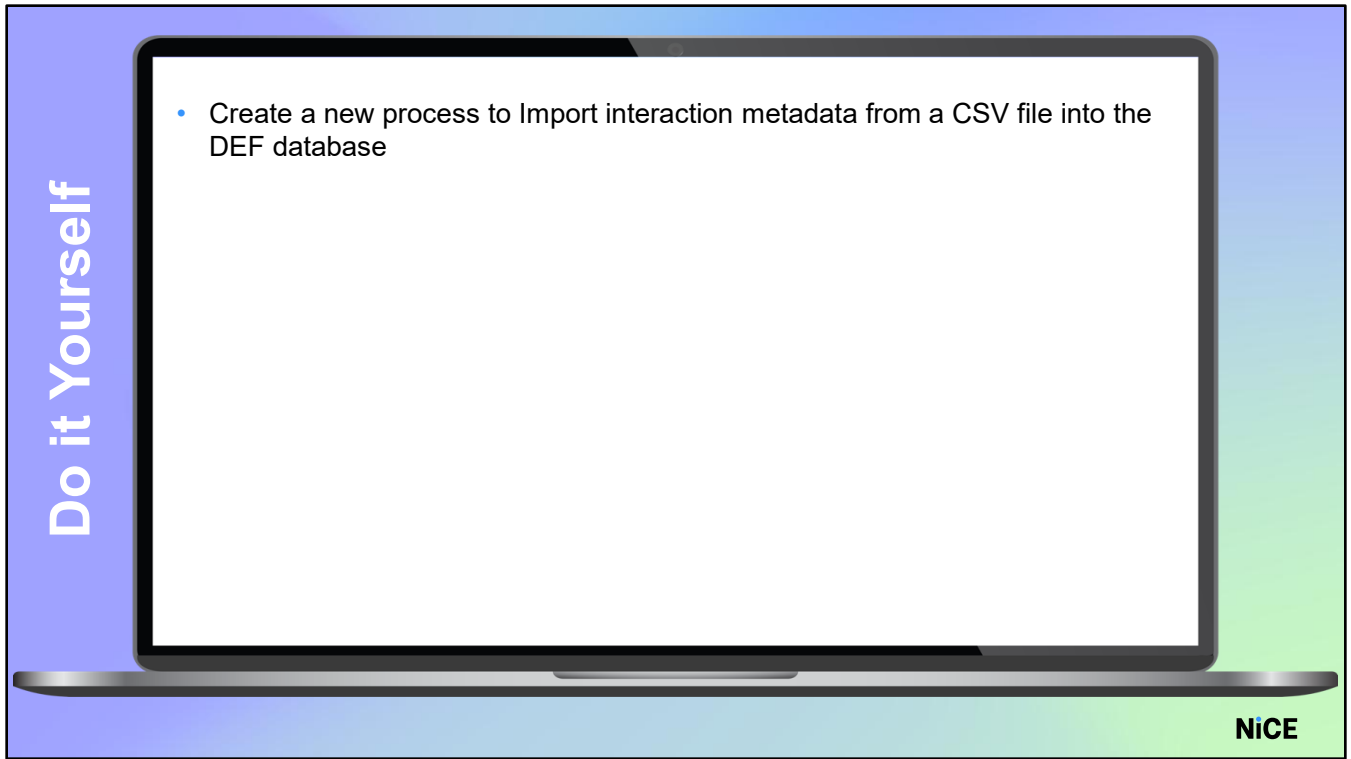
ACTION	IN P U T	OUTPUT	D E S C R I P T I O N
Agent Hierarchy	Fi le	N/A	F e

You can find the list of all the available actions online in the Help center



Process and
Working Table





Create a process that runs every hour and imports data from CSV files into the DEF database \ working table.

Assign a name to the Process Group

Working tables are database tables used by processes to record metadata and track the flow of records actively moving through them.



Creating Process

 Add a Process ✕

View Quantum

View Schedule

Process Template

?

Name

DEMO

Description

Start Time

05/01/2019 00:00



Interval

1

(min: 1; max: 60)

Interval Type

Hour

Quantum Stop Time Shift

+00:00:00

?

Process Group

DEMO

?

Submit

Schedule to run every hour

Specify the process group name

Create a new process, and set the process Schedule to run every hour, specify the process group name. Click Submit



Metadata Import Action

ACTION	INPUT RESOURCE	OUTPUT RESOURCE	DESCRIPTION
Metadata Import	File System or SFTP	n/a	Iterates a directory and takes all of the CSV files to a table for primary metadata. The attribute will be overwritten if the names are the same. You can specify metadata as either primary or secondary in the user interface.

No output resource is needed

Reminder- a process is a collection of data resources and actions.

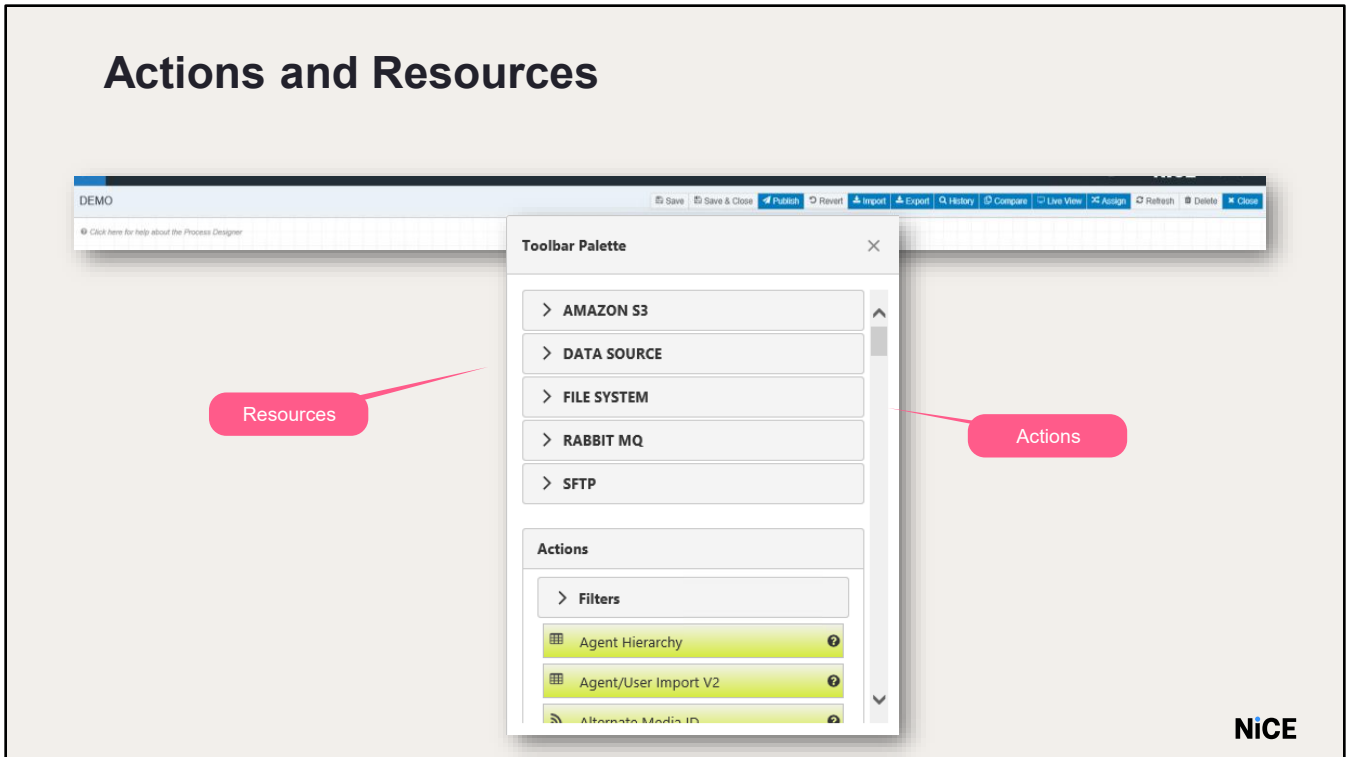
In this example, the process input resource (the location of the data) is a folder that contains the CSV file.

The action that the process needs to perform is to import metadata.

No output resource is needed, the metadata import action insert the data directly into the DEF database (working table)



Actions and Resources

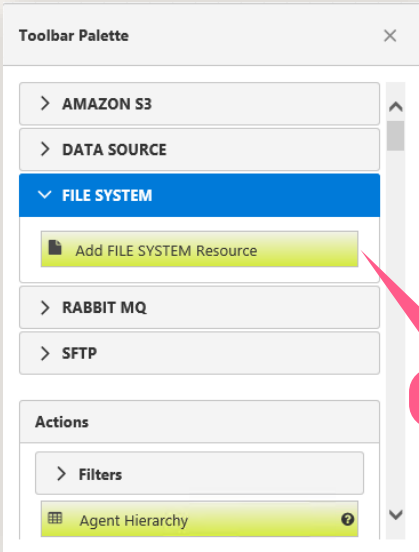


In the process's designer window right-click to display the floating toolbar palette.

A Process Designer contains two key components: Resources and Actions,

Using the Process Designer, we will add actions and resources to the grid and connect the items to produce a workflow

Create a File System Resource



The 'Resource - New Resource' dialog box is a form for creating a new resource. It has a title bar with 'Resource - New Resource' and a close button. Below the title bar are three tabs: 'Assign', 'Delete', and 'Close'. The 'Assign' tab is selected. The form has four fields: 'Name' (Input CSV), 'Path' (D:\input CSV), 'User Name' (empty), and 'User Password' (empty). Each field has a help icon (question mark) to its right. A 'Submit' button is located at the bottom right of the form.

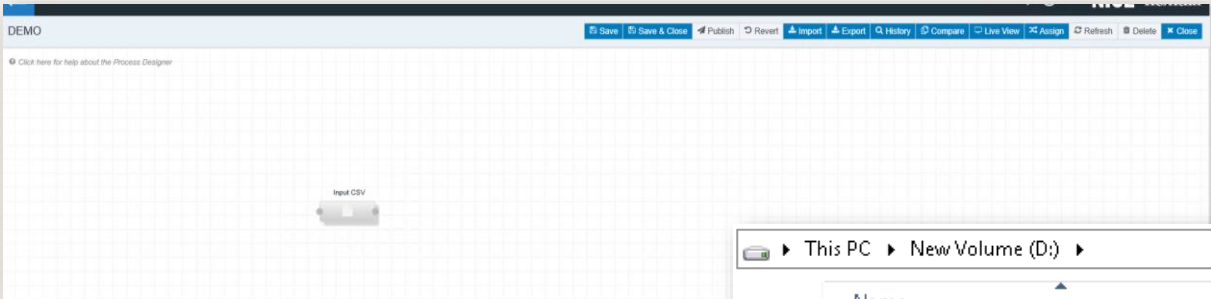
File System Resource

In our example, the data is in a CSV file, so our data source is a folder – the location of the CVS files. We need to start by defining the process data source

From the toolbar palette, expand the file system resource and click on the "Add File System Resource."



Create a File System Resource



DEMO

Save Save & Close Publish Revert Import Export History Compare Live View Assign Refresh Delete Close

Click here for help about the Process Designer

input CSV

Reminder – a local file system resource will be created on the NEST server.

This PC > New Volume (D:) >

Name
DEF_Data
Demo input
input CSV
nicetech
Program Files

The file resource was created in the NEST server



Metadata Import Action

The screenshot displays the 'Action - Metadata Import' configuration window. On the left, the 'Toolbar Palette' lists various actions, with 'Metadata Import' highlighted. The main configuration area includes fields for 'Process files from metadata' (false), 'File filter' (*.*), 'Import Type' (Primary), 'Import file key column', 'Required metadata', 'Import only required metadata' (false), 'Delete metadata inputs after import?' (false), and 'Fail if nothing is imported within # minutes' (0). A pink callout bubble explains that setting 'Delete metadata inputs after import?' to TRUE would delete the CSV file after processing. A pink arrow points to the 'Submit' button at the bottom right.

Now that we have our data source, the next step is defining the action we would like to perform on the data, in our example we want to import the data into the working table in the DEF DB.

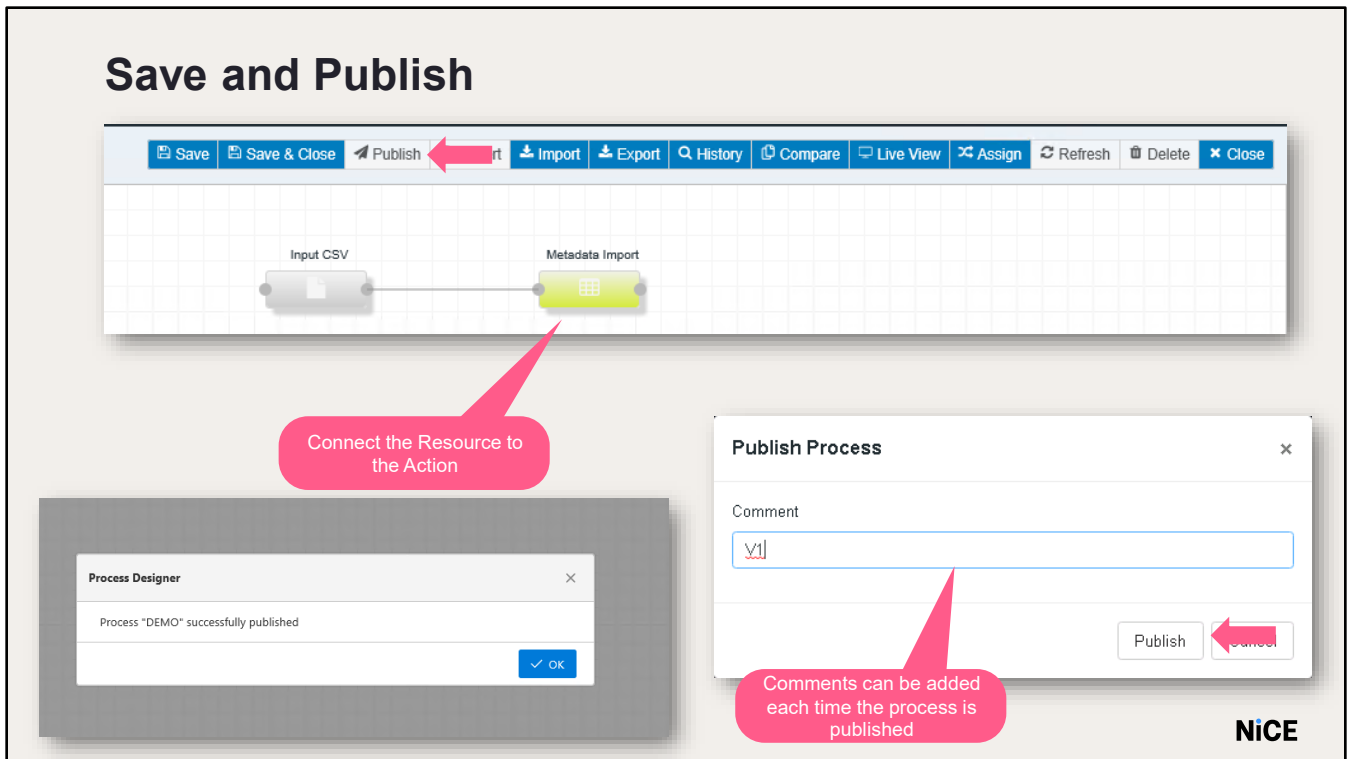
Metadata Import - iterates a directory and imports all the CSV files to a table for primary metadata.

From the toolbar palette, expand the actions tab and select the "Metadata Import" action.

Set the "Delete metadata inputs after import" parameter to false and click submit.

"Delete metadata input after import" field - When this is set to TRUE, the CSV file will be deleted after the process is completed.





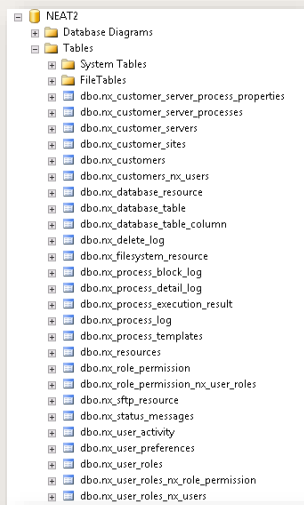
A process needs to be saved and published. The Publish action allows you to publish a process definition, including validation, and to make the process available to be scheduled for execution. Once the process is saved the Publish button is enabled. You will see the success message displayed. Once published Launch the process.

To view the current processing progress in a “live” environment, we will use the Live View button in the process launch window. You can optionally select the Live View button to see the following information which is refreshed automatically. When live is enabled, LIVE VIEW ON message flashes next to the Live View button. You can hover over this message to display the date and time stamp for the live view results that are showing.

Working Table
(Process Group)



Working Table



Working_table +
Process Group

Working_table +
Process Group+\$X

After the first run of the process, two tables are created in the DEF database :

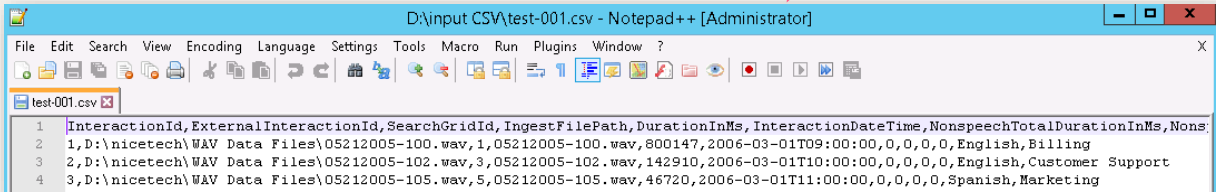
1. Working table + process group - the table contains all the data that need to be processed or failed to process
2. Working table + process groups\$X –" rollover" table that contains all the data that was successfully processed

By default, a process that has no process group defined uses the "working table." Processes with a defined process group use the "working_table_{process_group_name}."



Working table

Original CSV File

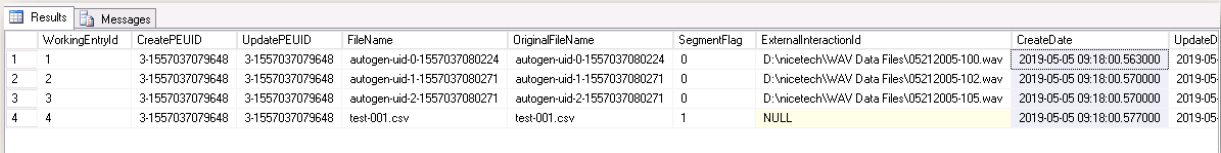


D:\input CSV\test-001.csv - Notepad++ [Administrator]

```

1 InteractionId,ExternalInteractionId,SearchGridId,IngestFilePath,DurationInMs,InteractionDateTime,NonspeechTotalDurationInMs,Nons:
2 1,D:\nicetech\WAV Data Files\05212005-100.wav,1,05212005-100.wav,800147,2006-03-01T09:00:00,0,0,0,0,English,Billing
3 2,D:\nicetech\WAV Data Files\05212005-102.wav,3,05212005-102.wav,142910,2006-03-01T10:00:00,0,0,0,0,English,Customer Support
4 3,D:\nicetech\WAV Data Files\05212005-105.wav,5,05212005-105.wav,46720,2006-03-01T11:00:00,0,0,0,0,Spanish,Marketing
  
```

Working Table



	WorkingEntryId	CreatePEUID	UpdatePEUID	FileName	OriginalFileName	SegmentFlag	ExternalInteractionId	CreateDate	UpdateD
1	1	3-1557037079648	3-1557037079648	autogen-uid-0-1557037080224	autogen-uid-0-1557037080224	0	D:\nicetech\WAV Data Files\05212005-100.wav	2019-05-05 09:18:00.563000	2019-05-
2	2	3-1557037079648	3-1557037079648	autogen-uid-1-1557037080271	autogen-uid-1-1557037080271	0	D:\nicetech\WAV Data Files\05212005-102.wav	2019-05-05 09:18:00.570000	2019-05-
3	3	3-1557037079648	3-1557037079648	autogen-uid-2-1557037080271	autogen-uid-2-1557037080271	0	D:\nicetech\WAV Data Files\05212005-105.wav	2019-05-05 09:18:00.570000	2019-05-
4	4	3-1557037079648	3-1557037079648	test-001.csv	test-001.csv	1	NULL	2019-05-05 09:18:00.577000	2019-05-

The CSV file contains three interactions. However, the process completed with four items processed why is there a difference? The four items that moved are the three interactions in the file and an additional row for the file name and details; this is due to the DEF internal management – it marks the file as completed by adding the file name after all items \ interactions. Under the Working Table, we have four entries: 3 rows for the interactions and an additional row for the file name. The metadata import is tracking the file name. This is how the process tracks which files have already been imported.



Metadata Import Action Configuration

Action - Metadata Import

Delete Close

Edit Properties Labels Logs

Process files from metadata false

File filter *. *

Import Type Primary

Import file key column

Required metadata

Import only required metadata false

Delete metadata inputs after import? false

Fail if nothing is imported within # minutes 0

The name of the column will be used as the Working Table key for the file

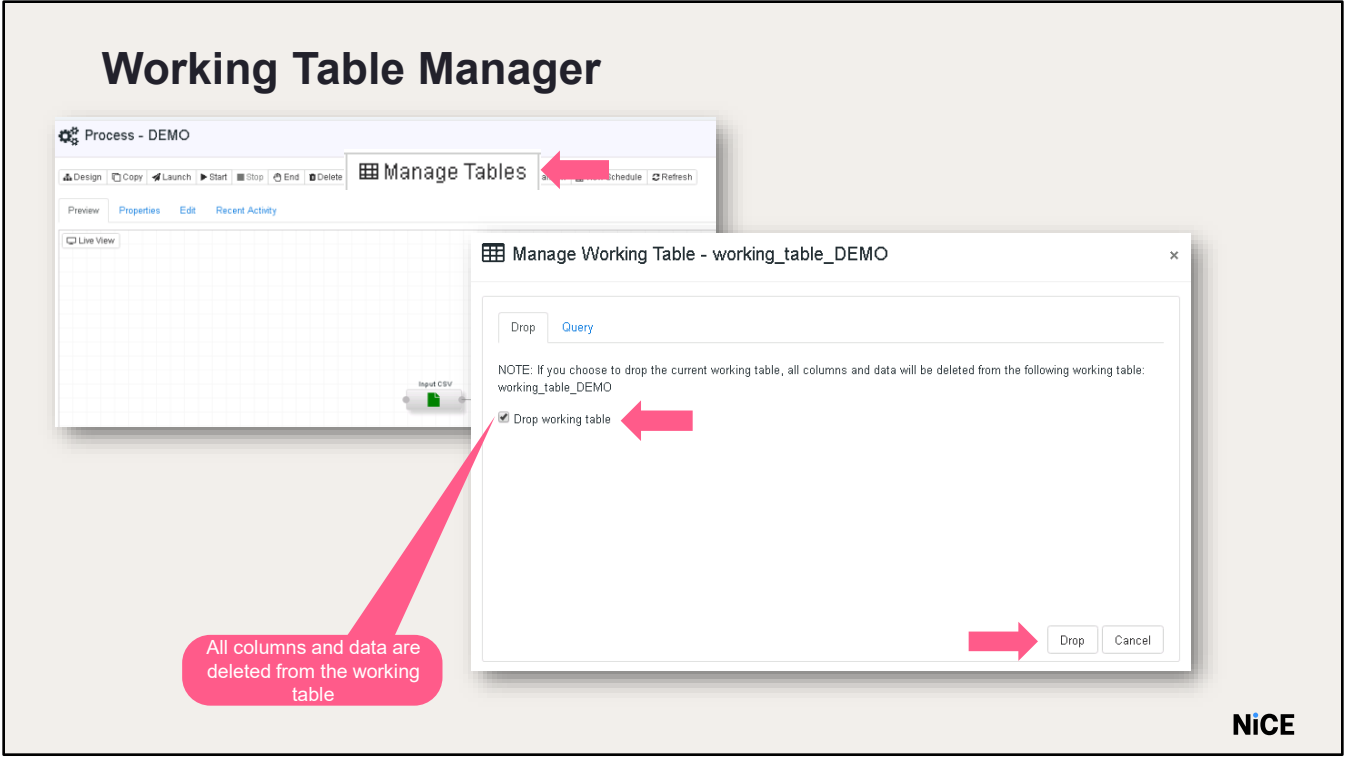
Import Type:

- Primary - each record will be imported into the working table; existing records will not be updated. Duplicates will be created if the same record already exists.
- Secondary - data will be joined, updating existing rows in the working table. However, you must enter the incoming columns with the matching columns to be updated. This option is more complex and not recommended unless you have only a few columns to import and update.

Import file key column field – The key column is the column in a CSV file that's going to be used to uniquely identify the data moving through the system.

What does it mean?

- When this field is left empty, the file name column will be used as the Working Table key.
- When this field is set with a specific column to uniquely identify the data, the column will be used as the Working Table key



The Working Table Manager provides administrator control over process group working tables. Access this window by selecting the Manger Tables option on the Processes Page. If you choose to drop the current working table, all columns and data are deleted from the working table



Do it Yourself

- Refer to Exercises #3, 4 and 5

summary

In this lesson we covered:

- How to create a process workflow using actions and resources
- The purpose of the working table





Thank You

Build And Edit Workflow - Data Folder

Data Exchange Framework v12.5
Fundamentals

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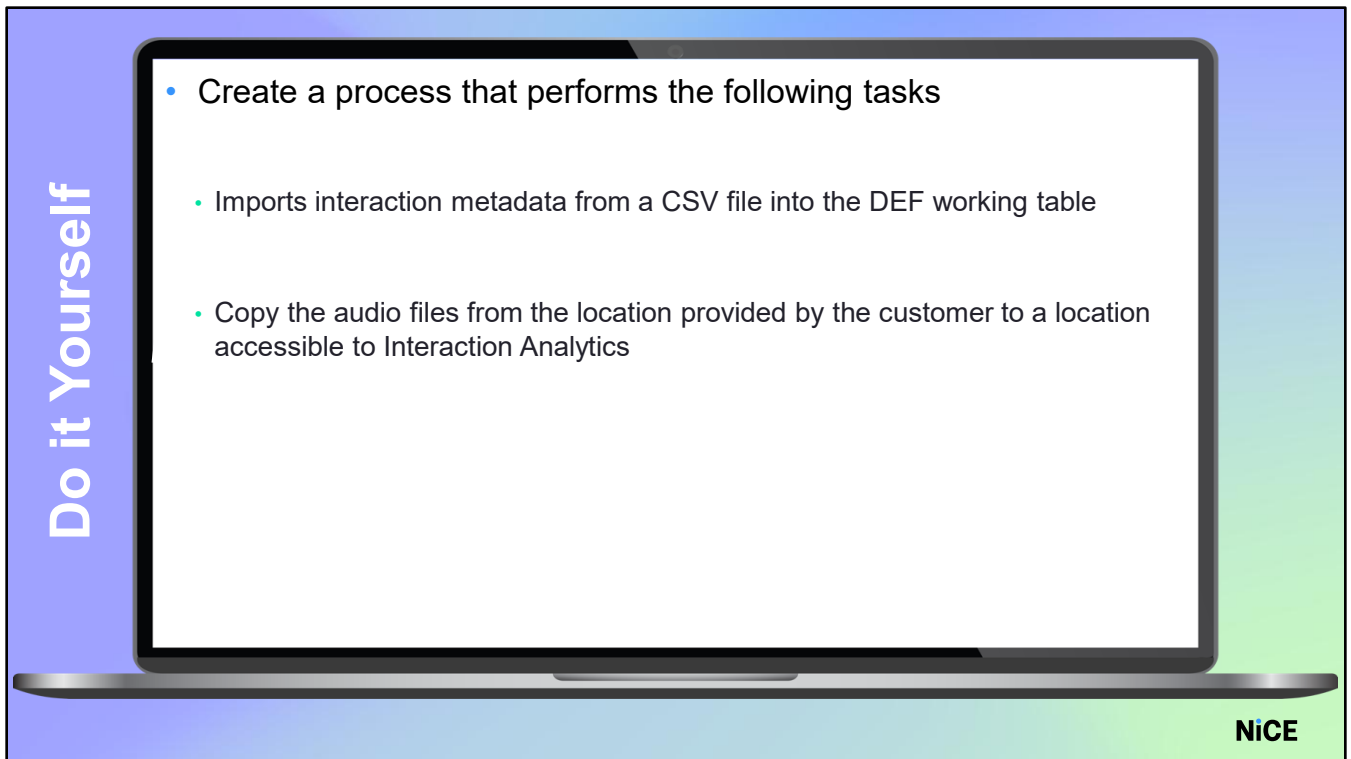
By the end of this lesson, you will be able to :

- Describe how to create a process workflow using actions and resources
- Describe how to create a process using file system and data source



Workflow- Data Folder





The customer, due to security and performance reasons, will not grant permission to allow the DEF server to access the recorder & database environment.

Instead, the customer extracts the audio files from the recorder and the metadata from the DB and places the files in a shared folder that the DEF can access.

Create a process that will import the interaction's metadata and audio file from the shared folder provided by the customer into the DEF server.



Metadata Import



Metadata File

Interaction ID		Audio file name							
A	B	C	D	E	F		K	L	
InteractionId	ExternalInteractionId	Search	IngestFilePath	DurationIn	InteractionDateTime		Language	Department	
1	D:\nicetech\WAV Data Files\05212005-100.wav	1	05212005-100.wav	800147	2006-03-01T09:00:00		English	Billing	
2	D:\nicetech\WAV Data Files\05212005-102.wav	3	05212005-102.wav	142910	2006-03-01T10:00:00		English	Customer Support	
3	D:\nicetech\WAV Data Files\05212005-105.wav	5	05212005-105.wav	46720	2006-03-01T11:00:00		Spanish	Marketing	
Audio file location						Audio file language			

The interaction’s metadata file contains the following information: Call-ID, audio file location, audio file name, call duration, interaction date, audio file language, and which department handles the call.



Metadata Import Action Configuration

Action - Metadata Import

DeleteClose

EditPropertiesLabelsLogs

Process files from metadatafalse?

File filter*.?

Import TypePrimary?

Import file key columnIngestFilePath?

Required metadataCallDateTime.timestamp without time zone.

Import only required metadatafalse?

Delete metadata inputs after import?false?

Fail if nothing is imported within # minutes0?

Submit

Show Advanced

Fields in bold are required

IngestFilePath will be used as the Working Table key for the file

Metadata elements that must be in the incoming data

If the columns are not there in the incoming data, the Action will Fail

In this example, we provide the "IngestFilePath" column, which contains the audio file name as the Working Table key.

Process files from metadata: Specify if you want to process files from metadata.

When you select "true," the application processes files from the process group's existing-working table.

When you specify false, the application processes files from attached input resources.

The required metadata field displays the selected metadata in the output file.

The required / expected metadata should be defined in a comma-separated format: column name1:datatype,column name2:datatype

Here is an example: CallDateTime.timestamp without time zone,InteractionID:bigint– if the columns are not there the action will fail.

We completed the first part of the process – Import the interaction's metadata.

Now we need to "build" the second part of the process – move the audio files from the location provided to us by the customer to the DEF server.

Do it Yourself

- Create a process that performs the following tasks
 - Import interaction metadata from a CSV file into the DEF database
 - Move\Copy the audio files from the location provided by the customer to a location accessible to Interaction Analytics

NICE

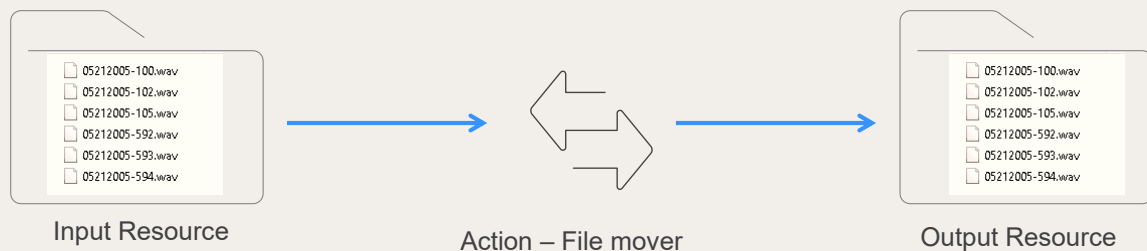
The Customer provided you with a folder with the interaction's audio files, import the audio file from the folder provided by the customer and place them into a folder that the NIA \ QC applications can access



Move / Copy Files



Import Audio File



- Copy and paste a file from a source to a destination
- Filter on type of file to move from location A to location B

We need to build a process that moves the audio files from the input resource that was provided by the customer to the output resource that it's accessible by the NIA\QC application.

File Mover creates a copy of the input file(s) and moves it to a destination location.

Using the File Mover action, you can specify to move a particular type of file such as *WAV from location A to location B.

Locations A and B are the resources that specify the original and destination directories.

The File Mover receives a file system or SFTP location as an input and returns results to a file system or SFTP location as the output resource.

If no output is provided, we will receive an error message when publishing.

File Mover Action Configuration

Process files from metadata: Processes files from attached input resources

File path column name: Process files from the working table

Delete inputs after move: Choose to delete the inputs after the move is complete

File path column name: Provide metadata column to determine the location of the files

Processes files from metadata: When you select "true," file mover will use the existing Working Table metadata to determine which file to move. Only files that appear in the table will be moved.

When you specify false, the file mover processes files from attached input resources.

File path column name: the name of the metadata column in the Working Table to use to determined the location of the files, this option appears only when “processes files from metadata” is set to “true.”

You can provide a column that contains a full absolute path to the file location or a partial path – each option affects the process design differently

Specify if you want the application to delete the inputs after they are moved. The valid options are "true" or "false".

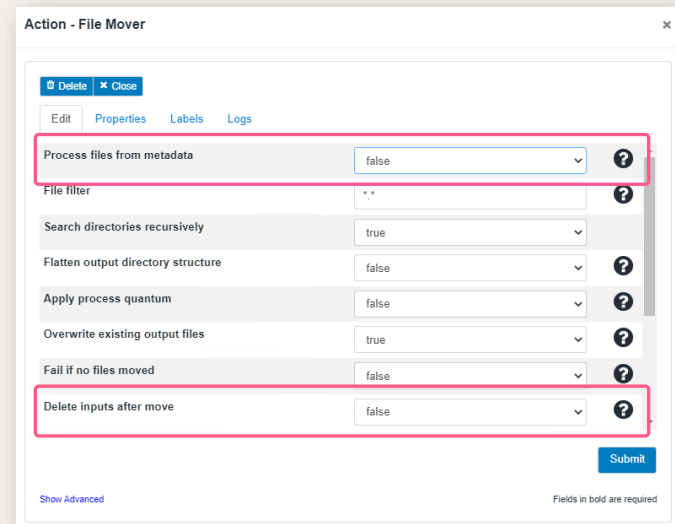
Note: in the lab set the " Delete inputs after move" parameter to false



Option 1:

Input Resource: File System

Process files from metadata = false



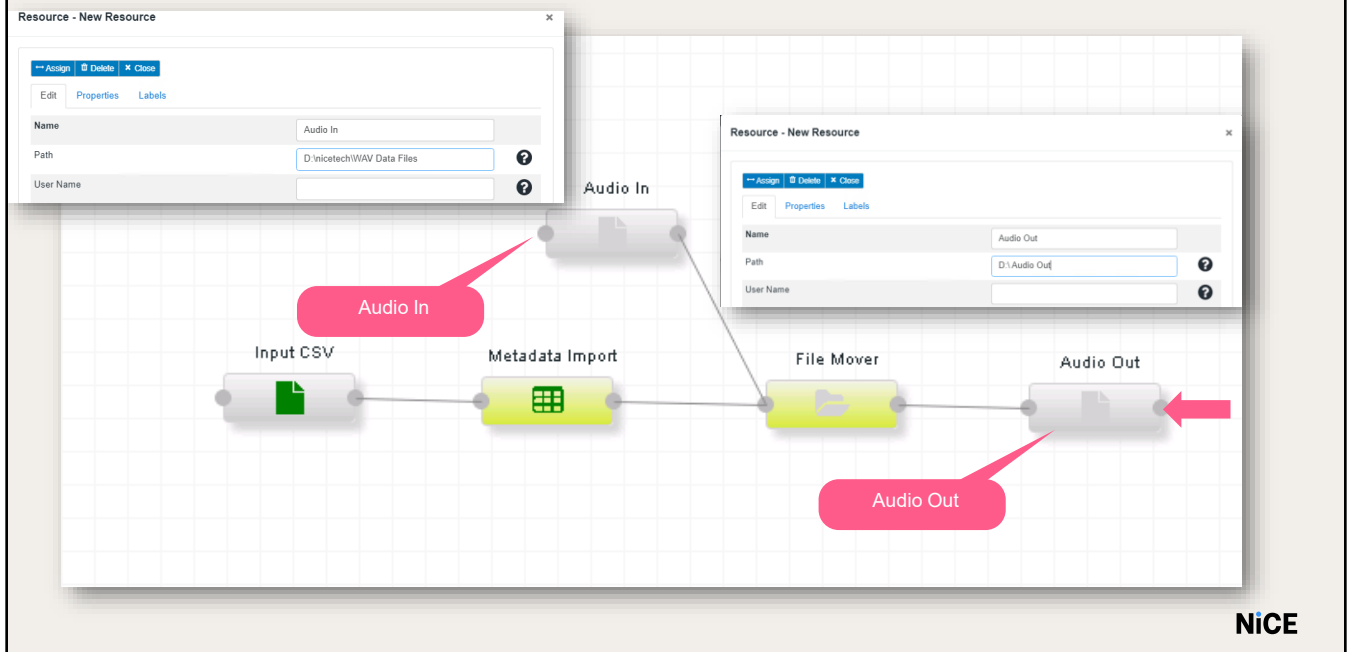
Parameter	Value	Required
Process files from metadata	false	Yes
File filter	*.*	Yes
Search directories recursively	true	No
Flatten output directory structure	false	No
Apply process quantum	false	No
Overwrite existing output files	true	No
Fail if no files moved	false	No
Delete inputs after move	false	No

We will set the Process file from metadata: parameter to “false,” we will need to provide the input resource from the type file system.

Important: In the 'file mover' configuration action window, set the "Delete inputs after move" parameter to false.

That way we will not delete the audio files from the input resource

Adding Input and Output

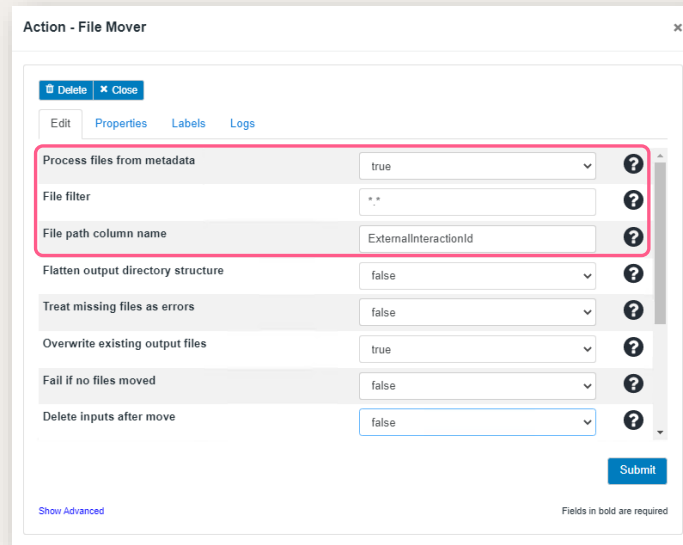


The location of the audio file in the training lab:
D:\nicetech\WAV Data Files
Drop the working tables and launch the process

Option 2:

Input Resource: Working Table

Process files from metadata = true



Action - File Mover

Delete Close

Edit Properties Labels Logs

Process files from metadata	true	?
File filter	* *	?
File path column name	ExternalInteractionId	?
Flatten output directory structure	false	?
Treat missing files as errors	false	?
Overwrite existing output files	true	?
Fail if no files moved	false	?
Delete inputs after move	false	?

Show Advanced

Submit

Fields in bold are required

Processing files from metadata - when set to "true," the application processes files from the process groups' existing working table.

File Path Column name: – the name of the metadata column in the Working Table used to determine the location of the files. This option appears only when “processes files from metadata” is set to true.

The “Flatten output directory structure” should be set to true.

True - move the files to a flat directory structure

False – Destination file structure will match the source file structure

File Mover – Understand Your Metadata

A	B	C	D	E	F	K	L
InteractionId	ExternalInteractionId	Search	IngestFilePath	DurationIn	InteractionDateTime	Language	Department
1	D:\nicetech\WAV Data Files\05212005-100.wav	1	05212005-100.wav	800147	2006-03-01T09:00:00	English	Billing
2	D:\nicetech\WAV Data Files\05212005-102.wav	3	05212005-102.wav	142910	2006-03-01T10:00:00	English	Customer Support
3	D:\nicetech\WAV Data Files\05212005-105.wav	5	05212005-105.wav	46720	2006-03-01T11:00:00	Spanish	Marketing

IngestFilePath column contains the .wav name

ExternalInteractionId column contains full absolute path

To design the process correctly, you should be familiar with the import data files and the metadata within them.

Using the import metadata action, we have the data located in the Working Table.

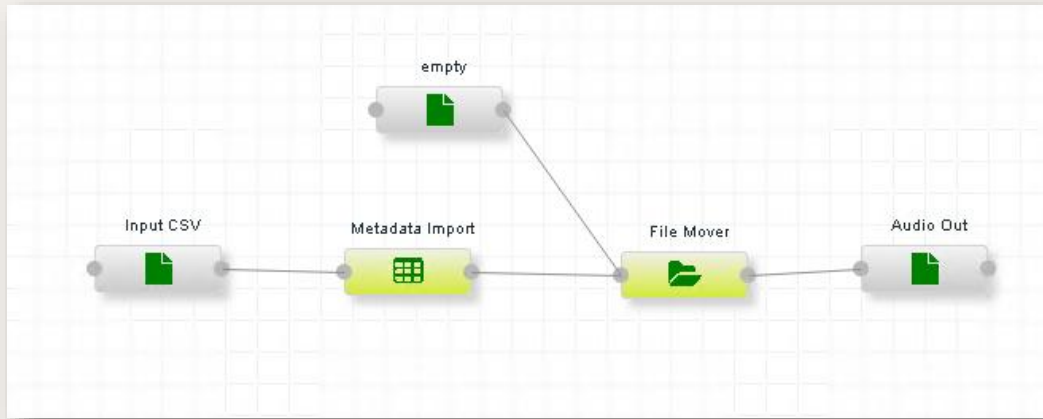
The full path to the files is stored under the “ExternalInteractionId” column, and the file name is located under the “IngestFilePath” column.

To make the file mover work with metadata, we need to modify the “Metadata Import” action.

Set the import file key column to 'ExternalInteractionId' in the metadata import configuration window



File Mover – Using Metadata



Create an input resource – file system type with an empty (blank) path.

In our example, the folder D:\nicetech\WAV Data Files contains six interactions, but only the three that appear in the Working Table were moved.

In this example, we set the “ExternalInteractionId” column as the file path column name.

Since “ExternalInteractionId” contains the full absolute path, we don’t need a file system folder as input for file mover action.

On the other hand, the file mover must have an input resource linked to it.

How is this conflict resolved?

Option 2:

Input Resource: Working Table +File system

Action - File Mover

DeleteClose

EditPropertiesLabelsLogs

Process files from metadatatrue

File filter**

File path column nameIngestFilepath

Flatten output directory structurefalse

Treat missing files as errorsfalse

Overwrite existing output filestrue

Fail if no files movedfalse

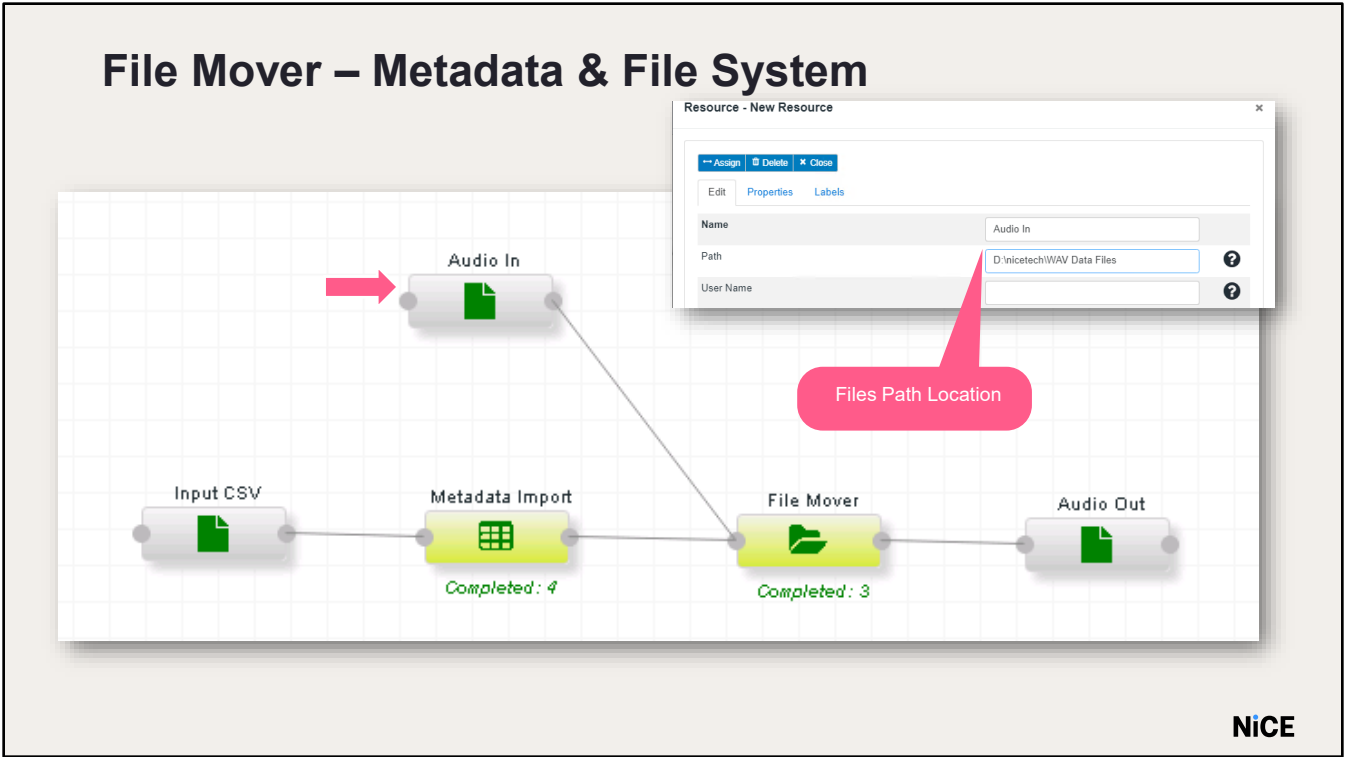
Delete inputs after movefalse

Show Advanced

Fields in bold are req

Submit

In this example, we will use the metadata from the column that contains the partial path to the location of the audio file and an input folder that will contain the second part of the file location. Combining both will produce a full path.



In this example, we set the “IngestFilePath” column as the file path column name.

Since it contains the file names, we need to provide the path for those files; therefore, create a file system folder as input for file mover action with the path for destination files.



Do it Yourself

- Refer to Exercises # 6 & 7



Parameterized Folder By Date

Parameterized folder path

Working table contains InteractionDate column

onId	ExternalInteractionId	SearchGridId	IngestFilePath	DurationInMs	InteractionDateTime	NonspeechTotalDurationInMs
1	D:\nicetech\WAV Data Files\05212005-100.wav	1	05212005-100.wav	800147	2006-03-01T09:00:00	0
2	D:\nicetech\WAV Data Files\05212005-102.wav	3	05212005-102.wav	142910	2006-03-01T10:00:00	0
3	D:\nicetech\WAV Data Files\05212005-105.wav	5	05212005-105.wav	46720	2006-03-01T11:00:00	0
4	NULL	NULL	NULL	NULL	NULL	NULL

We can direct each record to a specific folder based on existing metadata within the Working Table. This is called parameterization.

We can parameterize the structure of the Output data to fit the structure that is expected by the Interaction Analytics application. For example, customers will want to organize and analyze by a line of business, date, sites, organizational structure, etc.

We can set the desired structure using the file mover action.

The final output of the file mover can be parameterized, meaning we can direct each record to a specific folder base on existing metadata within the Working Table. For Example: If the metadata is LanguageID or CallType, we will direct the file to a specific language or call type folder.



- File resource path:
C:\Temp\nice\nia_ingest\audio\{languageID}
where call type can be English, Spanish ,French, Russian, etc.
- File resource path:
C:\Temp\nice\nia_ingest\audio\{CallType} where
call type can be voice, screen, chat, email, etc.

When metadata is a date type format (such as Start Date) we can parameterize the records to refer to a specific year\month\day\second folder

- File resource path:
C:\Temp\nice\{StartDate:yyyy}\{StartDate:mm}\{StartDate:dd}\{StartDate:hh}\{StartDate:mi}

Note: Substitution parameters are denoted using opening and closing { } brackets. If the parameter defined doesn't exist in your metadata, the action will fail. In our lab, we will parametrize based on interaction date as:
D:\AudioOut\{InteractionDateTime:yyyy}\{InteractionDateTime:mm}\{InteractionDateTime:dd}



Do it Yourself

- Refer to Exercise #8



summary

In this lesson we covered

- How to create a processing workflow using actions and resources
- How to create a process using file system a data source.





Thank You



Build And Edit Workflow - DB

Data Exchange Frameworkv12.5
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objectives

By the end of this lesson, you will be able to:

- Describe how to create a process using a database as the data source



Process Quantum



NiCE

Let's see now how to build a process using a database as our data source. Before we do this, let's first understand what a quantum is.



Process Interval and Quantum

Edit Process

Name: Table

Description:

Start Time: 02/26/2018 00:00

Interval: 1

Interval Type: Hour

Quantum (minutes; 0 = none): 1440

Quantum Lower Date Bound: 04/01/2017 00:00

Quantum Timezone Offset (minutes):

	Process Run				
Quantum	1	2	3	3	4
Current Date	2/26/2018 0:00	2/26/2018 1:00	2/26/2018 2:00	2/26/2018 3:00	2/26/2018 4:00
Quantum Size	0	0	0	0	0
Data portion\bulk Start time	04/01/2017 0:00	04/02/2017 0:00	04/03/2017 0:00	04/04/2017 0:00	04/05/2017 0:00
Data portion\bulk Stop time	04/02/2017 0:00	04/03/2017 0:00	04/04/2017 0:00	04/05/2017 0:00	04/06/2017 0:00

Let's look at the following example: we schedule the process to run every hour from February 26th, 2018, at midnight.

We set the quantum lower date to April 1st, 2017, which means the first time the process will run, it will retrieve information from that time.

The quantum is 1440 min (24hr), which means that the process will pull the data for 24 hours each run.

Each time the process runs, it tries to capture data between the last successful run and current time, but not more than the quantum size.

In our example, the first time the process will run will be on February 26th, 2018, at midnight (interval), and retrieve data from April 1st, 2017, 0:00 (quantum lower date) till April 2nd, 2017 0:00 (quantum size).

When we set the Quantum to 0, it's mean that it's disabled, so each time the process runs, it will try to capture data between the last successful run and the current time.

So the first time that our process will run, it will try to retrieve all the data from quantum lower date till current time.



Metadata Import + Audio Import

- Create a process that performs the following tasks:
 - Query the recording system database and import the interaction's metadata
 - Copy the audio files from the recording system to a location accessible to Interaction Analytics



The customer provides you with the DB connection details.
 Create a process that queries the customer's database then imports the interaction's metadata and audio files.

Process from Data Source

Add a Process

View Quantum

View Schedule

Name

Using Data Source

Description

Start Time

02/28/2023 00:00

Interval

1

(min: 1, max: 60)

Interval Type

Day

Blackout Start Time

00:00:00

Blackout End Time

00:00:00

Quantum (minutes; 0 = none)

1,440

Quantum Lower Date Bound

02/28/2018 00:00

Quantum Upper Date Bound

12/31/2100 00:00

Quantum Timezone Offset (minutes)

0

(min: -1440, max: 1440)

Quantum Start Time Shift

+00:00:00

Quantum Stop Time Shift

+00:00:00

Process Group

Data

Create new process

Specify the process group name



Database Resource

Resource - New Resource
x

← Assign
🗑 Delete
✕ Close

Edit
Properties
Labels

Name

Connection URL

User ID

Password

Submit

Fields in bold are required

Click to open the Connection Wizard

✎

?

The connection URL is a JDBC database connection string. DEF provides an internal wizard to create the JDBC connection string.

jdbc:sqlserver://Db server host
name;databaseName=nice_interactions;integratedSecurity=true;

Sample with Port specified:

jdbc:sqlserver://Db server
hostname;databaseName=nice_interactions;Port=1
433;integratedSecurity=true;

Note: in the training lab we will use DB1 as the DB server host name



Database Resource

Resource - New Resource

Assign Delete Close

Edit Properties Labels

Name Engage DB

Connection URL

JDBC Connection Wizard - Step 1 / 6 - Driver Type

Driver Type

SQL Server

Select the JDBC driver type.

Next Cancel

User ID

Password

Submit

Fields in bold are required

The SQL query to use in this demo located on NEAT1 server
D:\nicetech\SQL Queries\Exercise 9.txt

Step thru the wizard to define the DB Type, Hostname, Instance, Port and Database Name

Driver Type:

Hostname:

Instance:

Port :

Database Name:

Once connected to the customer's DB, query the DB, and extract the interaction's metadata. The "Database to CSV" action creates a CSV file based on database inquiry.

Database to CSV



Define Output

The SQL query is inserted to the "Select SQL" window
In the Pre-select SQL area - optionally, enter SQL script to execute before the main SQL query, for example, SQL scripts that create temp tables that will be used by the main SQL query, etc.

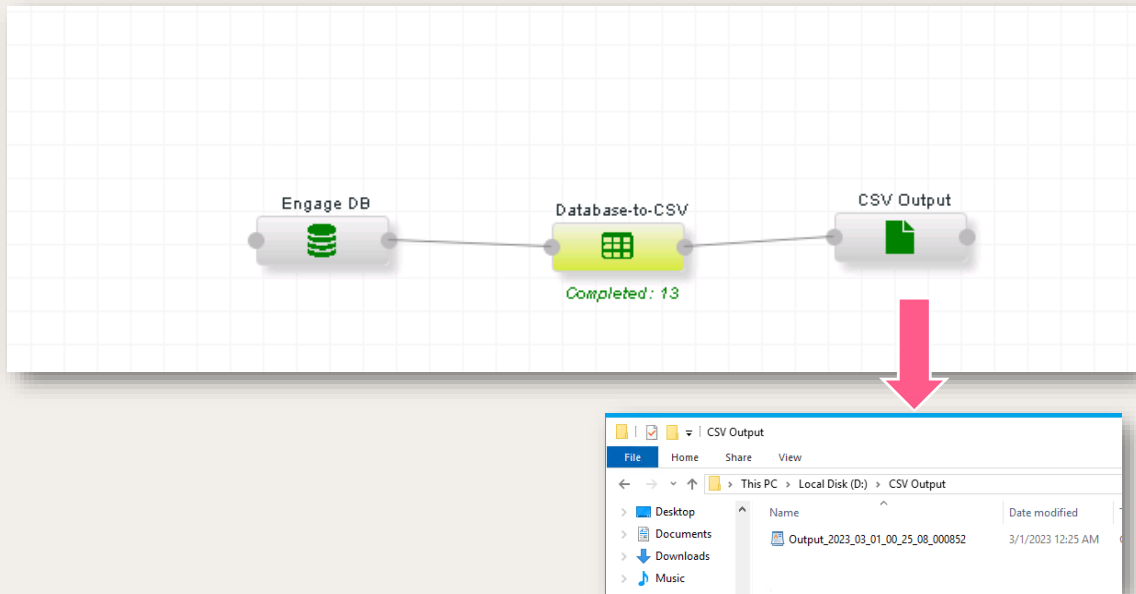
The Post-select SQL area, optionally, enter SQL script to execute after the main SQL query, for example, SQL scripts that perform data cleanup or delete the temp table etc.

Output file name – you can use a time stamp so it will be easier to track and debug, Optionally enter the output file name for the file being output. You can also include the "{TIMESTAMP}" parameter in the name to insert a time stamp. For example: agentRosterFile{TIMESTAMP}.csv.

If you leave this field blank, it is automatically generated.



Database to CSV



The "Database to CSV" action must be linked to an input resource. In our case, the input resource is Engage DB resource (DB Datasource). We also need to define where the CSV file will be placed - an output resource.

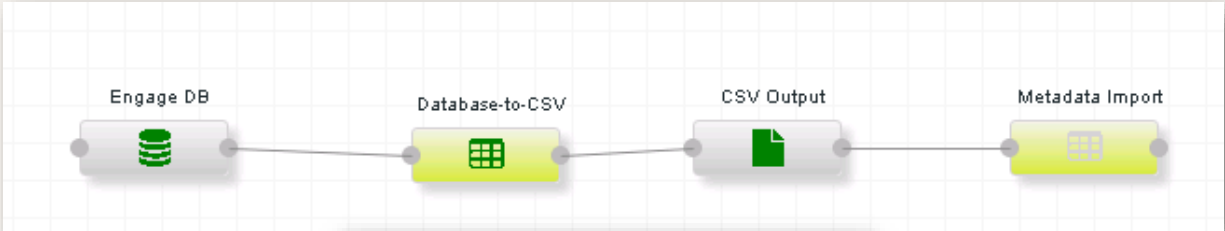
The date in the filename is the number of milliseconds since 1970. 31,556,952,000 mSec per year.

For this file $1,557,218,342,662 / 31,556,952,000 =$

49.3462848586264 years or 49 years

126 days 9 hours 27 minutes 19 seconds or May 7, 2019 09:27:19 AM

Metadata Import



DeleteClose

EditPropertiesLabelsLogs

Process files from metadatafalse?

File filter**?

Import TypePrimary?

Import file key columnInteractionID?

Required metadata?

Import only required metadatafalse?

Delete metadata inputs after import?true?

Fail if nothing is imported within # minutes0?

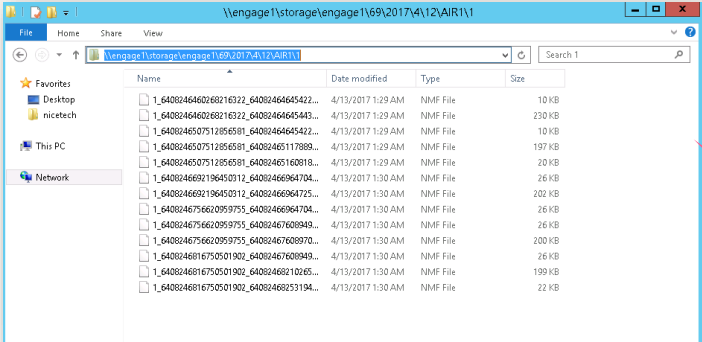
Submit

The next step is to insert the interaction’s metadata into the process’ Working Table through metadata input action.



Import Audio File

WorkingEntryId	CreatePEUID	UpdatePEUID	FileName	InteractionID	vcArchivePath
1	7-1557222301525	7-1557222301525	6408246460268216322	6408246460268216322	engage1\69\2017\4\12\AIR1\1\1_640824646026821632...
3	7-1557222301525	7-1557222301525	6408246507512856581	6408246507512856581	engage1\69\2017\4\12\AIR1\1\1_640824650751285658...
6	7-1557222301525	7-1557222301525	6408246692196450312	6408246692196450312	engage1\69\2017\4\12\AIR1\1\1_640824669219645031...
8	7-1557222301525	7-1557222301525	6408246756620959755	6408246756620959755	engage1\69\2017\4\12\AIR1\1\1_640824675662095975...
11	7-1557222301525	7-1557222301525	6408246816750501902	6408246816750501902	engage1\69\2017\4\12\AIR1\1\1_640824681675050190...
14	7-1557222301525	7-1557222301525	Output_1557222302013.csv	NULL	NULL



Partial path to the audio location

Full path to the audio location

In the Working Table and the metadata CSV file, we only have the partial path to the location of the audio files. We need to provide the full path to the files.



Do it Yourself

- Refer to Exercise #9



objectives

In this lesson we covered:

- How to create a process using a Database as the data source





Thank You





Data For NIA

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objectives

By the end of this lesson, you will be able to:

- Describe the data structure expected by NIA



Data For NIA

Option 1



Option 2



Interaction analytics server expects to receive the interaction's audio files and metadata files in one of the following ways:

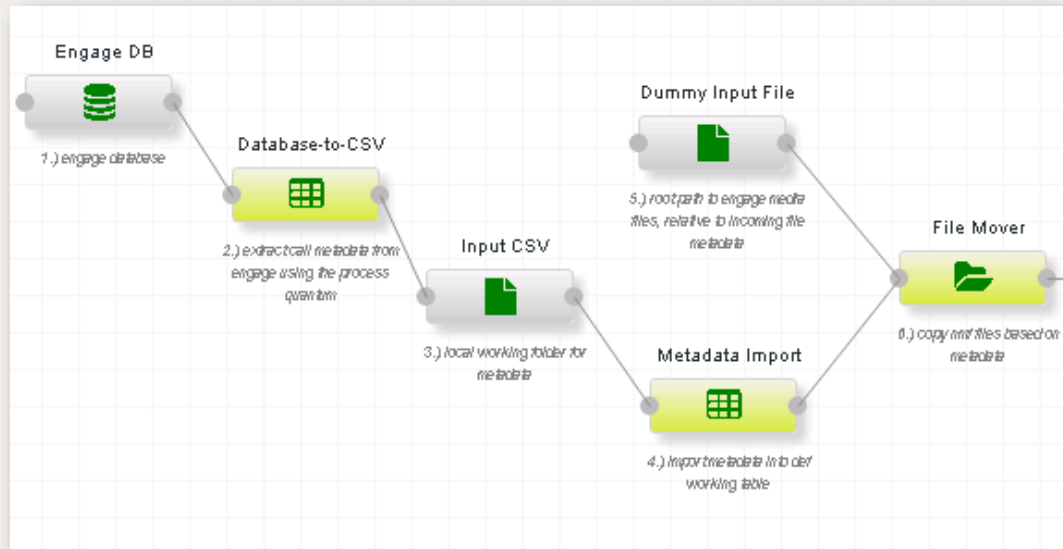
1. Both the audio file and the metadata file are placed in a folder called production media on the NIA server.
2. The audio file is placed in a folder called production media on the IA server, and the metadata is inserted to Interaction analytics DB's inbox table

Now that we know how to extract the audio files and the metadata files from the recording system, let's see how we transfer them into the Interaction analytics server.

**Option 1:
Media and Metadata
in the same Folder**



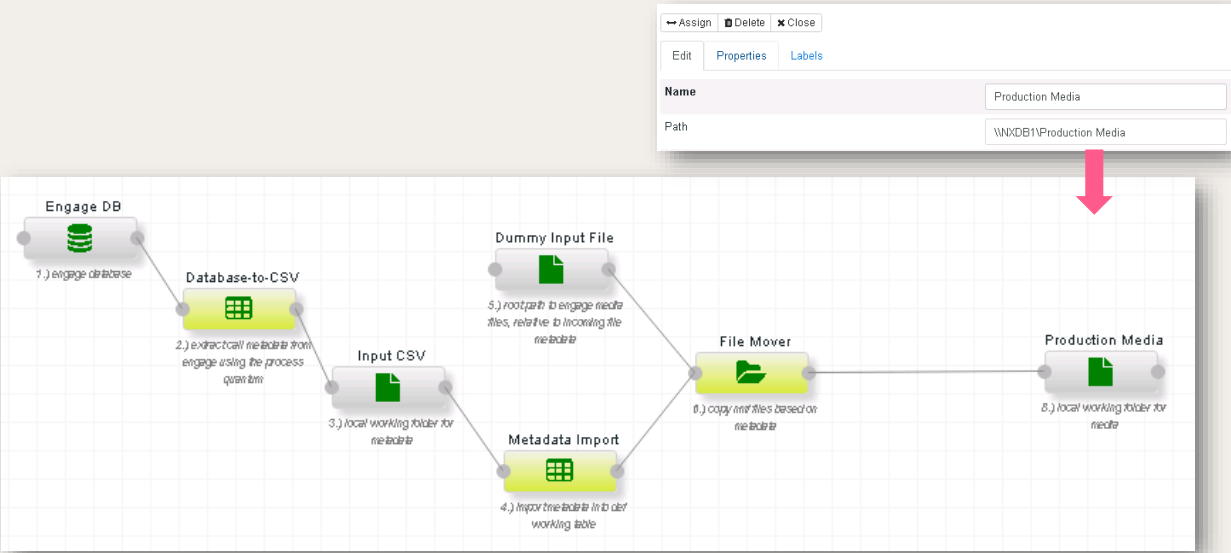
Both Files in the Production Media Folder



So far, we have built a process that extracts the metadata from the recorder database and inserts them to the DEF working table. The process also copies and moves the audio files to the folder that we specify.

Now we want both the metadata XML files and the audio file to be in the same folder. By convention, this folder is usually named "Production Media."

Both Files in the Production Media Folder



Let's create the Production Media folder as an output resource for the file mover. At this point, the production media folder contains the audio file. We need to extract the metadata from the working table into an XML file and transfer the XML file into the production media folder.



File Mover – External Media ID

Delete
Close

Edit
Properties
Labels
Logs

Overwrite existing output files	true	?
Fail if no files moved	false	?
Delete inputs after move	false	?
Capture ExternalMediaId metadata during move	true	?
Gets and sets dispositions	true	?
Update file key on move	true	?
Force to extension		?
Failure Threshold	0	?

Submit

Grabs the full path to the output location

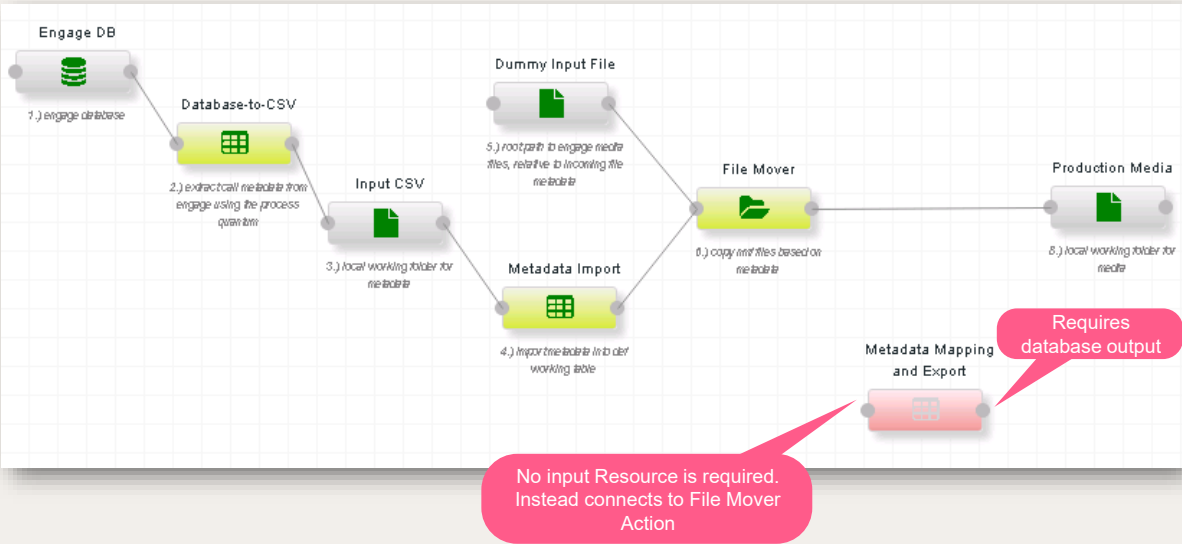
Only one File Mover in a process should have this parameter set to true

Set the “Capture External MediaId metadata during move “true” in the file mover action that transfers the audio to its final destination.

This parameter grabs the full path to the output location and adds it to the working table metadata.



Metadata Mapping and Export



The **Metadata Mapping and Export** action takes metadata from the working table, and exports it to either a CSV or XML file, and optionally remaps some of those fields to known NIA fields.



Both Files in the Production Media Folder

Action - Metadata Mapping and Export

DeleteClose

EditPropertiesLabelsLogs

Export type

xml

Generate ESM files

false

?

Include disposition data in output?

false

?

Metadata Tag Remapping

?

Export MappingImport Mapping

Default separator character

.

?

Default quote character

"

?

Default escape character

?

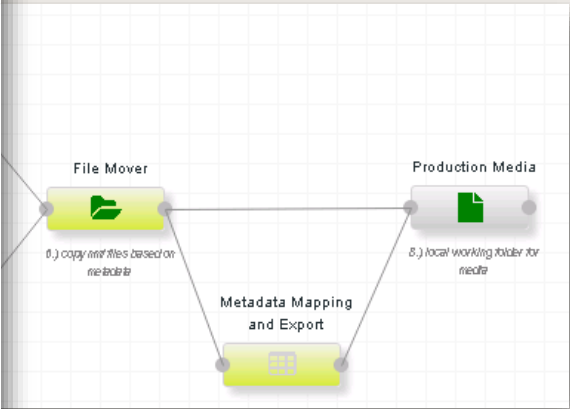
Default line end character

\n

?

Submit

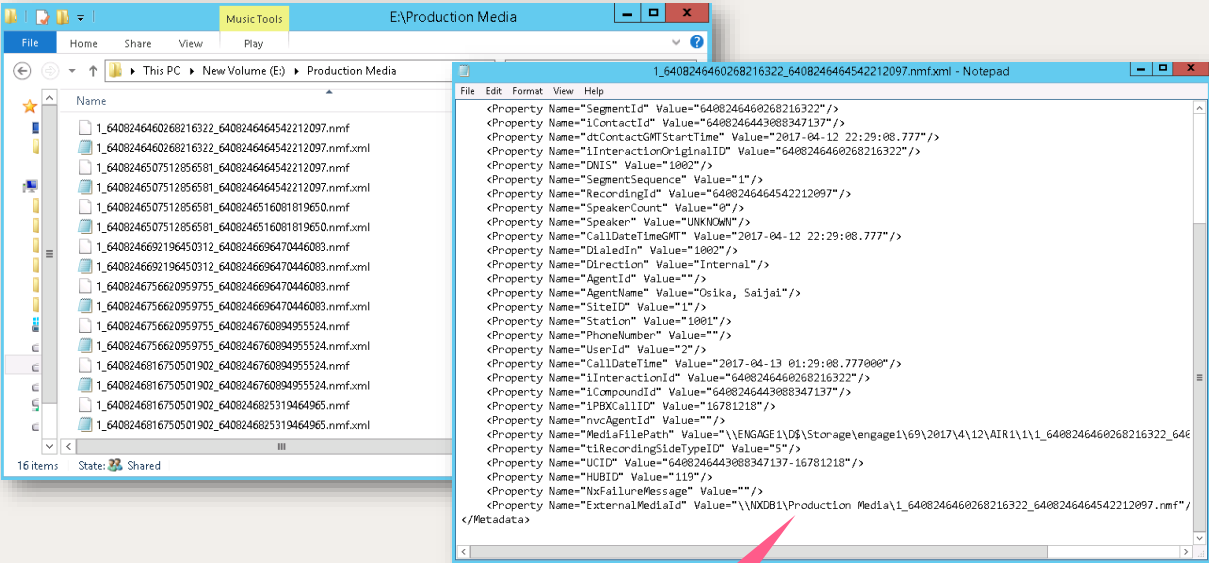
Leave the default setting



Leave the default setting



Production Media Folder



Audio location on the NIA server

Both the audio files and the metadata files are located in the production media folder. We can see that the Metadata XML file contains the audio location on the NIA server.



JSON File

⚙️ Add a Process

View Quantum

View Schedule

Process Template

Name

JSON

Description

Start Time

06/07/2019 00:00

Interval

1

Interval Type

Day

Blackout Start Time

00:00:00

Blackout End Time

00:00:00

Submit

Create Process

⚙️ Add a Process

View Quantum

View Schedule

Process Template

Predefined Process Templates are no longer pre-installed. These Processes are still available and documented on the DEF Confluence site and also downloadable as a complete package via SDC; any DEF Process can be imported below.

Process Library:
<https://thvconfluence01.nice.com/display/NIXDEF/Processes>

Select a Process Template (*.json)

Name

Description

Start Time

04/26/2022 00:00

Interval

1

Submit

Show Advanced

Fields in bold are required

Select JSON template

📄

We will use the Base.json Template

NiCE

DEF provides the option to import and export processes definition JSON file.

Create a new process, from the process design window, Click Select a Process Template and Browse to the location of the JSON file and Click Ok.

The NiCE logo, consisting of three blue dots of increasing size followed by a stylized black 'C' shape.

Do it Yourself

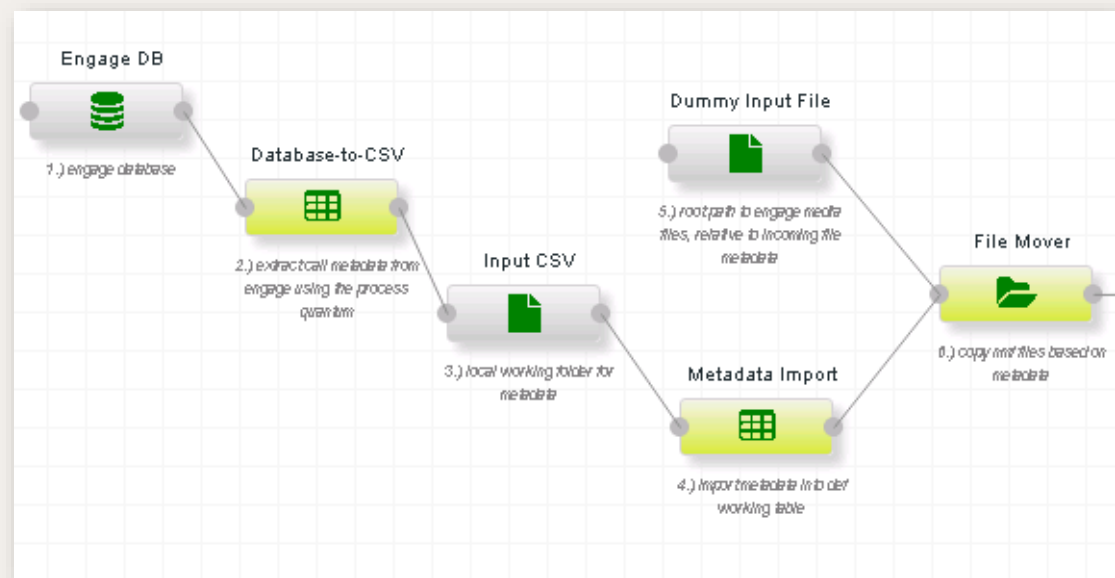
- Refer to Exercise #10A



Option 2:
Audio in “Production
Media”
Metadata in NIA DB

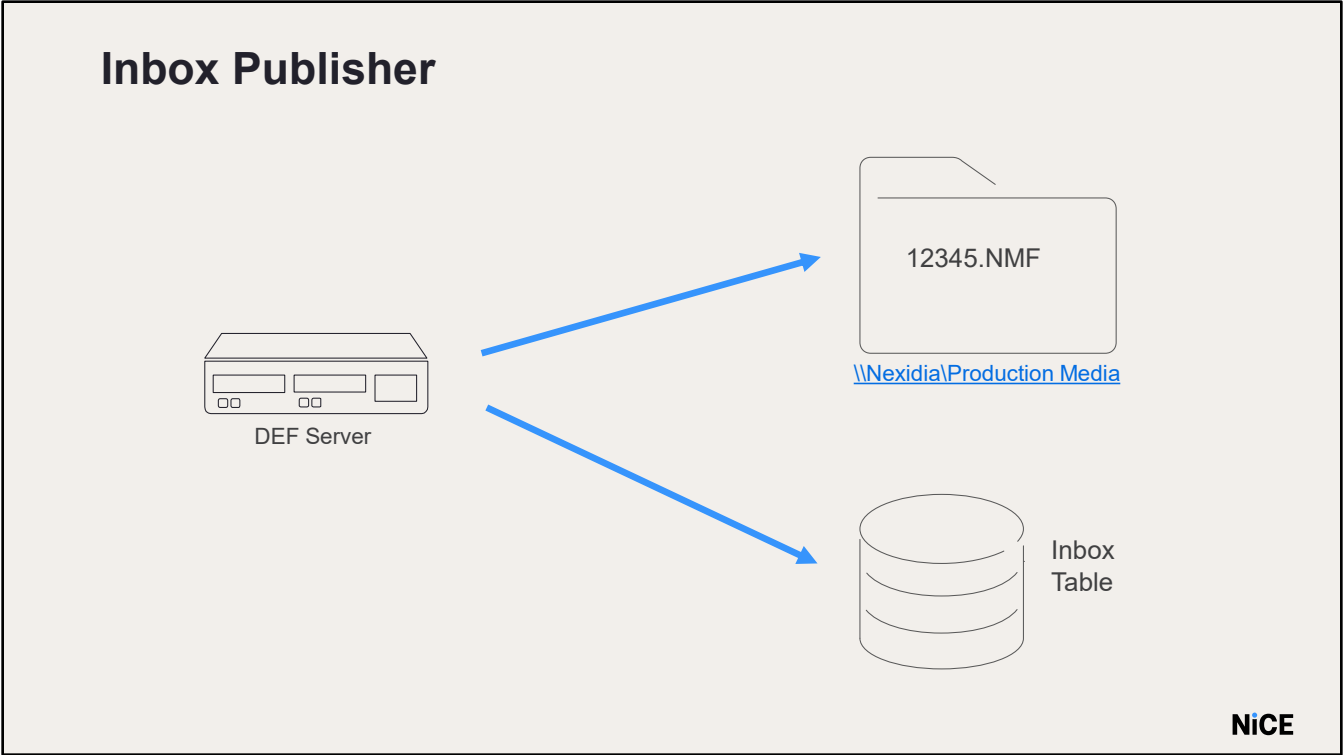


Inbox Publisher



We already know how to transfer the audio to the Production Media folder. Now we need to copy the metadata from the working table into the NIA database inbox table.

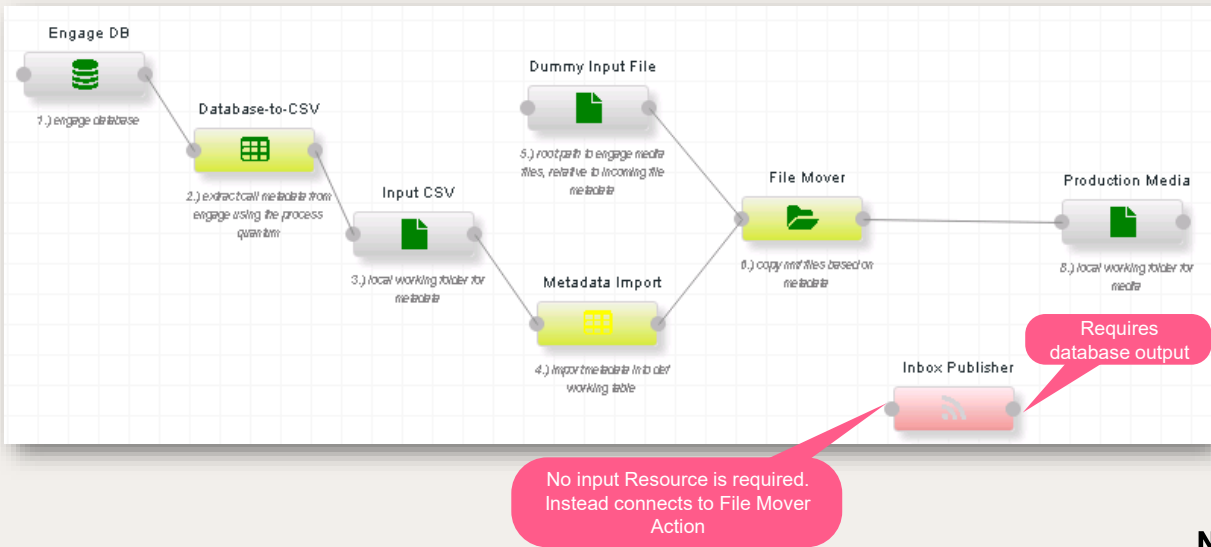




In this flow, the audio file is placed in a folder named “Production Media” on the NIA server, and the metadata is inserted into the Nexidia DBs inbox table.



Inbox Publisher



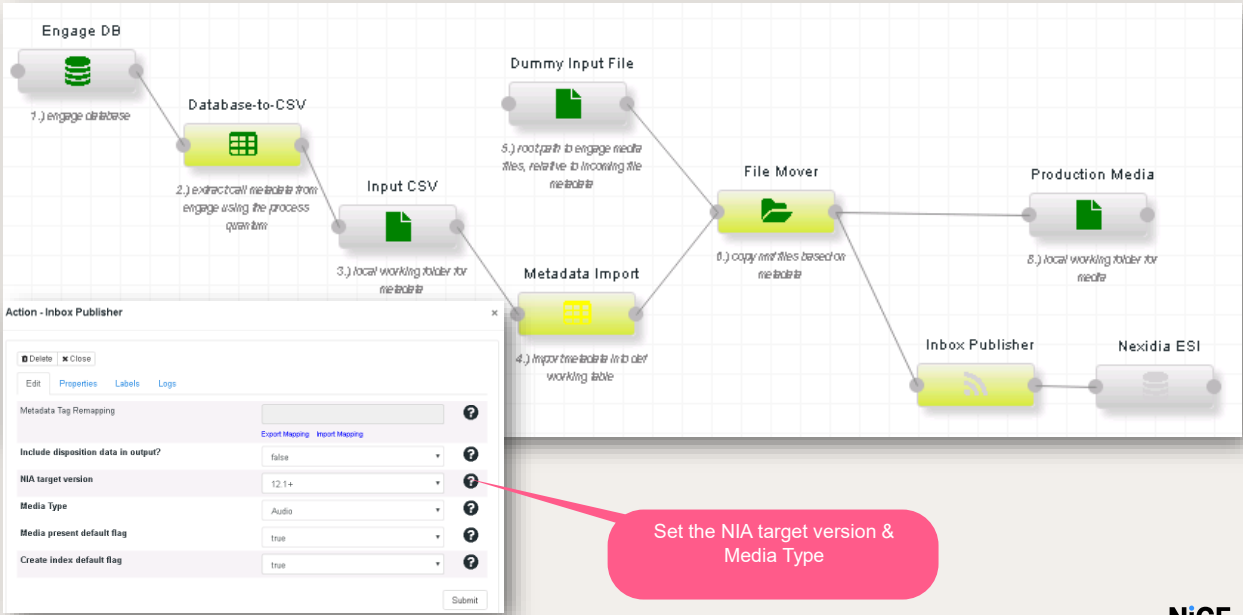
The Inbox Publisher – copy the metadata from the working table into the NIA Inbox table, optionally remapping some of the fields to known NIA fields.

Note: the incoming metadata is expected to contain an “ExternalMediaId” column that contains the full path to media files.

The inbox publisher requires a database output resource that points to NexidiaESI database.



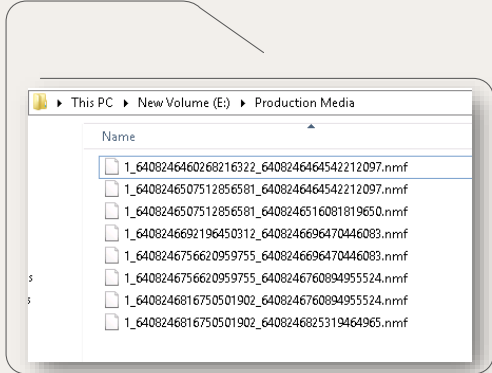
Inbox Publisher



Add the Inbox Publisher and connect it to the database output resource – the location of NIA database
jdbc:sqlserver://NXDB1;databaseName=NexidiaESI;integratedSecurity=true;



Inbox Publisher



\\Nexidia\Production Media



Inbox table

```
SELECT * FROM  
[NexidiaESI].[dbo].[SourceMediaInbox]
```

	IngestFilePath	Metadata<xml	RowId
1	\\N\>DB1\Production Media\6682640702407704583-668...	<?xml version="1.0" encoding="UTF-8"?> <Metada...	25
2	\\N\>DB1\Production Media\6682640702407704577-668...	<?xml version="1.0" encoding="UTF-8"?> <Metada...	26
3	\\N\>DB1\Production Media\6669966163670269953.nmf	<?xml version="1.0" encoding="UTF-8"?> <Metada...	27
4	\\N\>DB1\Production Media\6669967108563075075.nmf	<?xml version="1.0" encoding="UTF-8"?> <Metada...	28

Audio location on the NIA server

Let’s launch the process.
In the Inbox table, we see the audio location on the NIA server and the metadata XML.



Do it Yourself

- Refer to Exercise #10B



summary

In this lesson we covered:

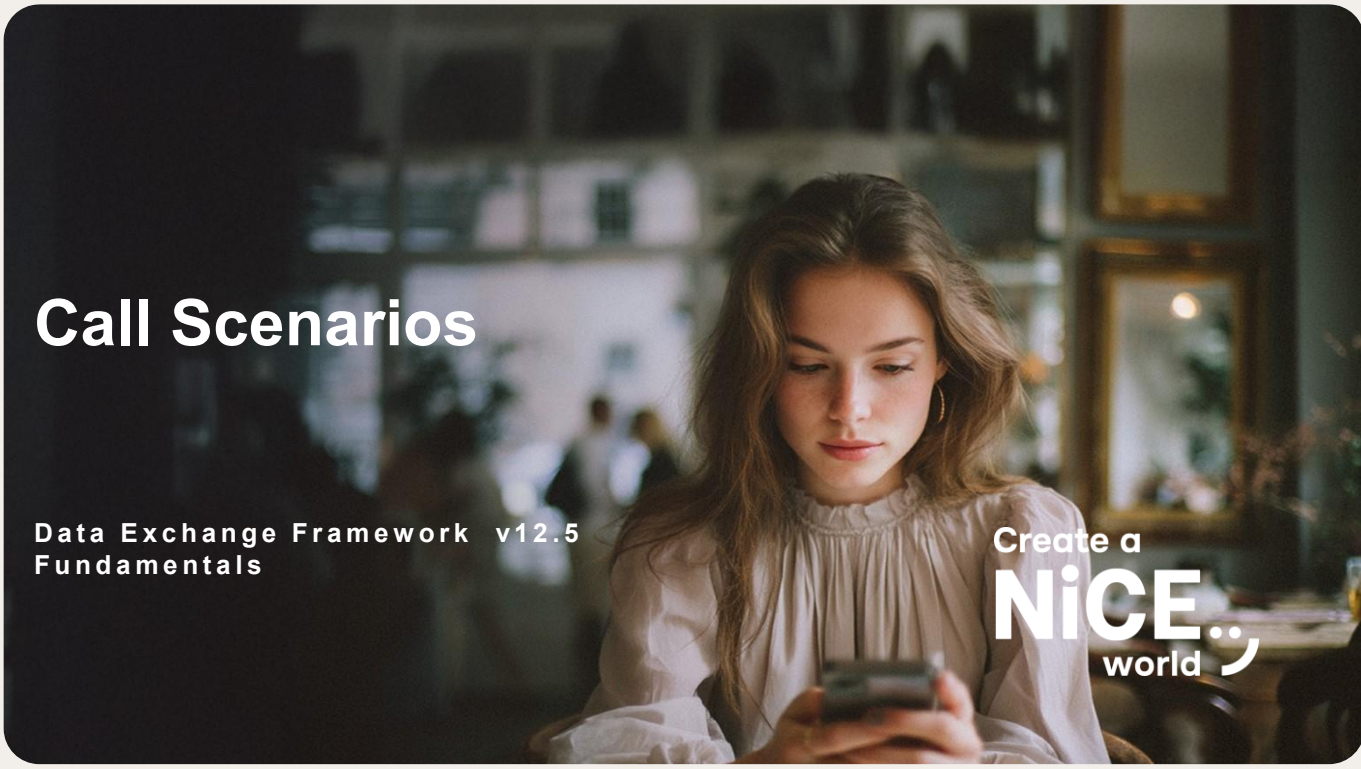
- Describe the data structure expected by NIA





Thank You





Call Scenarios

Data Exchange Framework v12.5
Fundamentals

Create a
NiCE
world 

objectives

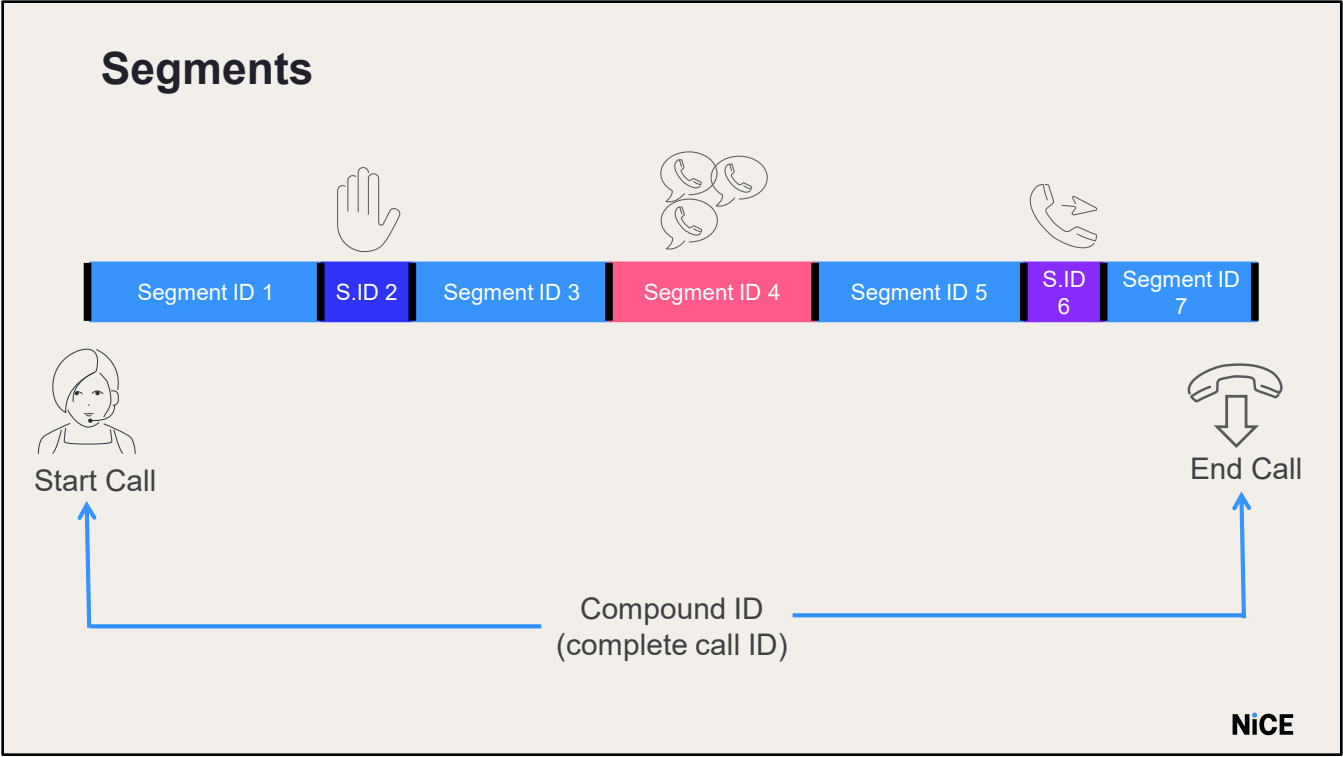
By the end of this lesson, you will be able to:

- Describe the processing flow for call with multi segments



Call Segments





Each call in the call center has a unique ID named compound ID.

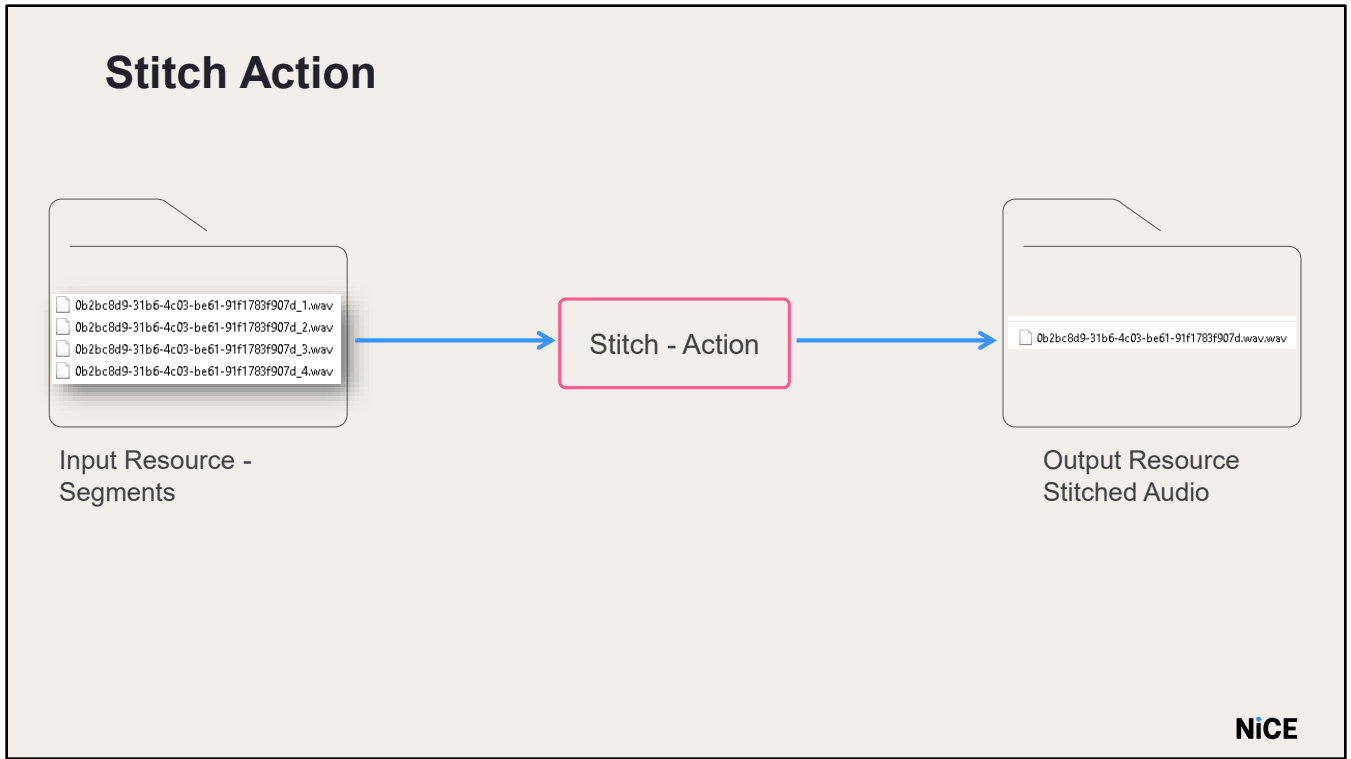
Certain actions performed by the phone switchboard during a call open new segment in the call :

action such as putting the customer on hold, conference the call with other agents, and transfer the call to a different agent.

Each segment will have a unique id (segment ID\interaction ID); some calls will have only one segment, and some calls will have multiple segments.

To identify all the segments that comprise the call from start to end, we will use the complete call id (compound ID)



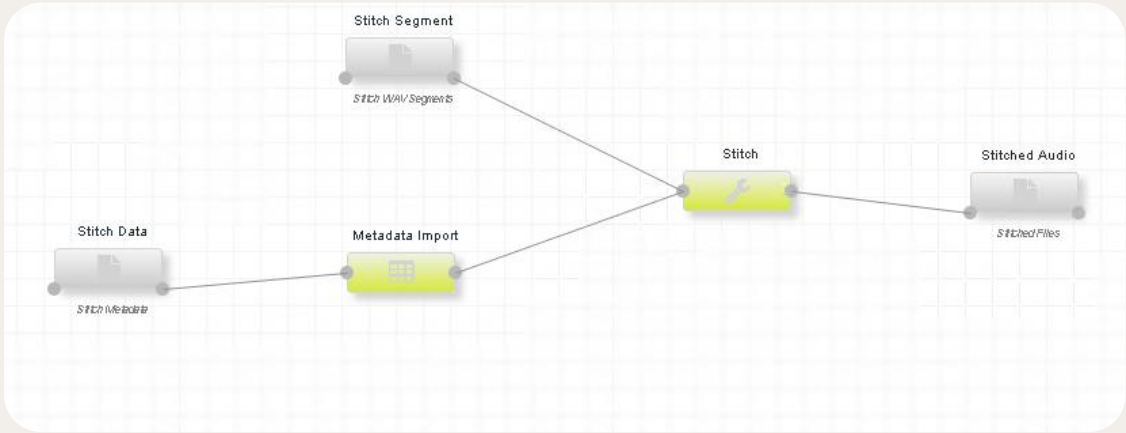


Interaction analytics application expects to receive a single audio file per interaction; this file should contain the full interaction from start to end.

Stitch Action - stitches multiple WAV or NMF files together for a single conversation based on metadata.

DEF will stitch all the segments to a single audio file and send it to Interaction analytics servers.

Stitch Action



This is how our process will look like



METADATA

Segment Order

CompoundID	SessionID	FQDN	SegmentNumber	CallDurati	HoldTime	CallStartD	CallEndDate
0b2bc8d9-31b6-4c03-be61-50b2bc8d9-31b6-4c03-be61-91f178	D:\DEF Test Files\WAV\0b2bc8d9-31b6-4c03-be61-91f178	D:\DEF Test Files\WAV\0b2bc8d9-31b6-4c03-be61-91f178	1	53	177	#####	#####
0b2bc8d9-31b6-4c03-be61-50b2bc8d9-31b6-4c03-be61-91f178	D:\DEF Test Files\WAV\0b2bc8d9-31b6-4c03-be61-91f178	D:\DEF Test Files\WAV\0b2bc8d9-31b6-4c03-be61-91f178	2	26	699	#####	#####
0b2bc8d9-31b6-4c03-be61-50b2bc8d9-31b6-4c03-be61-91f178	D:\DEF Test Files\WAV\0b2bc8d9-31b6-4c03-be61-91f178	D:\DEF Test Files\WAV\0b2bc8d9-31b6-4c03-be61-91f178	3	43	286	#####	#####
0b2bc8d9-31b6-4c03-be61-50b2bc8d9-31b6-4c03-be61-91f178	D:\DEF Test Files\WAV\0b2bc8d9-31b6-4c03-be61-91f178	D:\DEF Test Files\WAV\0b2bc8d9-31b6-4c03-be61-91f178	4	31	747	#####	#####

Call ID

This is the metadata we will use for this demonstration



Stitch Action

Action - Stitch

Delete Close

Edit Properties Labels Logs

Media file type WAV

Stitching method Sequence

Call duration correction Duration Column

Call ID column name CompoundID

Segment ID column name

Segment sequence column name Segment Number

Call duration column

Delete segments after stitching? true

Submit

Show Advanced

Fields in bold are required

Media Type: WAV or NMF

Stitching Method: sequence, when all segments of a single call can be ordered by a single column of data. Use chained stitching when the order of segments of a call is derived from two columns, one identifying the current segment, the next a pointer to the ID of the next segment.

Call Duration Method: which column to use to measure the call duration, the action will add silence to the end of the call to match the metadata. A duration column is expected to be represented in seconds.

Call ID column name: Use to determine files to stitch from existing working table metadata, and will be used to name the resulting output file. If you enter multiple columns, you should identify multi-column uniqueness with a plus mark (+) between the column names. For example: CompoundId+PBXId;

Segment sequence column name: the metadata column that identifies the sequence of segments to stitch



Do it Yourself

- Refer to Exercise #11



summary

In this lesson, we covered:

- Processing flow for call with multi segments



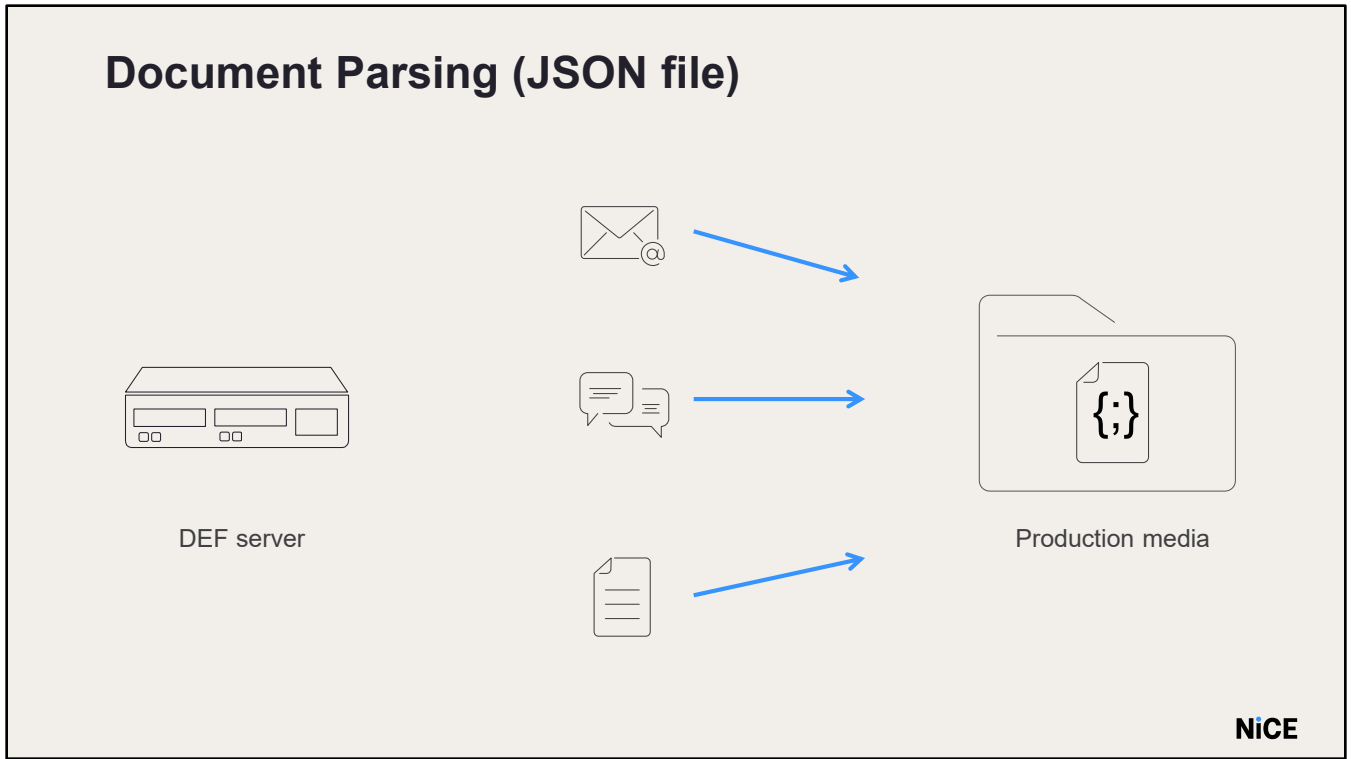


Thank You



Appendix :
Text Interactions





Interaction analytics text analysis engine (text search grid) expected to receive data to process in a JSON format.

DEF will process text interactions: emails, chats, and text files and convert them to a JSON file and placed it in the production Media folder.

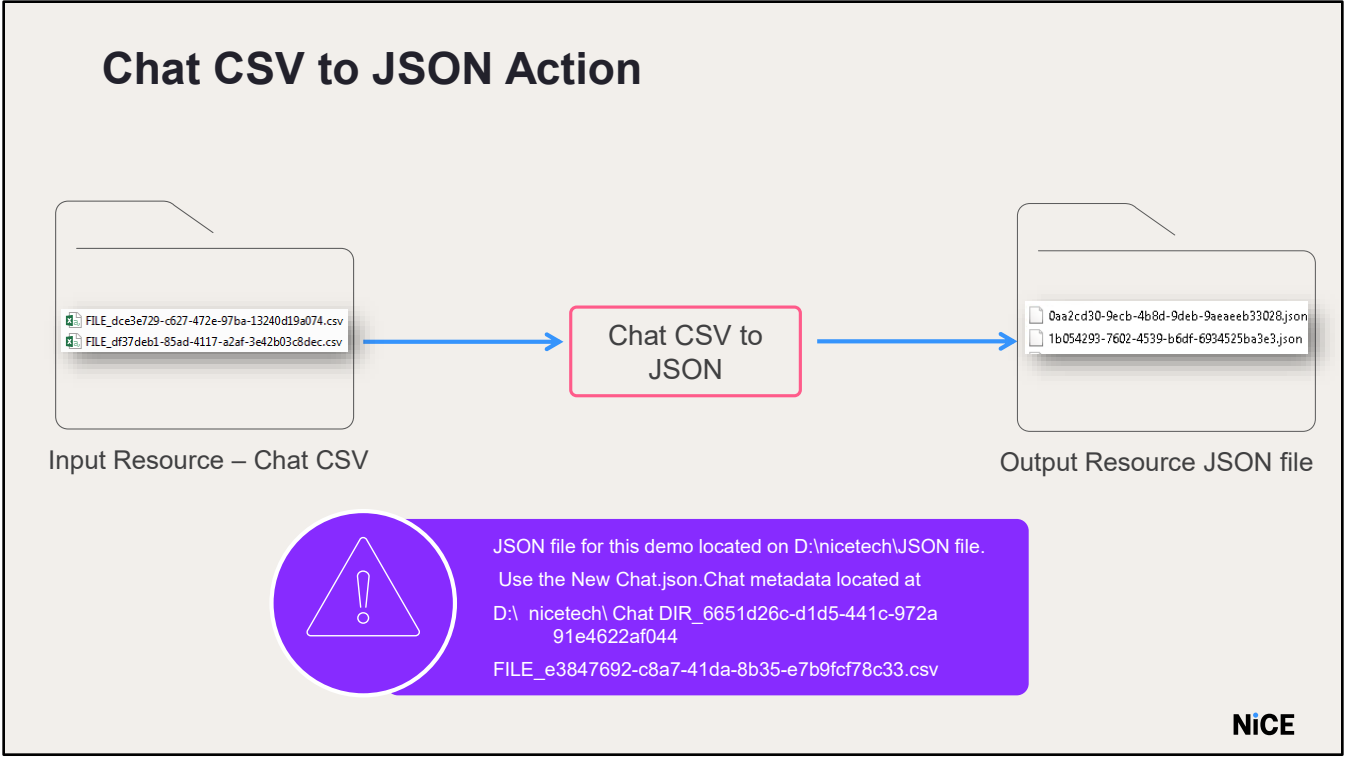
JSON -is short for JavaScript Object Notation and is a way to store information in an organized, easy-to-access manner. In a nutshell, it gives us a human-readable collection of data that we can access in a logical manner.

JSON File

```
{
  "chat_id": "21b9a63a-7a84-4bff-824f-d97fa490fbd5",
  "excerpted": false,
  "start_time": "2015-09-20T09:23:00.0000000Z",
  "end_time": "2015-09-20T09:35:00.0000000Z",
  "participants": [
    {
      "participant_id": "Cliff",
      "role": "Agent",
      "name": "Cliff"
    },
    {
      "participant_id": "Barb_",
      "role": "Customer",
      "name": "Barb_"
    }
  ],
  "messages": [
    {
      "content": "Thank you for contacting APS Cable.",
      "participant_id": "Cliff",
      "timestamp": "2015-09-20T09:23:00.0000000Z"
    },
    {
      "content": "I understand that you have a question about your bill?",
      "participant_id": "Cliff",
      "timestamp": "2015-09-20T09:24:00.0000000Z"
    },
    {
      "content": "I am trying to get the free cell phone I was promised when I signed up for DSL service.",
      "participant_id": "Barb_",
      "timestamp": "2015-09-20T09:24:00.0000000Z"
    },
    {
      "content": "Alright. I can help you with that",
      "participant_id": "Cliff",
      "timestamp": "2015-09-20T09:25:00.0000000Z"
    }
  ]
}
```

Here is an example for JSON file of chat interaction: We can see when the chat took place, who are the agent and customer who participate in the conversation, and the chat body the message itself





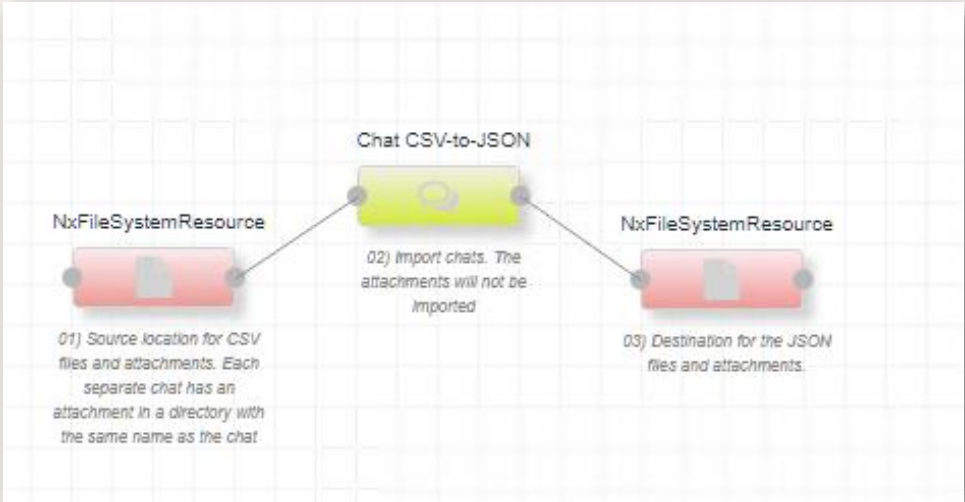
This action will convert chat-based CSV files to Text Grid's expected JSON format.

Transforms web chats into an expected format so that Nexidia Analytics and Interaction analytics Text Grid can process web chats.

The action requires both input and output resources that can be either a file system or SFTP types



Chat CSV to JSON Action



Metadata

Unique Chat ID

Message Text

Order of the message

ChatID	Source	ChatDate	MsgText	International	F	TurnID	AgentLast	AgentFirst	AgentID	CustomerI	CustomerI
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Customer	3/30/2019	Back To Square One. To go back to the beginning; back to	Ukrainian	ĐĐ	0	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Agent	3/30/2019	Be afraid be very afraid To gild refined gold to paint the lily	Et Duumpien	f	1	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Customer	3/30/2019	Leopard cannot change its spots - A Fools rush in where	Thai à	gà àà à	2	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Agent	3/30/2019	Agree to disagree	—	HiZi/fi	3	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Customer	3/30/2019	Moving finger writes - The Run the gauntlet	Latvian	MaziA	4	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Agent	3/30/2019	Be still my beating heart	Icelandic	How	5	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Customer	3/30/2019	Beating a Dead Horse To bring up an issue that has alrea	All MÄnsch k		6	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Agent	3/30/2019	Where do random thoughts come from?	Random Swec		7	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Customer	3/30/2019	Beating Around The Bush. Someone who is beating aroun	Finnish: Miele		8	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Agent	3/30/2019	Twelve good men and true	Random Swec		9	Dean	Joyce	528-78-60	Harvey	Robert
8b3b8f8c-6e3f-40b1-abb4-d8580fe5326a	Customer	3/30/2019	Hit the nail on the head	Frisian (West)		10	Dean	Joyce	528-78-60	Harvey	Robert
e6a2d162-fa11-486d-9a8d-c483bfe35283	Agent	3/30/2019	Derring-do The whole shebang	Tibetan: Nga	f	0	Williams	Daniel	720-13-47	Boyd	Billy
e6a2d162-fa11-486d-9a8d-c483bfe35283	Customer	3/30/2019	Know The Ropes. Having a familiarity or understanding of	Cherokee: áá		1	Williams	Daniel	720-13-47	Boyd	Billy

Participant Role

Agent and Customer Info

NiCE

This process will read chat data from a CSV file and push the separate chats to Nia. The data in the CSV file must be in double-quotes. The CSV file must have columns identifying the chat ID, the message text, the time of the message or a sort column, participant role, and role to participant mapping. The chat ID must be unique for each chat but not for each turn in the chat.

The message time or sort column will be used to identify the order the turns appear in the chat and each must be unique. The role to participant mapping is used by the text grid to present the data in NIA.



Metadata

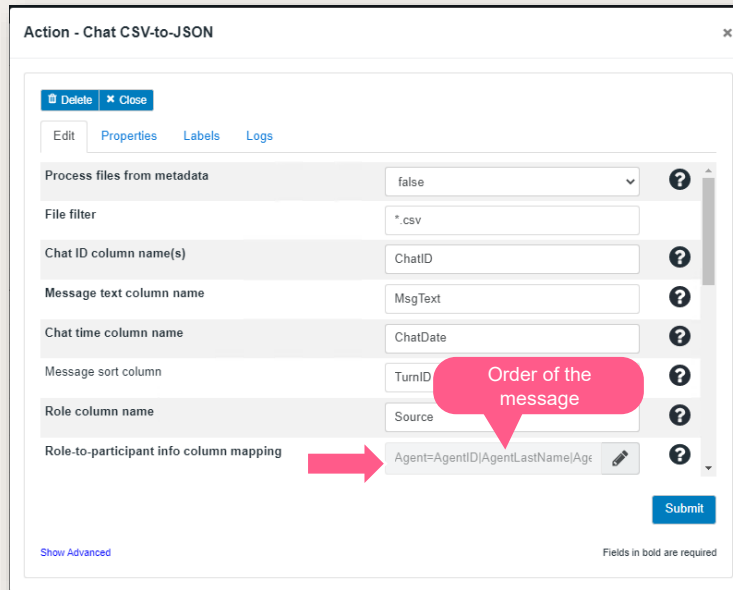
Attachment Location

R	S	T	U	V	W	X	Y	Z	AA	AB	AC
Project	City	Classificat	Misc	Attachmen	AttachmentLocation						
Internation:	Tianjin	Orbiter sp	Vega 1 Ver	FALSE	DIR_3e2131c0-fc8f-421d-b2fa-70b393c89581\	FILE_0ea9672b-09a1-4c0d-b8f5					
Internation:	Tianjin	Orbiter sp	Vega 1 Ver	FALSE							

If the chat contains attachments, a column that indicates the relative path to the attachment file must be provided. The contents of the attached file will be read and will be appended to the chat message for that turn. If configured, the attached files will be saved to the NIA ingest folder.



Chat CSV to JSON Action



Action - Chat CSV-to-JSON

Delete Close

Edit Properties Labels Logs

Process files from metadata false ?

File filter *.csv

Chat ID column name(s) ChatID ?

Message text column name MsgText ?

Chat time column name ChatDate ?

Message sort column TurnID ?

Role column name Source ?

Role-to-participant info column mapping Agent=AgentID|AgentLastName|Age ?

Submit

Show Advanced

Fields in bold are required

Enter the optional name of a numeric sequence column to sort messages by in the "Message sort column name" field. If you do not enter a name, the message timestamp is used instead.

Role column-name - The name of the column that identifies the role of each participant in the conversation.

Role to participant info column mapping- mapping of expected roles to the names of columns holding identifiers (ID name, Display Name)for those roles

Chat CSV to JSON Action

 Role-to-participant Info Column Mapping ×

#	Role ?	ID ?	Name ?	Display Name ?	Text Grid Role ?	
0	Agent	AgentID	AgentLastName	AgentFirstName	Agent	+ Add Delete
1	Customer	CustomerID	CustomerLastName	CustomerFirstName	Customer	+ Add Delete

Clicking in this field, the Chat Mapper window opens. You will use this window to map chat role entries by specifying the following values:

Role Value - The role name as defined in the CSV file's role column.

ID - The CSV column is containing the participant identification for the role.

Name - The CSV column containing the participant name for the role.

Display Name - The CSV column is containing the participant display name for the role.

Text Grid Role - The Text Grid role that the CSV role name maps to. The valid options are Automation, Agent, or Customer.



Chat CSV to JSON Action

Action - Chat CSV-to-JSON

DeleteClose

EditPropertiesLabelsLogs

Role-to-participant info column mappingAgent=AgentID|AgentLastName|Age

Parse attachmentstrue

Message attachment column nameAttachmentLocation

Append parse text label to message bodyfalse

Attachment typesMicrosoft Office Documents^applicat

Copy attachments to output foldertrue

Delimiter.

Delete files after processingtrue

Submit

Show Advanced

Fields in bold are required

Select if the attachment should be parsed to JSON file or not, select the attachment type to parse. Enter the name of the column that holds the location of the file attached to the message.



Chat CSV to JSON Action

Action - Chat CSV-to-JSON

DeleteClose

EditPropertiesLabelsLogs

Role-to-participant info column mappingAgent=AgentID|AgentLastt

Parse attachmentstrue

Message attachment column nameAttachmentLocation

Append parse text label to message bodyfalse

Attachment typesMicrosoft Office Documents

Copy attachments to output foldertrue

Delimiter.

Delete files after processingtrue

Show Advanced

Attachment Types

1Emailmessage/rfc822

2JSONapplication/json

3☒ Microsoft Office Documentsapplication/vnd.openxmlformats-officedocument.wordprocessingml.document,application/vnd.openxmlformats-officedocument.spreadsheetml.sheet,application/vnd.openxmlformats-officedocument.presentationml.presentation,application/msword,application/vnd.ms-

4☒ PDFapplication/pdf

OKCancel

Select the attachment type to parse, double click on the attachment types field and select the relevant options



NiCE

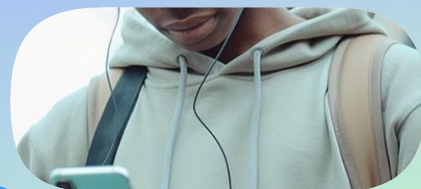
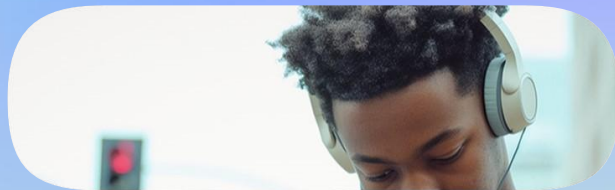
Here is an example of a JSON file with parsed attachment, the parsed attachment is placed in the order that it was attached during the chat conversation

Do it Yourself

- Refer to Exercise #17



Appendix : Extract and Prepare Screens



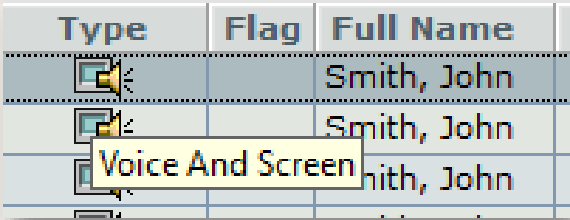
Extract and Prepare Screens in DEF

- The process is used to Extract Screen recordings from NICE Engage and export them to NIA



The screenshot shows the NICE Business Analyzer interface. The 'Interactions' section is active, displaying a table of records. The table has columns: Type, Flag, Full Name, Segment Start Time, Segment Stop Time, Segment Duration, Voice Archive Status, and Screen Archive. The first row is highlighted in red.

Type	Flag	Full Name	Segment Start Time	Segment Stop Time	Segment Duration	Voice Archive Status	Screen Archive
EQ		Smith, John	9/11/2023 8:34:51 AM	9/11/2023 8:35:12 AM	00:00:21	04	04
EQ		Smith, John	9/11/2023 8:34:19 AM	9/11/2023 8:34:48 AM	00:00:29	04	04
EQ		Smith, John	9/11/2023 8:30:59 AM	9/11/2023 8:31:24 AM	00:00:25	04	04
EQ		Smith, John	9/11/2023 8:30:12 AM	9/11/2023 8:30:55 AM	00:00:43	04	04
EQ		Smith, John	9/11/2023 8:25:03 AM	9/11/2023 8:25:34 AM	00:00:31	04	04
EQ		Smith, John	9/11/2023 8:24:28 AM	9/11/2023 8:24:58 AM	00:00:30	04	04
EQ		Smith, John	9/11/2023 8:20:23 AM	9/11/2023 8:20:45 AM	00:00:22	04	04
EQ		Smith, John	9/11/2023 8:19:48 AM	9/11/2023 8:20:21 AM	00:00:34	04	04
EQ		Smith, John	9/11/2023 8:16:05 AM	9/11/2023 8:19:44 AM	00:03:39	04	04
EQ		Smith, John	9/11/2023 8:14:33 AM	9/11/2023 8:14:59 AM	00:00:26	04	04
EQ		Smith, John	9/11/2023 7:38:43 AM	9/11/2023 7:41:28 AM	00:02:45	04	04



This is a close-up of the table from the screenshot above. It shows the header row and the first three data rows. A tooltip is visible over the third row, displaying the text 'Voice And Screen'.

Type	Flag	Full Name
EQ		Smith, John
EQ		Smith, John
EQ		Smith, John



Extract and Prepare Screens in DEF

- Download the latest process JSON



DEF_Extract and prepare screens.json

- Create a new process in DEF and upload the JSON



Add a Process

View Quantum

View Schedule

Process Template

Predefined Process Templates are no longer pre-installed. These Processes are still available and documented on the DEF Confluence site and also downloadable as a complete package via SDC; any DEF Process can be imported below.

Process Library:

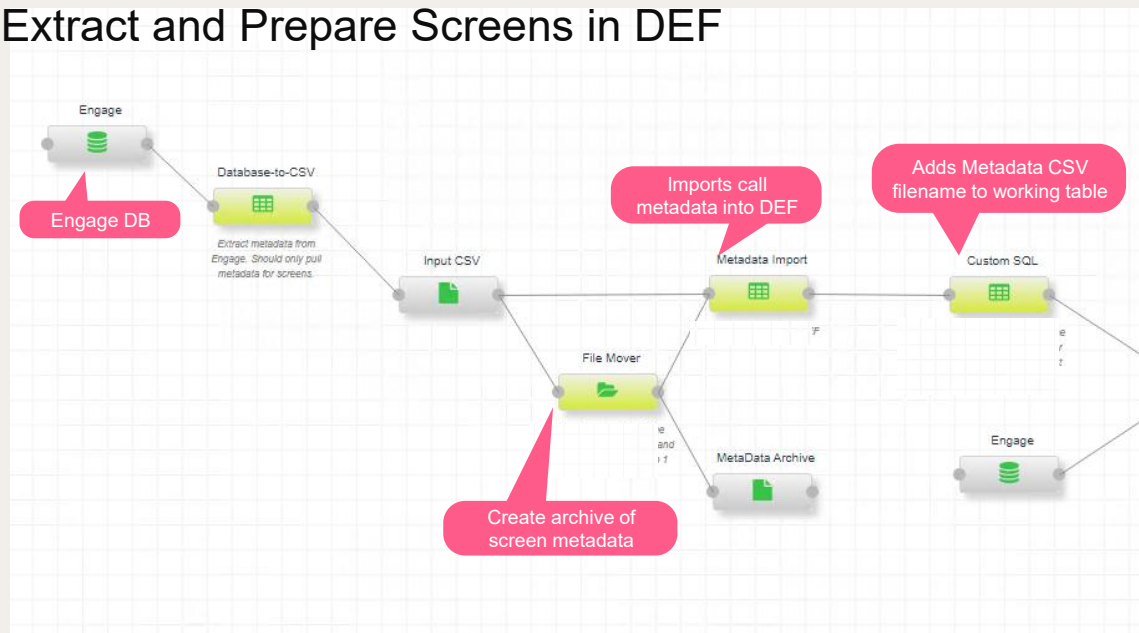
<https://tlvconfluence01.nice.com/display/NXDEF/Processes>

Select a Process Template (*.json)

Set the Quantum Lower Bound Date to August 08, 2023



Extract and Prepare Screens in DEF

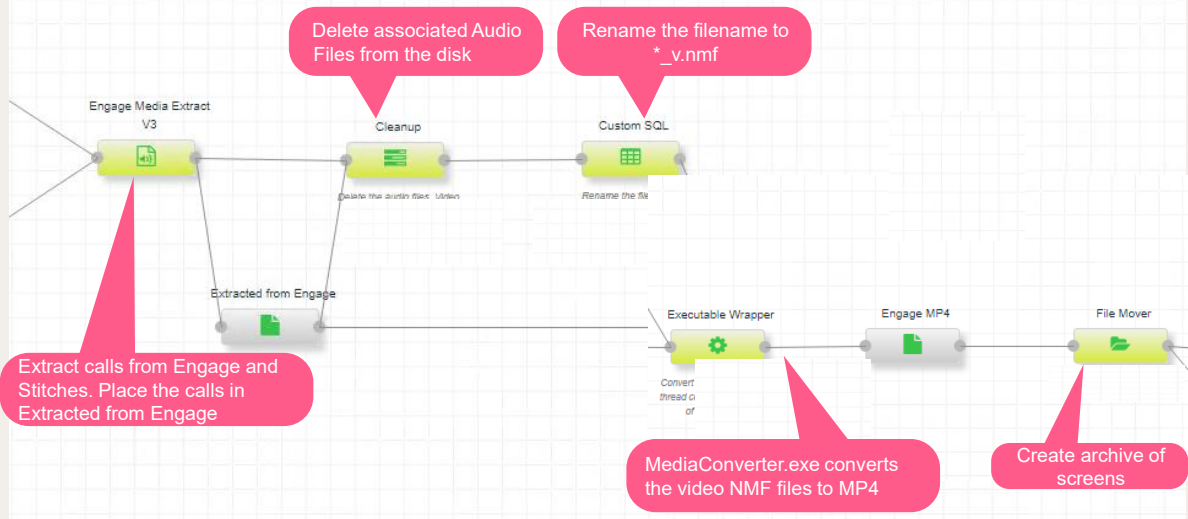


Engage is queried and only calls with screens are pulled within the quantum window.

An archive copy of the CSV file is created. This is limited to one thread and one file. Metadata is imported into the working table and some extra columns are created. The name of the metadata file is added to each row in the working table.



Extract and Prepare Screens in DEF

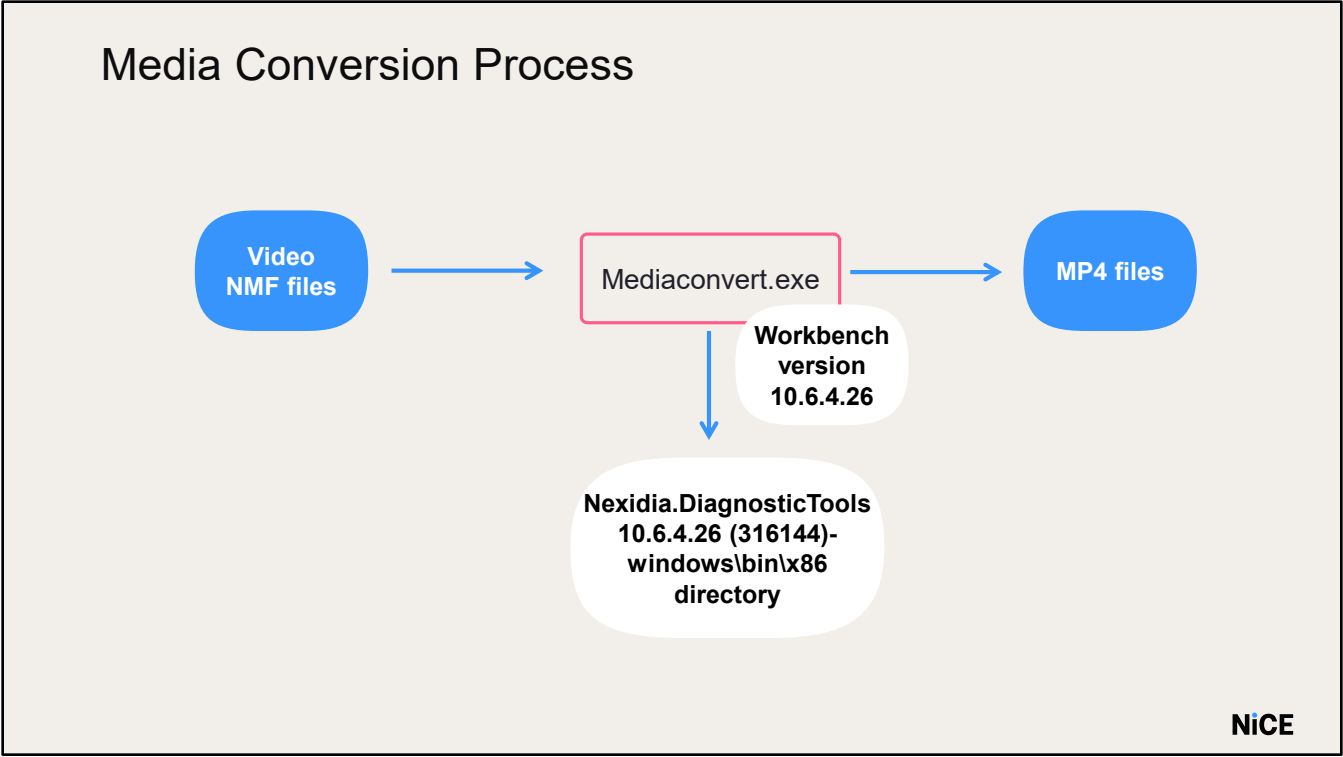


The audio and screen files are extracted and stitched.

Audio files are deleted from the disk.

We rename the FileName to *_V.nmf in the working table.

Executable Wrapper uses mediaconvert.exe to convert video NMF files to MP4 files.



This process requires a 32-bit version of mediaconvert.exe that supports converting video NMF files to MP4 files. The latest available version is with Workbench version 10.6.4.26. The executable is available in the **Nexidia.DiagnosticTools10.6.4.26 (316144)-windows\bin\x86 directory**.
Copy the EXE and all other files. .



Media Conversion Process – Batch File

Header

- Filename of the file
- Taken from Working Table
- DEF variable name %FILE_KEY_BATCH%

Header

- The path of the file resource on the input side of the EXE Wrapper action
- DEF variable name %INPUT_PATH%

Header

- The path of the file resource on the output side of the EXE Wrapper action
- DEF variable name %OUTPUT_PATH%

Header

- The directory where the MediaConvert.exe file is located
- Used by batch file
- Not a DEF variable



Make sure that there are no spaces, missing commas and spelling errors in the file path specified.

NiCE

The mediaconvert.exe will be called by a batch file that is wrapped by the DEF EXE Wrapper action. Some edits to this file will be required based on how the system is configured.

There are four inputs to the batch file.

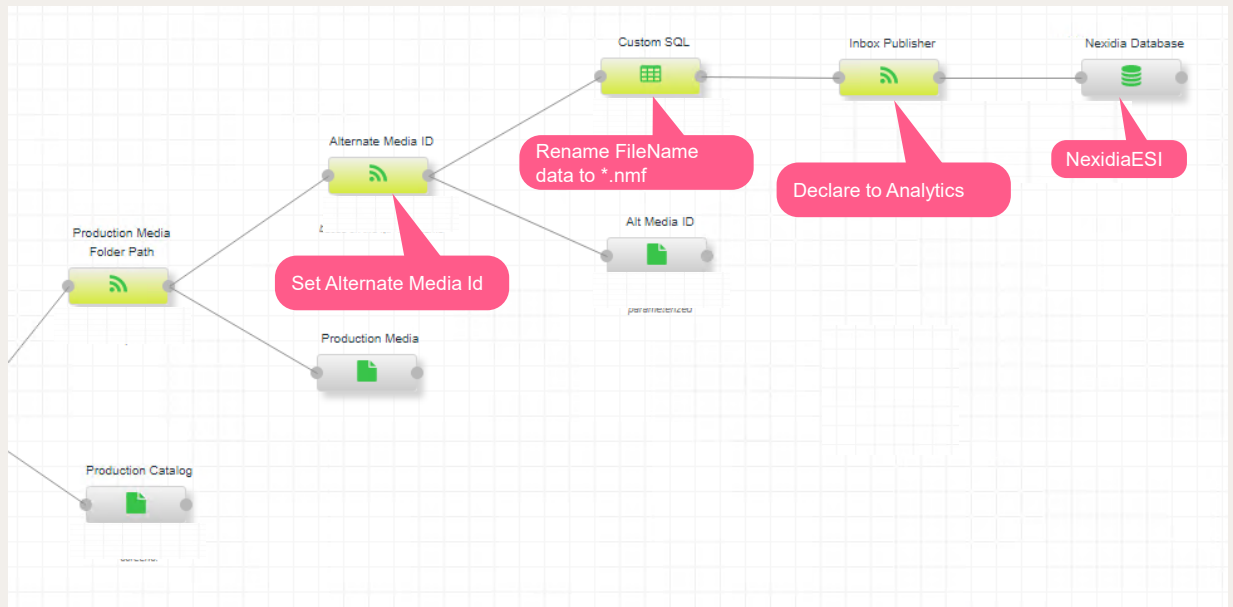
1. %1 – The filename of the file. This is taken from the working table. DEF variable name %FILE_KEY_BATCH%
2. %2 – The path of the file resource that is on the input side of the EXE Wrapper action. DEF variable name %INPUT_PATH%
3. %3 - The path of the file resource that is on the output side of the EXE Wrapper action. DEF variable name %OUTPUT_PATH%
4. %4 – The directory where the MediaConvert.exe file is located. This is used by the batch file and is not a DEF variable



The batch file will validate the file exists for any file passed. If the file exists, the `mediaconvert.exe` will run and transcode the file. If the file does not exist an error will be raised, and the action will show error in the log. The command line used to launch `MediaConvert.exe` sets the thread count to 1 for the executable. This must not be changed as DEF will launch multiple copies of the executable based on the thread's property for the EXE Wrapper action. The last step in the batch file is to delete the original `*_V.nmf` file. There is a timeout set to allow `MediaConvert` to release the file handle so DEF can delete the file. This may need to be increased if the files are on a network share. This conversion is time consuming. The thread count for the EXE Wrapper action should be manually set to 1.5 the number of physical CPU cores on the machine.



Extract and Prepare Screens in DEF



NiCE

The mp4 file is copied to the network prod folder. The process will not move beyond this point if the associated audio file has not been ingested in Nia. Production media folder path is calculated. The alternate media specific metadata is captured. The working table filename is change back to .nmf. This is so Nia will reprocess the metadata properly.

Inbox Publisher Push collected working table metadata about processed files to the NIA Rich Inbox, optionally remapping some of those fields to target NIA fields.

Provides various options for media type, media presence, how to declare media, etc. Note that incoming metadata is expected to contain an 'ExternalMediaId' column that contains the full path to media files. That column should be auto-populated by any single File Mover earlier in the process when it moves media to its final ingest location. The data is pushed to Nia.



Action- Metadata Import

Action - Metadata Import

DeleteClose

EditPropertiesLabelsLogs

Process files from metadatatrue?

File filter*.csv?

Import TypePrimary?

Import file key columnMediaFileName?

Required metadataiCompoundID,iInteractionId,dtInteractionGl?

Import only required metadatafalse?

Delete metadata inputs after import?true?

Fail if nothing is imported within # minutes0?

Submit

Show AdvancedFields in bold are required

This imports the metadata of the calls to DEF



Action- Custom SQL

Action - Custom SQL

DeleteClose

EditPropertiesLabelsLogs

SQL

/*Logic to Fetch SourceMetadataFileName
--DEF 5.7 and before , need to set the
--Metadata Import should be working
--The feature of automatically capturing
Click here to edit in a larger window

Fail if nothing is updated

false

Execute against working table

true

Resolve SQL

Resolve SQL

Reconciliation Identifier

Default input disposition handling

OR

Input Disposition(s)

Hide Advanced

Fields in bold are required

SQL

EditFindReplaceUndo

/*Logic to Fetch SourceMetadataFileName*/
--DEF 5.7 and before , need to set the FileMover Action that catalogs the metadata file to have MaximumFilesToProcess=1 and Threads=1
--Metadata Import should be working after FileMover action and the below SQL running in CustomSQLAction after the MetadataImport
--The feature of automatically capturing the filename is in DEF R&D list, so when its available this workaround is not needed
--IF OBJECT_ID('dbo.(WORKING_TABLE)\$X') IS NOT NULL
--BEGIN

--update w set ModManifestFileName=x.(FileName),
--'[OUTPUT_DISPOSITION]'=getdate()
--from (WORKING_TABLE) w join (WORKING_TABLE)\$X x on x.CreatePEUID=w.CreatePEUID
--WHERE x.(FileName) like '%.csv'
--[DISPOSITION_AND_CLAUSE w]

--END
--ELSE

BEGIN
update w set ModManifestFileName=x.(FileName),
'[OUTPUT_DISPOSITION]'=getdate()
from (WORKING_TABLE) w join (WORKING_TABLE) x on x.CreatePEUID=w.CreatePEUID
WHERE x.(FileName) like '%.csv'
[DISPOSITION_AND_CLAUSE w]
END

NiCE

Action- Engage Media Extract V3

DeleteClose

EditPropertiesLabelsLogs

Use CTI datatrue?

Skip Invalid Callsfalse?

Failure Threshold0?

Packet size maximum (kB)1,024?

Extended Storage TypeNone?

Include Screenstrue?

Internal Call Screen Fallbackfalse?

Include H264 Screenfalse?

Submit

Hide Advanced

Fields in bold are required

Action - Engage Media Extract V3

DeleteClose

EditPropertiesLabelsLogs

Trace packet filteringfalse?

Recording duration minimum (s)1?

Interaction duration minimum (s)2?

Set Channel0true?

Flip channels for Channel0 Agenttrue?

Channel0 For Internal CallsCustomer?

Force Mono (Testing Only>false?

Allowed Packet Types Audio1/1, 1/5, 4/0, 4/3?

Allowed Packet Types Screen1/1, 4/101, 4/102, 4/103, 4/111?

Add Audio Packets To Screentrue?

Remove Duplicate Storage Center Pathsfalse?

Remove duplicate internal recordingsfalse?

Reconciliation Identifier?

Default input disposition handlingAND?

Input Disposition(s)?

Thread count?

Submit

Hide Advanced

Fields in bold are required

Extract media from Engage storage for interactions rolling through in current metadata, stitching where appropriate. Include screens field is set to 'true' to include screens in the extraction; 'false' to gather only audio



Action- Cleanup

Action - Cleanup

DeleteClose

EditPropertiesLabelsLogs

Cleanup typeFiles from current disposition?

File filter**?

Allow multiple input resourcesfalse?

Directory cleanup strategyRecursive?

Preserve assigned/reserved mediafalse?

Failure Threshold0^v?

Reconciliation Identifier?

Maximum files to process0^v?

Submit

Hide AdvancedFields in bold are required

Handles cleanup of DEF working tables, DEF internal logs, and input resources based on a configurable number of hours, in many different configurations. Cleanup is absolutely necessary in all systems to insure that the database and/or file system doesn't grow unchecked. The files are deleted from the physical disk and not from the working table.



Action- Custom SQL

Action - Custom SQL

DeleteClose

EditPropertiesLabelsLogs

SQL

update (WORKING_TABLE)
SET FileName = SUBSTRING(FileName,1
,MP4Filename = SUBSTRING(FileName,1
,"(OUTPUT_DISPOSITION)" = GETDATE()

Click here to edit in a larger window

Fail if nothing is updated

false

Execute against working table

true

Resolve SQL

Resolve SQL

Reconciliation Identifier

Default input disposition handling

AND

Input Disposition(s)

Submit

Hide Advanced

Fields in bold are required

Rename FileName from *.nmf to *_V.nmf.



Action- Executable Wrapper

DeleteClose

EditPropertiesLabelsLogs

Process files from metadatatrue?

File filter**?

Full path to executable"D:\nicetech\Batch Files Rev 10.2.3.47\Bati"?Parameters"%FILE_KEY_BATCH%" %INPUT_PATH%"?

Number of files to pass to %FILE_BATCH% param in batch1(min: 1)?

Check exit code for success/failure checktrue?

Success exit code0?

Executable success response regex?

Submit

Show AdvancedFields in bold are required

Full path to Executable:

"D:\nicetech\Batch Files Rev
10.2.3.47\Batch
Files\ConvertToMP4.bat"

Parameters:

%FILE_KEY_BATCH%
%INPUT_PATH%
%OUTPUT_PATH%
"D:\nicetech\Batch Files Rev
10.2.3.47\Batch Files"

Refer slides for Media Conversion Process



Action- File Mover

Action - File Mover

DeleteClose

EditPropertiesLabelsLogs

Process files from metadatatrue?

File filter~.nmf, *.nmf.mp4?

File path column nameFileName?

Flatten output directory structurefalse?

Treat missing files as errorsfalse?

Overwrite existing output filestrue?

Fail if no files movedfalse?

Delete inputs after movefalse?

Submit

Show AdvancedFields in bold are required

File mover replaces the .nmf file with .nmf.mp4 in the working table



Action- Alternate Media ID

Alternate Media ID:

In some cases, more than one type of media file may optionally arrive alongside the main media file for each row of incoming interaction metadata.

For example, there may be cases where recorded calls are *.wav files, but corresponding *.mp4 files might also arrive that show what was happening on the agent's screen during the interaction.

Alternate media ID identifies the presence of these alternate media files, and tracks them by adding specific metadata fields that NIA will use to ingest. It can be configured to do this by looking either for specific metadata field values in incoming metadata, or by testing against the file system, using a file filter pattern matching the type of additional media to look for.



It will then add appropriate metadata and expiry info into the working table.

If alternate media is found, it will always populate an indexed HasAlternateMedia column with a Yes/No value in the working table. It will also include AlternateMediaSourcePath and AlternateMediaExpirationDateUTC metadata fields indicating the location of the media and time of expiry.

That data will then be passed on to NIA for processing. Aligning alternate media working table entries with corresponding main media metadata rows may require the use of file filter subtractors when processing from metadata ('-.wav,*.mp4', e.g., if main media is *.wav, and alternate media to be discovered and matched to the .wav row is .mp4).

Requires an input resource if not processing from metadata.

Requires an output resource for AlternateMediaSourcePath resolution. (Note that it outputs nothing to this folder ... it only uses it for path info in building AlternateMediaSourcePath.)



Action- Custom SQL

Action - Custom SQL

DeleteClose

EditPropertiesLabelsLogs

SQL

update (WORKING_TABLE)
SET FileName = SUBSTRING(FileName,
,"(OUTPUT_DISPOSITION)" = GETDATE
WHERE HasVideo = 'true'

Click here to edit in a larger window

Fail if nothing is updated

false

?

Execute against working table

true

?

Resolve SQL

Resolve SQL

?

Reconciliation Identifier

?

Default input disposition handling

AND

?

Input Disposition(s)

?

Submit

Hide Advanced

Fields in bold are required

The working table filename is change back to .nmf.



Action- Inbox Publisher

Action - Inbox Publisher

DeleteClose

EditPropertiesLabelsLogs

Metadata Tag Remapping

ilInteractionId=InteractionId;AlternateName

Export MappingImport Mapping

Include disposition data in output?

false

NIA target version

12.1+

Media Type

Audio

Media present default flag

true

Create index default flag

true

Allow Duplicate Mapping

false

Submit

Show Advanced

Fields in bold are required

Inbox Publisher Push collected working table metadata about processed files to the NIA Rich Inbox, optionally remapping some of those fields to target NIA fields.

Provides various options for media type, media presence, how to declare media, etc. Note that incoming metadata is expected to contain an 'ExternalMediaId' column that contains the full path to media files. That column should be auto-populated by any single File Mover earlier in the process when it moves media to its final ingest location.



Summary: Extract and Prepare Screens in DEF

- Engage DB is queried
- Calls with screens and within quantum window are pulled and archived
- Metadata is imported into the working table .Extra columns are created
- The name of the metadata file is added to each row in the working table
- The audio and screen files are extracted and stitched
- Audio files are deleted from the disk
- Rename the Filenames to *_v.nmf in the working table
- Wrapper action converts from _v.nmf video files to MP4
- The mp4 file is copied to the network prod folder
- Production media folder path is calculated
- The alternate media specific metadata is captured
- The working table filename is change back to .nmf for Nia to reprocess the metadata properly. The data is pushed to Nia



Do it Yourself

- Refer to Exercise #18





Hierarchy Import

Data Exchange Framework v12.5
Fundamentals

Create a
NiCE..
world ☺

objectives

By the end of this lesson, you will be able to:

- Understand Hierarchy Import of Agents and Users from Engage



Hierarchy Basics



Terminology

- Agent
 - Interactions are recorded and transferred to QC
- Supervisor
 - The formal manager for an agent
 - An agent can have only a single supervisor



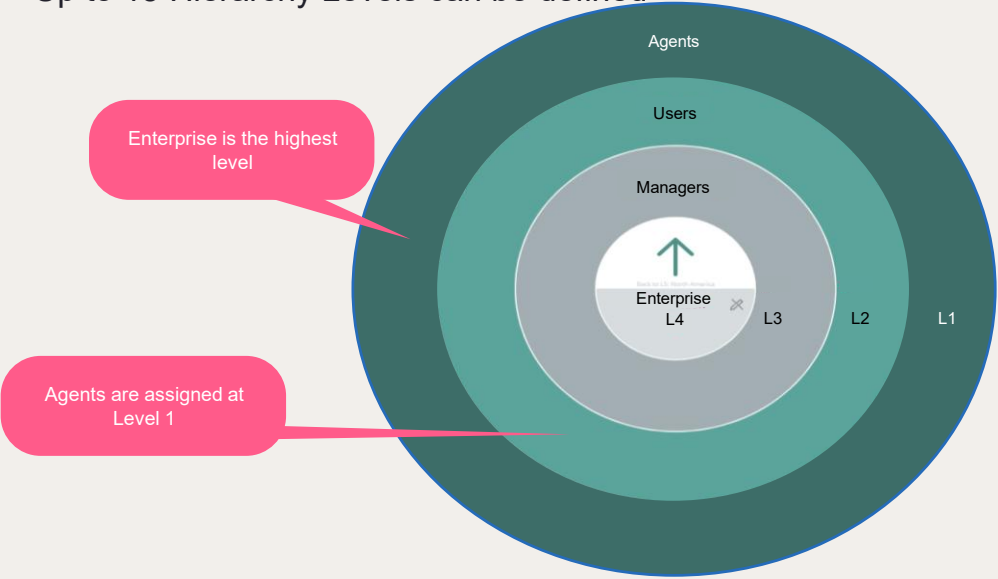
Terminology

- Team Lead
 - Senior agent that can assist the supervisor to evaluate other agents
 - Defined as both an agent and a user in QC
 - Can be assigned to multiple L1 groups in the hierarchy
- System User
 - Engage users that work with QC, but they are not an Agent, Supervisor or Team Lead



Hierarchy Levels

- Up to 15 Hierarchy Levels can be defined



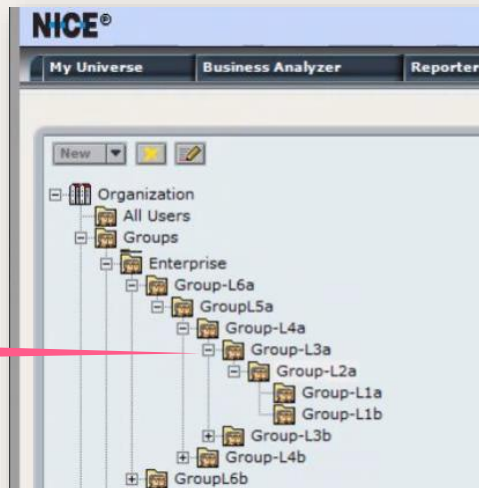
Engage
Configuration



Engage Hierarchy Groups

- Build the Organizational Hierarchy in Engage

Start with the highest level
"Enterprise"



NiCE

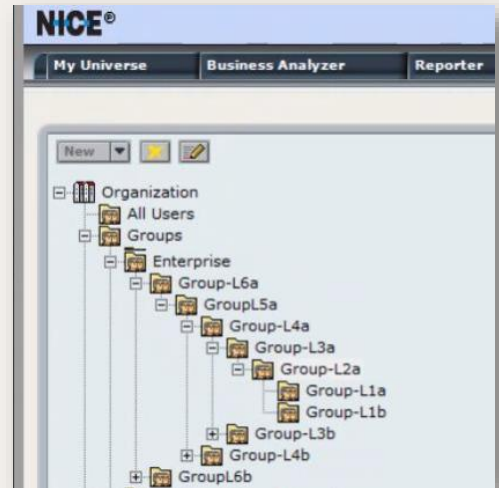
In the Engage User Admin, build the organizational hierarchy groups. Start with the highest level "Enterprise"

Each sub-group below "Enterprise" will represent another level.
NOTE: Multiple groups can exist on the same level, except for the enterprise level:

The level 1 groups are where the agents will be assigned.
Reminder – Agents can only be associated with a single level 1 group

Engage Hierarchy Groups

- Agents are assigned to the level 1 groups
- Agents can only be assigned to a single level 1 group
- Each level 1 group should have a single supervisor assigned (Marked as a group manager in Engage)



The level 1 groups are where the agents will be assigned.

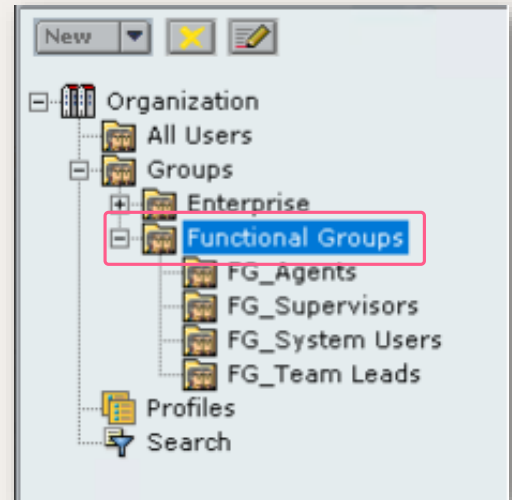
Reminders:

- Agents can only be associated with a single level 1 group
- Agents can only have 1 assigned supervisor
- Supervisors should be assigned to the level 1 group and marked as a group manager



Engage Functional Groups

- Build 4 Functional Groups in Engage (Not part of the hierarchy groups)
 - FG_Agents
 - FG_Supervisors
 - FG_System Users
 - FG_Team Leads



NiCE

Build 4 Functional Groups in Engage

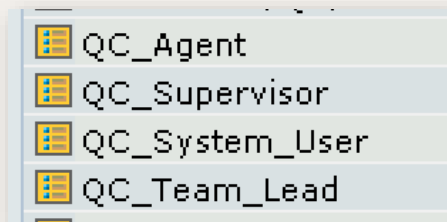
- FG_Agents
- FG_Supervisors
- FG_System Users
- FG_Team Leads



Engage Functional Groups

- Build 4 Profiles and assign to the respective Functional Group
 - QC_Agent
 - QC_Supervisor
 - QC_System_User
 - QC_Team_Lead

Create these exactly as you see here



No permissions need to be assigned to the profiles, unless the users will be accessing Engage

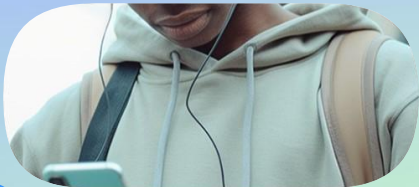
Build 4 Functional profiles in Engage

- QC_Agent
- QC_Supervisor
- QC_System_User
- QC_Team_Lead

NOTE: Permissions do not need to be assigned to these profiles, unless the users will be logging in to Engage

Assign each profile to their respective Functional Group – QC_Agent → FG_Agents, QC_Supervisor → FG_Supevisors, etc.)

Interaction analytics
Configuration



Interaction analytics Hierarchy Configuration

- Set the number of levels for the hierarchy
- Mark the levels you want visible in NIA/QC
- Set the display name
- NOTE: Additional configurations

FileToolsHelp

MetadataConfiguration ParametersHierarchyJobs

Levels: 3SET

Name	Visible	Display Text	Control Type
Hierarchy L1	☒	L1	MultiSelectPopup
Hierarchy L2	☒	L2	MultiSelectPopup
Enterprise	☒	Enterprise	MultiSelectPopup
Hierarchy L4	☐	Hierarchy L4	MultiSelectPopup
Hierarchy L5	☐	Hierarchy L5	MultiSelectPopup
Hierarchy L6	☐	Hierarchy L6	MultiSelectPopup
Hierarchy L7	☐	Hierarchy L7	MultiSelectPopup
Hierarchy L8	☐	Hierarchy L8	MultiSelectPopup
Hierarchy L9	☐	Hierarchy L9	MultiSelectPopup
Hierarchy L10	☐	Hierarchy L10	MultiSelectPopup
Hierarchy L11	☐	Hierarchy L11	MultiSelectPopup
Hierarchy L12	☐	Hierarchy L12	MultiSelectPopup
Hierarchy L13	☐	Hierarchy L13	MultiSelectPopup
Hierarchy L14	☐	Hierarchy L14	MultiSelectPopup
Hierarchy L15	☐	Hierarchy L15	MultiSelectPopup



Interaction analytics Hierarchy Configuration

Metadata Configuration Parameters Hierarchy Jobs		
FIND/REPLACE >>		
Valid	Name	Value
✔	Nexidia.Hierarchy.HierarchyDataRestrictionLevel	0
✔	Nexidia.Hierarchy.UseSupervisor	true
✔	Nexidia.Hierarchy.Viewer.VisibleLevelCount	3

Controls access restrictions in NIA/QC

Sets supervisor based on hierarchy or data extraction

Sets the number of levels in the hierarchy



Hierarchy Process in DEF



Hierarchy Process in DEF

- Download the latest process JSON

 Engage - Import Hierarchy Agent and User (V2) to Nia Rev B.json

- Create a new process in DEF and upload the JSON

 Add a Process

View Quantum

View Schedule

Process Template

Predefined Process Templates are no longer pre-installed. These Processes are still available and documented on the DEF Confluence site and also downloadable as a complete package via SDC; any DEF Process can be imported below.

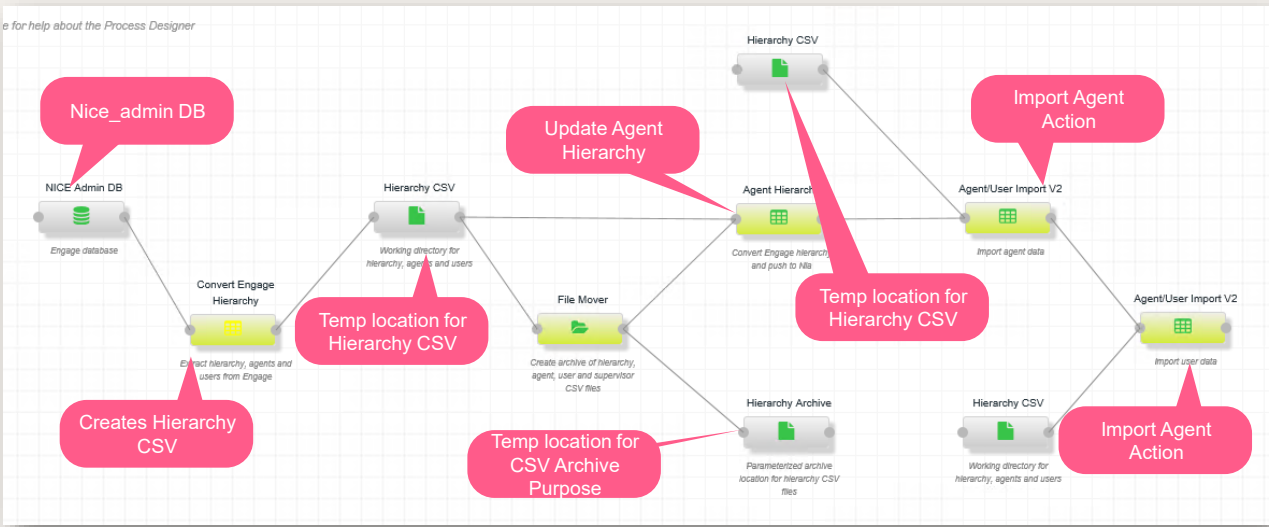
Process Library:
<https://tlvconfluence01.nice.com/display/NXDEF/Processes>

✓ Select a Process Template (*.json)

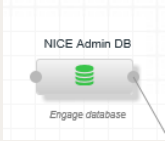


Hierarchy Process in DEF

For help about the Process Designer



Engage DB Resource



NICE Admin DB
Engage database

Resource - NICE Admin DB

AssignDeleteClose

EditPropertiesLabels

NameNICE Admin DB

Connection URLjdbc:sqlserver://db1;databaseName=nice_admin;integratedSecurity=true

User IDnicetraining\niceadmin

Password.....

Submit

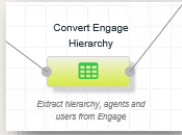
Fields in bold are required

Engage DB resource should connect to the nice_admin database in Engage

Connection string -
jdbc:sqlserver://db1;databaseName=nice_admin;integratedSecurity=true



Convert Engage Hierarchy Action



Delete

Close

Edit

Properties

Labels

Logs

Organizational Hierarchy Root Identification

Hierarchy Name

?

Organizational Hierarchy Root Node Name

Enterprise

?

Build Agent Supervisor CSV

true

?

Inactive Users Node

Inactive Users

?

Integrated Authentication

false

?

Agent Identification

"QC_Agent" Profile

?

Exclude nodes from Hierarchy Identification

Hierarchy Name

?

Exclude nodes from Hierarchy-Including All Children

?

Submit

Hide Advanced

Fields in bold are required

Advanced Options

The Convert Engage Hierarchy Action contains several advanced options for Hierarchy. These include:

- Specifying the root node (Either by name or ID)
- Including an Inactive Users Node
- Specifying the Agent Identification method
- Selecting nodes to be excluded from Hierarchy

Hierarchy CSV Resource



Resource - Hierarchy CSV

← Assign Delete × Close

Edit Properties Labels

Name

Hierarchy CSV

Path

D:\Program Files\Data Exchange Framework

?

User Name

?

User Password

?

Submit

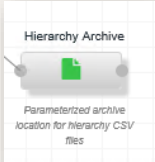
Fields in bold are required

Temporary storage for the Hierarchy CSV

Hierarchy CSV Resource is a temporary storage location for the CSV file generated by the Convert Engage Hierarchy action



Hierarchy CSV Resource



Resource - Hierarchy Archive

AssignDeleteClose

EditPropertiesLabels

Name

Hierarchy Archive

Path

e:\hierarchyArchive

?

User Name

?

User Password

?

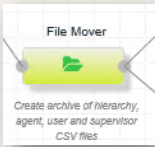
Submit

Fields in bold are required

Temporary storage for the Hierarchy CSV for archival purpose



File Mover



Action - File Mover

DeleteClose

EditPropertiesLabelsLogs

Process files from metadata	false	?	^
File filter	*.CSV	?	
Search directories recursively	true		
Flatten output directory structure	false	?	
Apply process quantum	false	?	
Overwrite existing output files	true	?	
Fail if no files moved	false	?	
Delete inputs after move	false	?	^

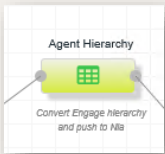
Submit

Show Advanced

Fields in bold are required



Agent Hierarchy Action



DeleteClose

EditPropertiesLabelsLogs

Process files from metadatafalse?

File filter*_Hierarchy_*.csv

Delimiter,

Host NameNIA?

Fail if nothing is imported within # minutes0?

Force file encoding?

Delete files after processingtrue?

Reservation Duration Minutes60?

Submit

Show AdvancedFields in bold are required

Set the hostname or FQDN for the NIA Applications Server

The Agent Hierarchy Action feeds the CSV data to the NIA hierarchy web service to update the hierarchy data



Agent / User Import v2



Action - Agent/User Import V2

DeleteClose

EditPropertiesLabelsLogs

Process files from metadatafalse?

File filter*_Agent_*.csv

Delimiter,

FlowAgent?

Host NameNIA?

Default Integrated Authentication in NIAfalse?

Integrated Authenticationtrue?

Fail if nothing is imported within # minutes0?

Submit

Show AdvancedFields in bold are required

Set Agent or User Flow

The Agent Hierarchy Action feeds the CSV data to the NIA hierarchy web service to update the hierarchy data



Do it Yourself

- Refer to Exercises # 12



summary

In this lesson we covered:

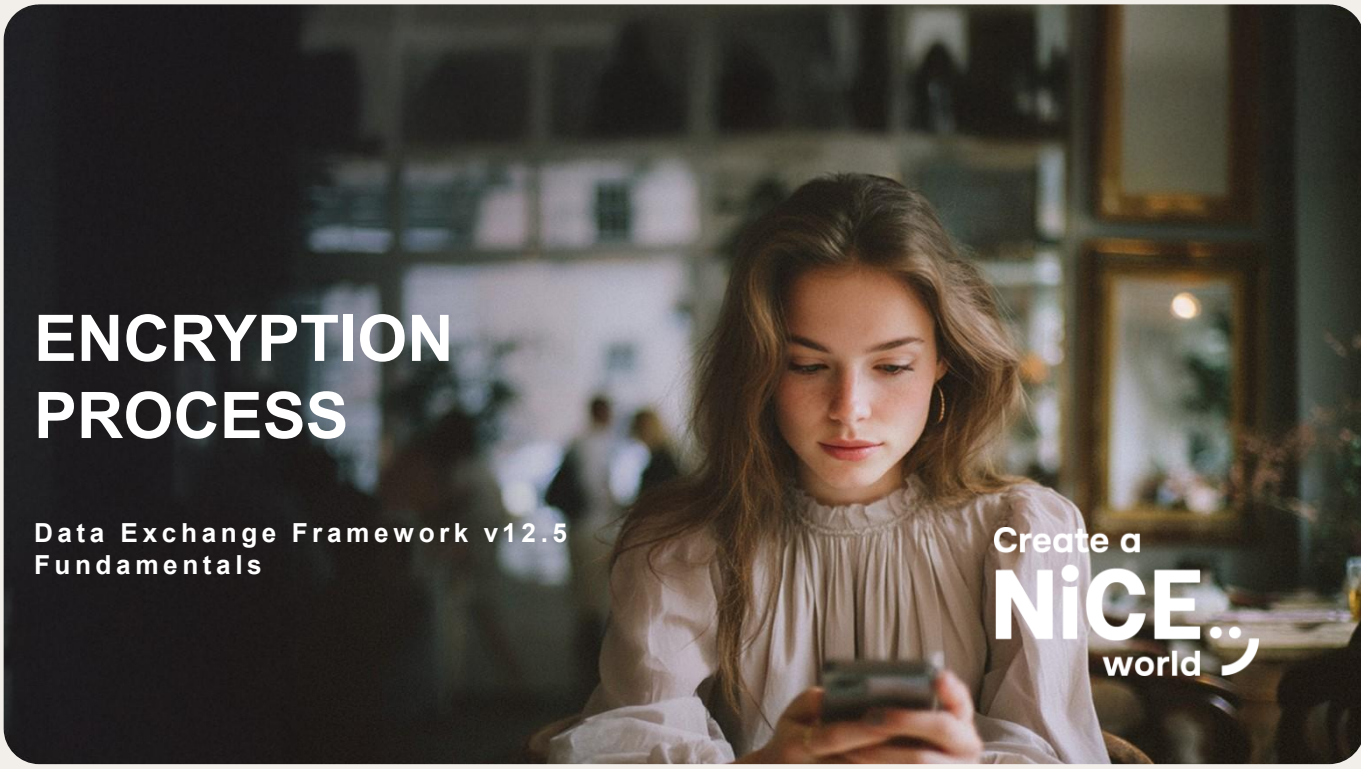
- Hierarchy Import of Agents and Users from Engage





Thank You





ENCRYPTION PROCESS

Data Exchange Framework v12.5
Fundamentals

Create a
NiCE..
world ☺

objectives

By the end of this lesson, you will be able to:

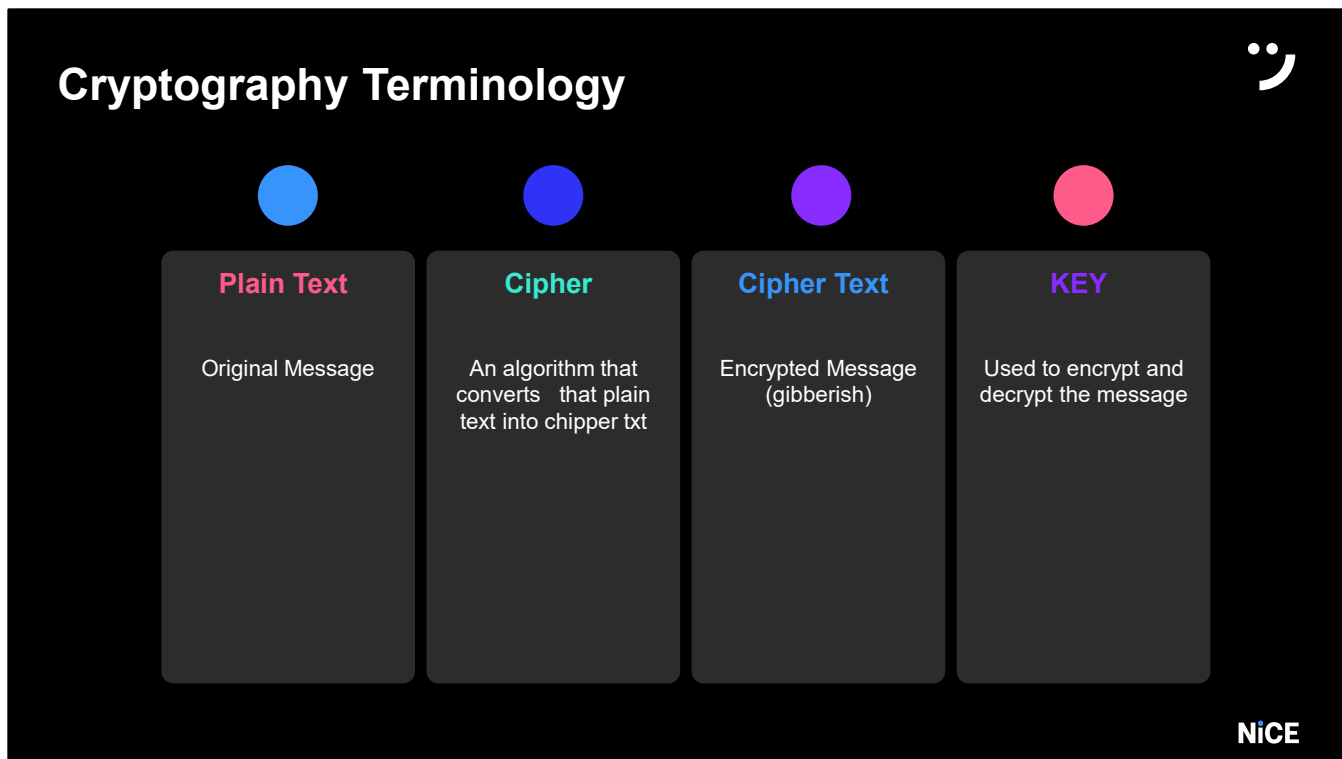
- Describe how to configure the 3 decryption templates:
 - Generate Protected Key (GPK)
 - Key Extraction
 - Load Key



Encryption

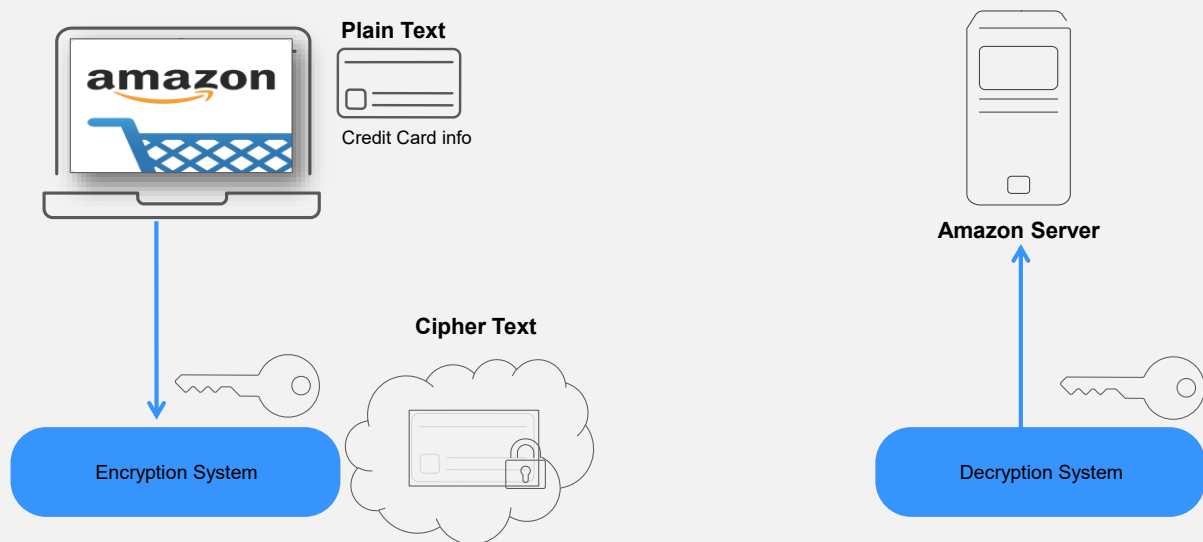


What is encryption and why it is needed



The word Cryptography is combined from the words crypto and graph, which means secret writing.

Cryptography - Symmetric Key



Let's look at an example of encryption; we will review the process of encryption using a Symmetric Key=Same key is used to encrypt and decrypt the message.

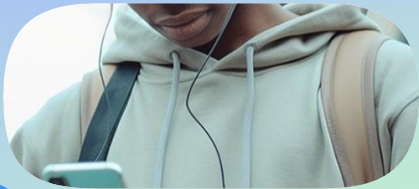
When we are shopping online, for example, on Amazon, we would like to secure our credit card information from the moment we typed it on our laptop till it reaches the Amazon server.

We have a cipher algorithm (the most common one is the ASE) that encrypts the credit card information.

The encrypted credit card information reaches the Amazon server; then, the message is decrypted; the credit card is validated and stored securely within the Amazon server.

For Amazon to be able to decrypt the credit card information, we need to transfer to it the key we used in the encryption process.

Engage – DEF
Decryption process



Engage System

Encrypted File

nice_crypto DB

	uGuid	vbEncryptionKey
1	7F5D0A91-02C7-477E-8EDF-03AFA0AEE015	0x85D0CCF5C39C7C286566644372F8C1BA8A93F7870775C4B6...
2	CACFDDEED-A2B5-47D4-8127-07354B226C7D	0xD5D044C1DA5A9D0C6AA046227DD8C33341328D3F159E4F6...
3	1DB52C2B-E0EB-46FD-BCE0-073D5AAC594A	0x8B86241911A0C6BD659E6438643B2AE86241A6FFB8AE0FFE...
4	8BE1ADCE-DB26-4303-8ED4-0AE31497F7B5	0xCE6A8F3AD5E2D1A942628A1F6D877ECF9B9F8FFC49F585B...
5	11F15E6D-ED17-46C8-BC20-10205DD3295E	0x36CABA7814551001994292E62A5DD0E13C826B46E7CD4F46...

The calls in Engage are saved as an encrypted file, to playback a call; we need to use the key that decrypts the call. Each call requires a decryption key. The keys are stored in nice_crypto database.

Engage DEF Decryption Process

Engage Server

- Encrypted Audio Files

Extraction Server

- Extracts from Engage the keys that are needed to decrypt the audio files
- For security reason is installed on a dedicated server

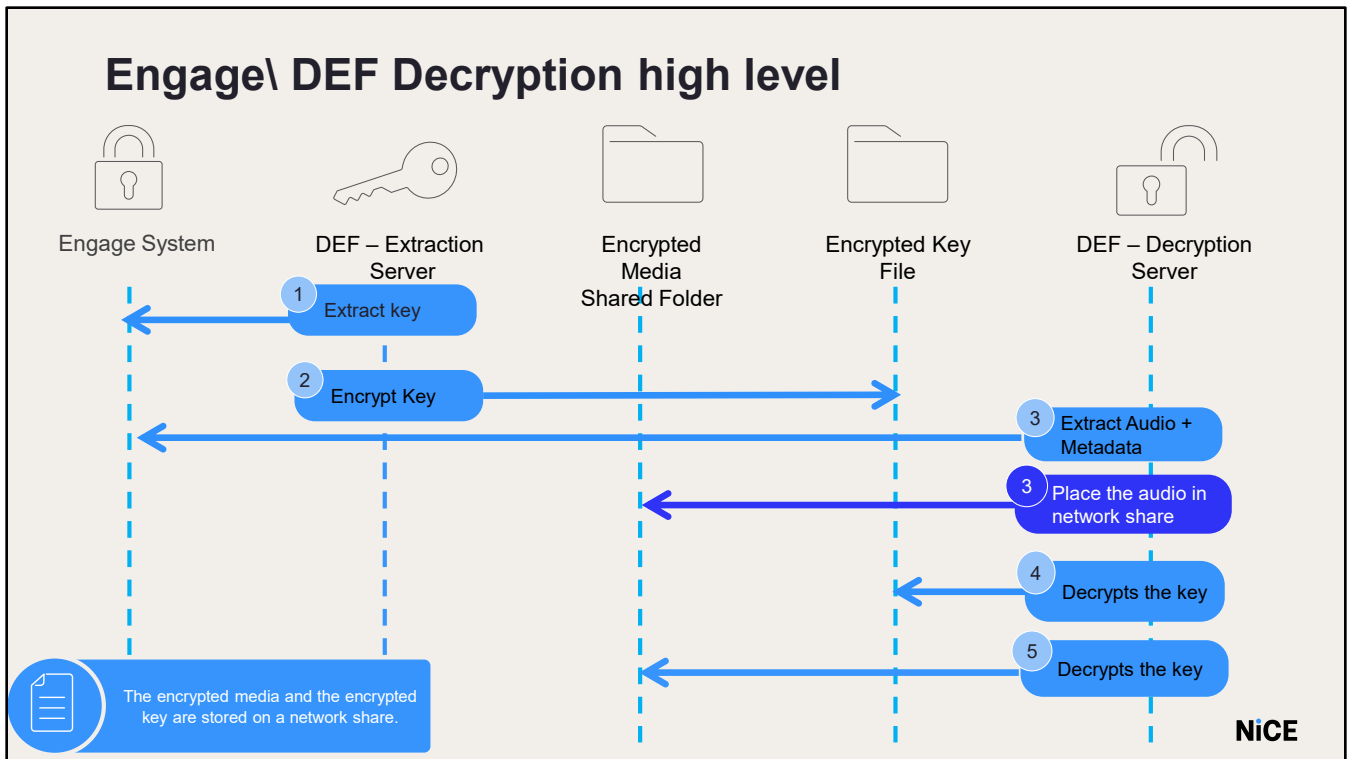
Decryption Server

- Generate protected keys (DEF internal symmetric keys)
- Decrypts the audio files

NMF file decryption takes place on two separate DEF servers. The first server is used to extract keys from the recorder, encrypt them with a symmetric key, and write the contents to an encrypted CSV file. This server will be referred to as the Extraction server.

The second server is used to generate a symmetric key for encrypting and decrypting keys, decrypt and load the keys into the DEF database and decrypt the NMF interaction files. This server will be referred to as the Decryption server.





1. The extraction server will query the engage DB for new encrypted interactions and extract the relevant keys from nice_crypto database.

2. The extraction server encrypts the keys that were extracted from Engage with DEF internal symmetric key and write the contents to an encrypted CSV file

3. The decryption server extract from Engage the encrypted audio files and the relevant metadata, it places the encrypted audio in a network share and metadata in the DEF database

4. The decryption server decrypts the encrypted key files using the DEF internal symmetric key (the same key that was used to encrypt the key that was extracted from Engage in step 2) and loads them into the DEF database

5. The decryption server encrypts the media file using the key that was extracted from Engage

Note: The encrypted media and the encrypted key are stored on a network share.





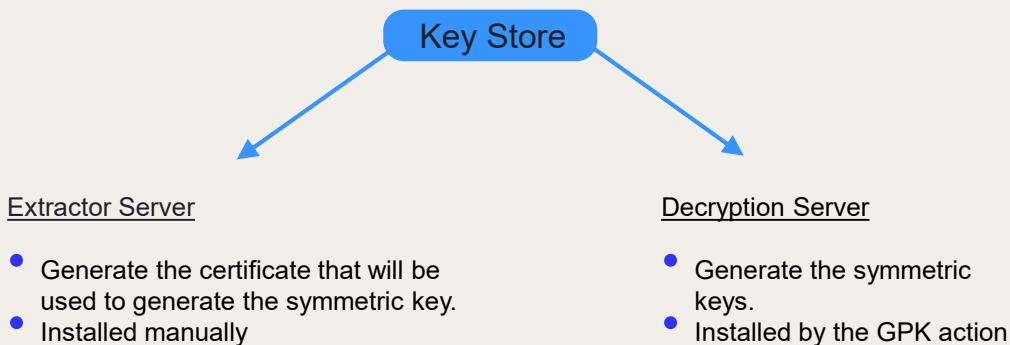
Before we can extract the encrypted audio files from Engage, a Java KeyStore should be installed on the extraction server as preliminary one time step. A KeyStore is a repository where private keys, certificates, and symmetric keys are stored; it stores the client certificates of the CAs it is willing to trust.

1. We start by installing the KeyStore on the extraction server and generating a certificate and copy it to the decryption server.
2. The generated protected key process uses the certificate to create symmetric keys used to encrypt and decrypt key information pulled from the nice_crypto database. This process runs only once when we initiate the system.
3. The extraction server will query the engage DB for new encrypted interactions, and it will extract the relevant keys encrypting them with the symmetric key from step 2, and writing the encrypted contents to a CSV file.
4. The decryption server will use the symmetric key to decrypt the key from the extraction server and load them into the DEF database, and then use this key to decrypt the audio file and push it to NIA



Prerequisite - Java KeyStore

A KeyStore is a repository where private keys, certificates, and symmetric keys are stored. It part of Java Kit



The Key Extraction action must have a self-signed certificate manually created as a preliminary step.

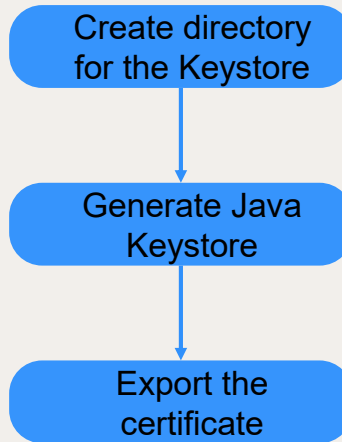
In the Engage DEF decryption process, two Keystore are used:

- The first Keystore is installed manually on the Extractor server, and creates the certificate that is used to generate the symmetric keys.
- The second Keystore is installed by the Generate Protected Key action on the Decryption server, and is used to create the symmetric keys that encrypt and decrypt key information pulled from the nice_crypto database.

Installing Keystore on the Extraction Server



Installing Keystore



NiCE

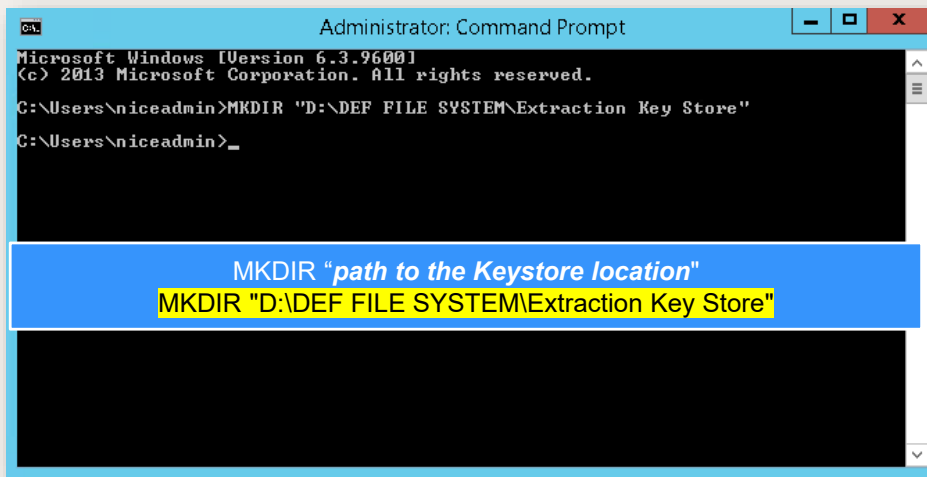
A KeyStore and certificate must be created on the server hosting the Key Extraction process that will be used by the Generate Protected Key process running on a decryption server.

The steps are:

- Create a directory for the Keystore
- Generate Java Keystore
- Export the certificate



Create directory for the Keystore



NICE

The following commands must be executed in a command prompt with administrative privileges. On a production server, the resources should be created on a network share.

Create a directory for the Keystore. Open a CMD prompt with admin privileges.

MKDIR "path to the Keystore location."

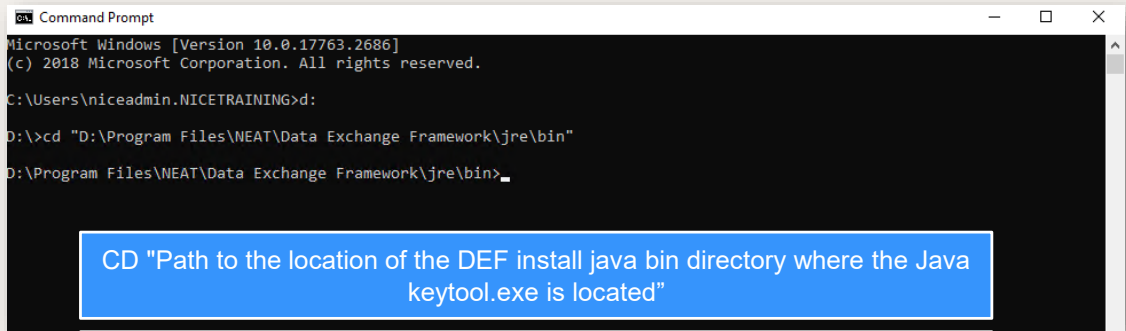
In the training lab:
MKDIR "D:\DEF FILE SYSTEM\Extraction Key Store"

Note: If you copy and paste this command into the CMD prompt you will need to manually edit the command and update the double quotes (") and hyphen (-) characters.

The characters used in Word\PDF will cause a failure in the CMD prompt.



Generate Java Keystore



```

Command Prompt
Microsoft Windows [Version 10.0.17763.2686]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\niceadmin.NICETRAINING>cd
D:\>cd "D:\Program Files\NEAT\Data Exchange Framework\jre\bin"
D:\Program Files\NEAT\Data Exchange Framework\jre\bin>
  
```

CD "Path to the location of the DEF install java bin directory where the Java keytool.exe is located"

CD "D:\Program Files\NEAT\Data Exchange Framework\jre\bin"

The keytool utility can be found in the jre/bin folder of the DEF install directory.

change the directory location to the location of the DEF install java bin directory where the Java keytool.exe is located, or the bin directory of your installed JRE, if you are not using the embedded

CD "D:\Program Files\NEAT\Data Exchange Framework\jre\bin "



Generate Java KeyStore

```
Administrator: Command Prompt - keytool -storetype JCEKS -keystore "D:\DEF...
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\niceadmin.NICETRAINING>d:

D:\>cd "D:\Program Files\NEAT\Data Exchange Framework\jre\bin"

D:\Program Files\NEAT\Data Exchange Framework\jre\bin>
D:\Program Files\NEAT\Data Exchange Framework\jre\bin>
>keytool -storetype JCEKS -keystore "D:\DEF FILE SYSTEM\Extraction Key Store\ExtractionkeyStore.ks" -genkey -keyalg RSA -keysize 2048 -validity 3650 -alias ExtractAlias
Enter keystore password:
keytool -storetype JCEKS -keystore "directory for the keystore you
created in the 1st step\keystore name.ks" -genkey -keyalg RSA -keysize
2048 -validity 3650 -alias AliasName
keytool -storetype JCEKS -keystore "D:\DEF FILE SYSTEM\Extraction Key
Store\ExtractionkeyStore.ks" -genkey -keyalg RSA -keysize 2048 -validity
3650 -alias ExtractAlias
```

Create the Keystore:

Run the following command and edit the Keystore location, name and alias name

keytool -storetype JCEKS -keystore " path to the directory for the keystore you created in the 1st step\keystore name.ks" -genkey -keyalg RSA -keysize 2048 -validity 3650 -alias AliasName

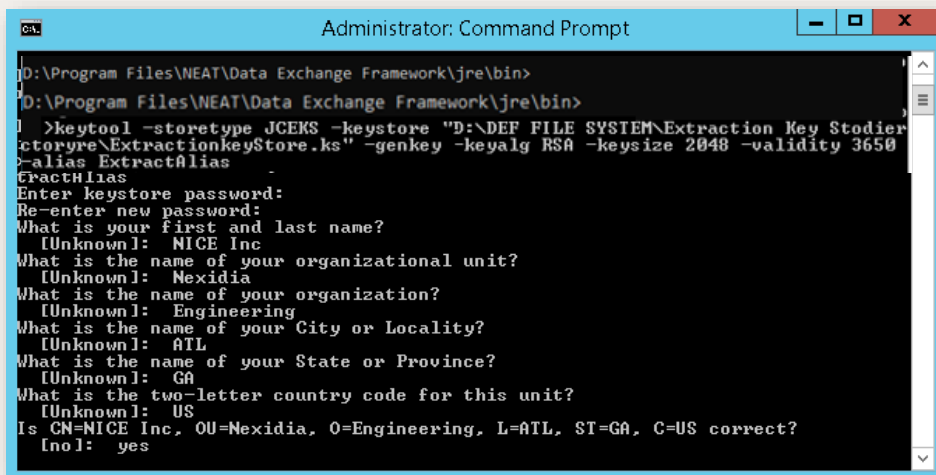
In the training lab:

keytool -storetype JCEKS -keystore "D:\DEF FILE SYSTEM\Extraction Key Store\ExtractionkeyStore.ks" -genkey -keyalg RSA -keysize 2048 -validity 3650 -alias ExtractAlias

Note: The alias for the Keystore can be set to any value, but this alias is used later in the process and must be documented., Make a note on the Keystore location, password, and alias



Generate Java KeyStore



```
Administrator: Command Prompt

D:\Program Files\NEAT\Data Exchange Framework\jre\bin>
D:\Program Files\NEAT\Data Exchange Framework\jre\bin>
>keytool -storetype JCEKS -keystore "D:\DEF FILE SYSTEM\Extraction Key Stodier\ExtractionKeyStore.ks" -genkey -keyalg RSA -keysize 2048 -validity 3650
>-alias ExtractAlias
ExtractAlias
Enter keystore password:
Re-enter new password:
What is your first and last name?
[Unknown]: NICE Inc
What is the name of your organizational unit?
[Unknown]: Nexidia
What is the name of your organization?
[Unknown]: Engineering
What is the name of your City or Locality?
[Unknown]: ATL
What is the name of your State or Province?
[Unknown]: GA
What is the two-letter country code for this unit?
[Unknown]: US
Is CN=NICE Inc, OU=Nexidia, O=Engineering, L=ATL, ST=GA, C=US correct?
Inol: yes
```



Note:

During routine security audits, a certificate without valid information can be removed, which disables the decryption process (do not leave blank info)

NICE

You will be prompted to create a password during the creation of the key store. This password should meet the security requirements of the customer.

There is no recommended default password.

During the generation of the Keystore, information about the signing entity must be entered.

When prompted for first and last name enter: Nice Inc.

When prompted for the organizational unit enter: Nexidia

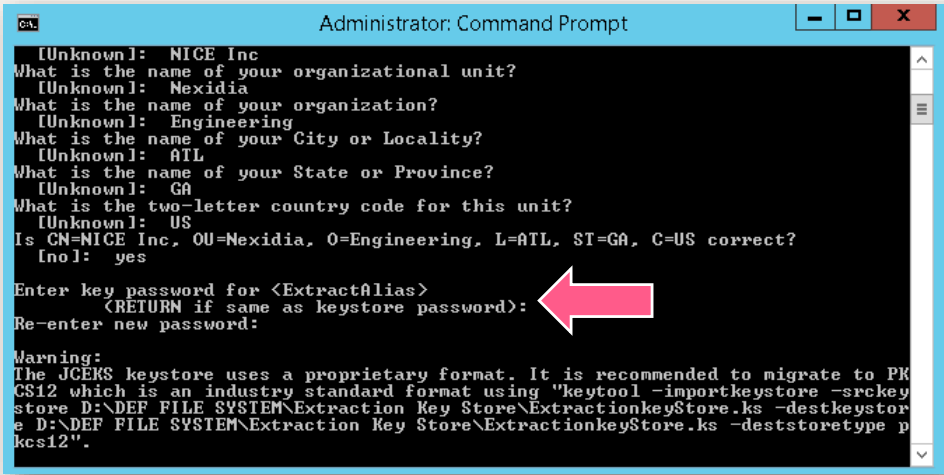
When prompted for the organization enter the division: Engineering

Enter the City, State, and Country code when prompted: ATL, GA, US

Important: During routine security audits, a certificate without valid information can be removed, which disables the decryption process. (do not leave blank info)



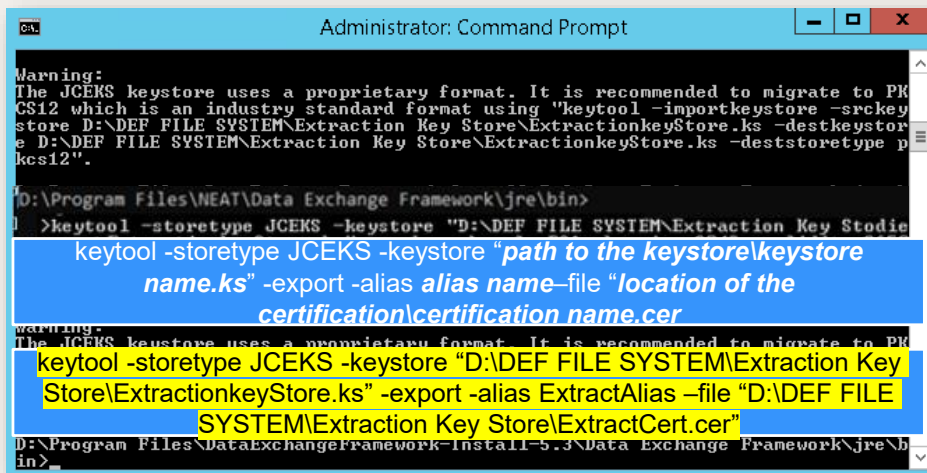
Generate Java KeyStore



Enter the password created in the previous step when prompted
Ignore the warning message



Export the Utility Certificate



```

Warning:
The JCEKS keystore uses a proprietary format. It is recommended to migrate to PKCS12 which is an industry standard format using "keytool -importkeystore -srckeystore D:\DEF FILE SYSTEM\Extraction Key Store\ExtractionkeyStore.ks -destkeystore D:\DEF FILE SYSTEM\Extraction Key Store\ExtractionkeyStore.ks -deststoretype pkcs12".

D:\Program Files\NEAT\Data Exchange Framework\jre\bin>
>keytool -storetype JCEKS -keystore "D:\DEF FILE SYSTEM\Extraction Key Store\ExtractionkeyStore.ks" -export -alias ExtractAlias -file "D:\DEF FILE SYSTEM\Extraction Key Store\ExtractCert.cer"

```

NiCE

After the Keystore has been created, use the following command to export the public key from the certificate into a .cer file that can be consumed by the Generate Protected Key action on the Decryption server.

Note: the alias used for the exporting the certificate was specified when the Keystore was created in the previous step.

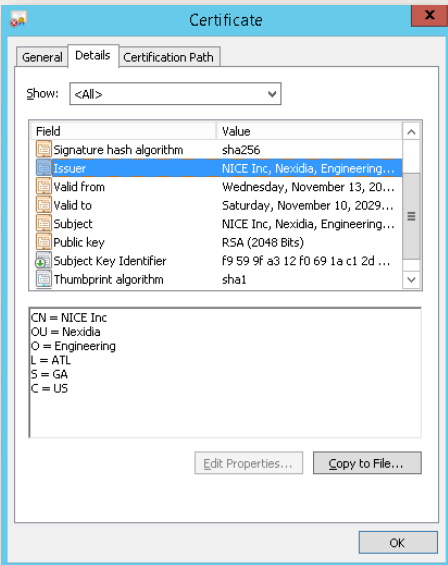
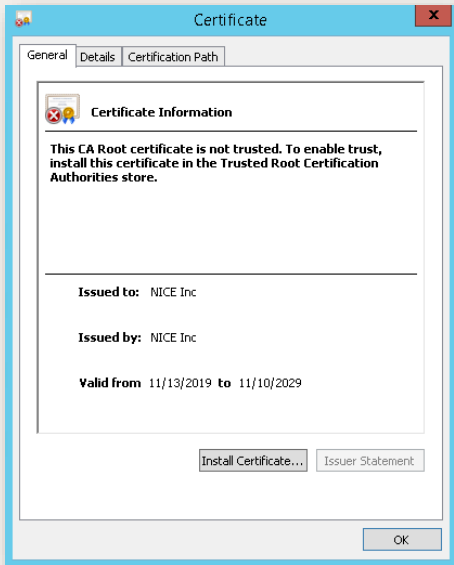
```
keytool -storetype JCEKS -keystore "path to the keystore\keystore name.ks" -export -alias alias name -file "location of the certification\certification name.cer"
```

In the training lab:

```
keytool -storetype JCEKS -keystore "D:\DEF FILE SYSTEM\Extraction Key Store\ExtractionkeyStore.ks" -export -alias ExtractAlias -file "D:\DEF FILE SYSTEM\Extraction Key Store\ExtractCert.cer"
```



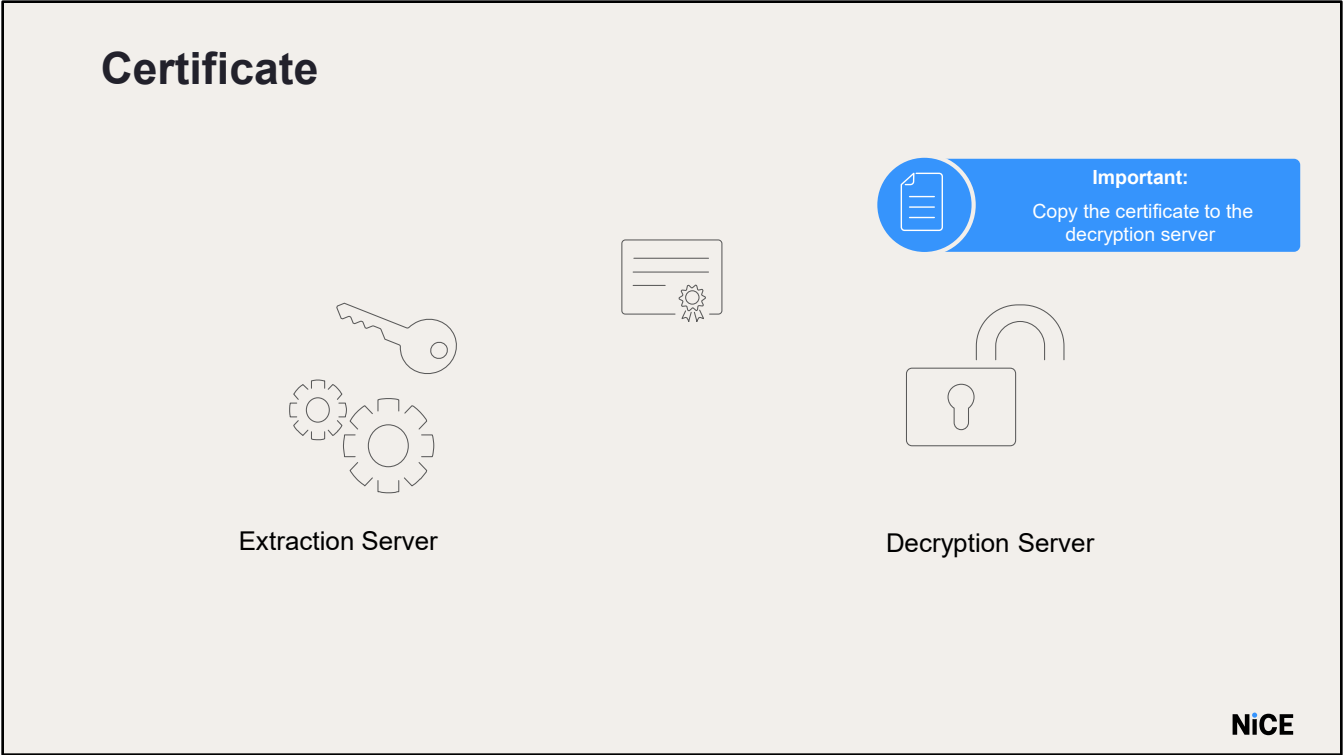
Certificate



Open Windows Explorer and navigate to the directory where the certificate was created and double click on the certificate and validate the information

D:\DEF FILE SYSTEM\Extraction Key Store





Copy the certificate to the decryption server



Utilize the
Decryption
templates



NiCE

The DEF Decryption templates are used to decrypt the encrypted files of Engage calls

Engage\DEF Decryption Templates

Generate Protected Key

- One time process run on demand
- Generate symmetric keys use to encrypt and decrypt the key information pulled from Engage (nice_crypto database)

Key Extraction

- Extract keys from the recorder, encrypting them with the symmetric key from the previous action, and writing the encrypted contents to a CSV file

Load Key & File Decryption

- Decrypt the audio files and load them to NIA

There are three Decryption templates:

Generate Protected Key - GPK-responsible for generating the symmetric keys used to encrypt and decrypt key information pulled from the nice_crypto database. It's a one-time process run on demand

Key Extraction - Extract keys from the recorder, encrypting them with the symmetric key from the previous action, and writing the encrypted contents to a CSV file. This process runs on a dedicated server

Load Keys - Decrypt the audio files and load them to NIA



Generate Protected Key (GPK) (On the Decryption Server)



NiCE

This is a one-time process that runs on-demand launched for initial key creation, and subsequent key rotation.

It generates symmetrical keys use to encrypt and decrypt the key information pulled from Engage and is done on the Decryption server.

Generate Protected Key (GPK)

Add a Process

View Quantum

View Schedule

Process Template

NICE Decryption - Generate Protected Key

?

Name

01-NICE Decryption - Generate Protected Key

Description

Generates a protected and encrypted symmetr

Start Time

11/17/2019 00:00

Interval

1

(min: 1; max: 60)

Interval Type

Day

Blackout Start Time

00:00:00

?

Blackout End Time

00:00:00

?

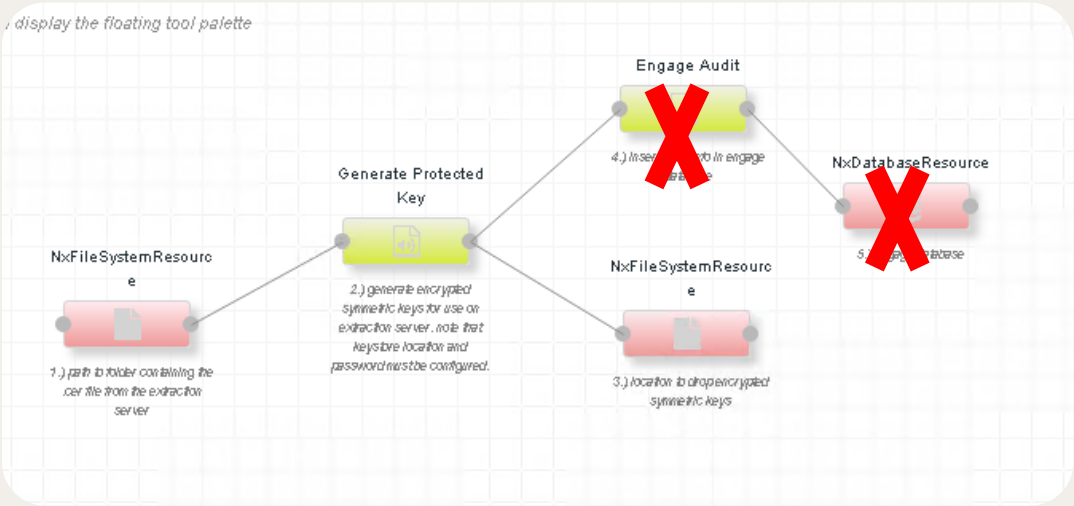
Submit

Rename this process as "01-NICE Decryption Generate Protected Keys"

Tip: When creating processes it is helpful to preface the name with a two-digit number that indicates where this process falls in the DEF flow, rename this process as
01- NICE Decryption Generate Protected Keys



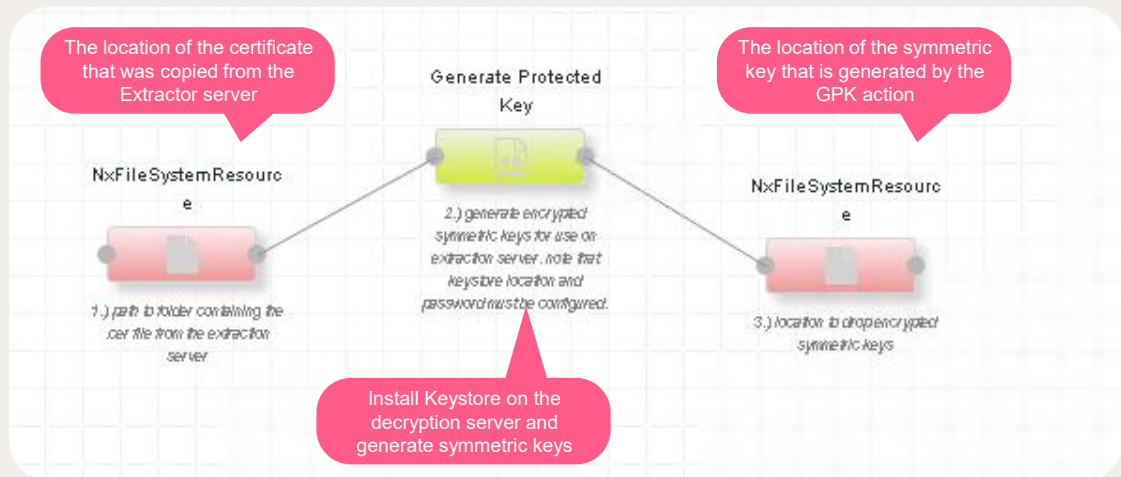
Generate Protected Key (GPK)



The flow was created with the Engage Audit action, delete the Engage Audit action, and the associated DB resource.



Generate Protected Key (GPK)



Note: all resources should be created on network shares whenever possible

Input NxFileSystemResource :Folder where the certificated that was copied from the Extractor server is stored

Output NxFileSystemResource :Folder where the Generate Protected Key process will drop the Symmetric key. This resource is used by the Key Extraction and Load Keys process

Generate Protected Key (GPK)

Enter the path to the Keystore\keystore name.ks

Process files from metadata: false

File filter: *.cer

Keystore location: D:\GPK keystore\GPK_keystore ks

Keystore password: *****

Delete certificate files after processing: false

Failure Threshold: 0

Submit

Show Advanced

Fields in bold are required

To avoid confusion, assign different names to the two Keystore in the system: extraction server Keystore, decryption server Keystore

The GPK action performs two actions:

- Installs the Keystore on the decryption server
- Uses the Keystore to generate symmetric keys used to encrypt and decrypt the key information pulled from Engage

The Generate Protected Key action generates a symmetric key that's been encrypted against the certificate of the remote key extraction server, and stores it in both the output resource and in a local, configurable Keystore.

In the Keystore location field enter the path to the location where the GPK action would install the keystore and the keystore name.

Note: To avoid confusion assign different names to the two Keystore in the system: extraction server Keystore, decryption server Keystore



Generate Protected Key (GPK)

Delete
Close

Edit
Properties
Labels
Logs

Process files from metadatafalse?

File filter*.cer?

Keystore locationD:\GPK keystore\GPK_keystore.ke?

Keystore password*****

Delete certificate files after processingfalse?

Failure Threshold0

Submit

Show Advanced

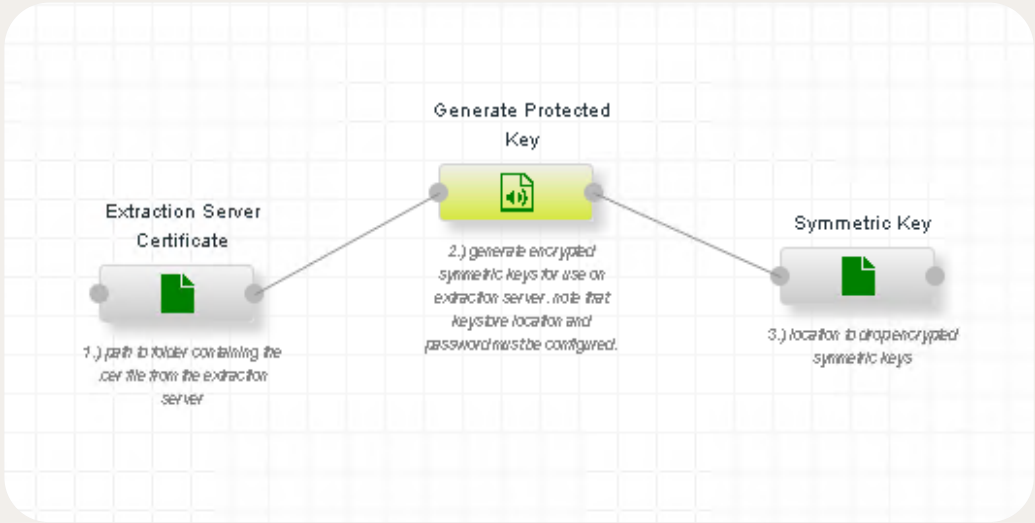
Fields in bold are required

Create in advance the folder where the GPK action would install the Keystore

This is how the GPK action should look like

Important: you need to create in advance the folder (GPK Keystore in NEST1) where the GPK action would install the Keystore

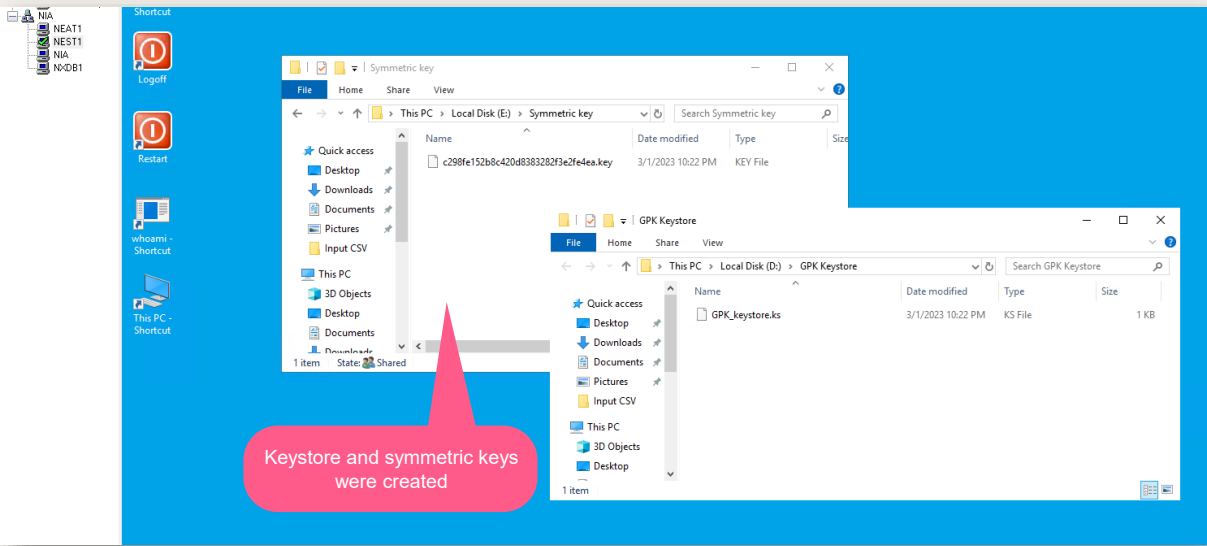
Generate Protected Key (GPK)



This is how the configured flow should look like



Generate Protected Key (GPK)



Do it Yourself

- Refer to Exercises #13



Key Extraction (On the Extraction Server)



This process is done on the Extractor Server and Extract keys from the recorder, encrypting them with the symmetric key from the GPK process, and writing the encrypted contents to a CSV file.

For security reasons, this should be the only process running on the DEF server

Key Extraction

Add a Process

View Quantum View Schedule

Process Template: NICE Decryption - Key Extraction

Name: 02-NICE Decryption - Key Extraction

Description: Extracts keys from Engage, and writes them to

Start Time: 11/21/2019 00:00

Interval: 15 (min: 1; max: 60)

Interval Type: Minute

Blackout Start Time: 00:00:00

Blackout End Time: 00:00:00

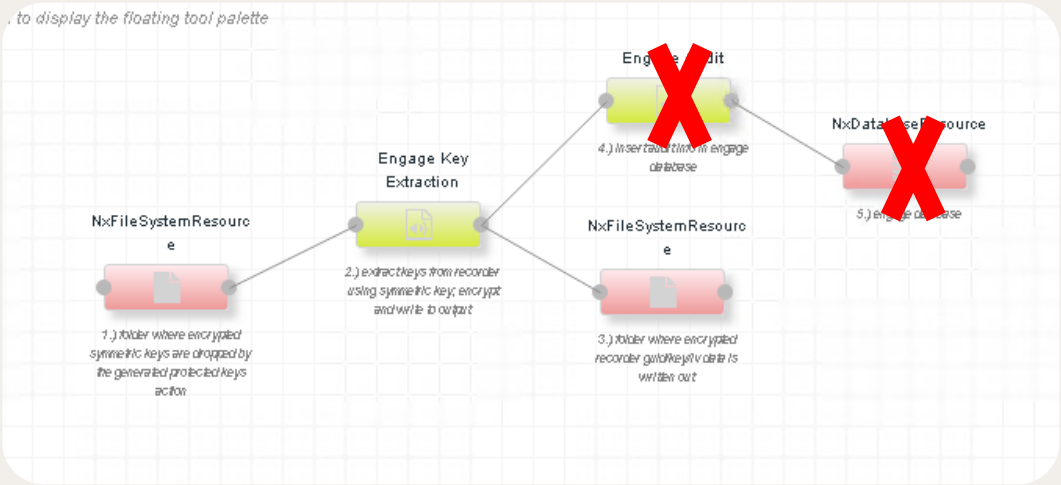
Submit

For security reasons, this should be the only process running on the Extractor server

Tip: When creating processes it is helpful to preface the name with a two-digit number that indicates where this process falls in the DEF flow, rename this process as

02-NICE Decryption - Key Extraction

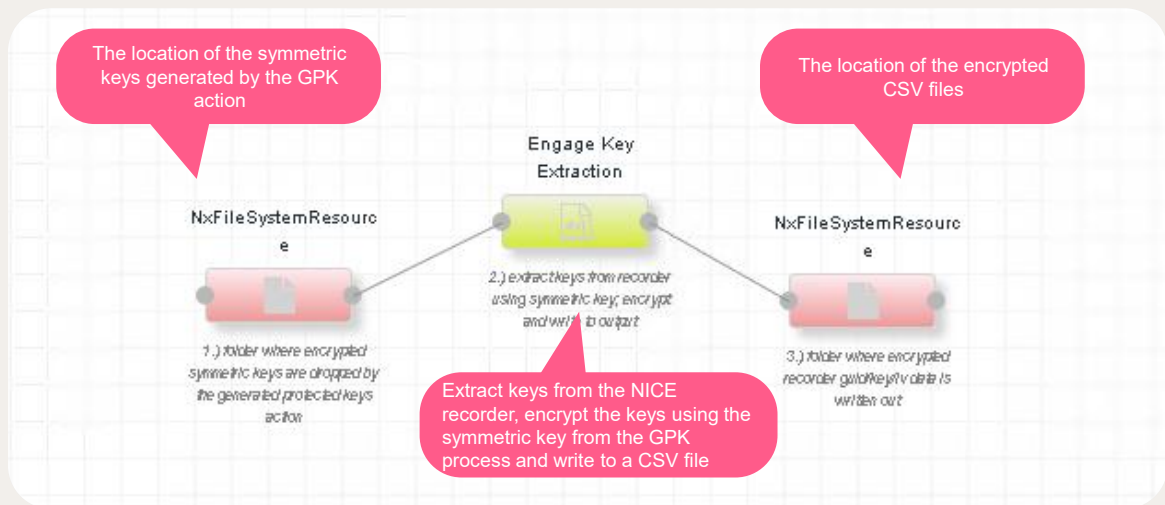
Key Extraction



The flow was created with the Engage Audit action, delete the Engage Audit action, and the associated DB resource.



Key Extraction



Input NxFileSystemResource : The folder where encrypted symmetric keys are dropped by the generated protected keys action. Point to the same location as the resource that was created on the GPK action. New resource needs to be created since this is a different DEF server (each server has its own resources)

Output NxFileSystemResource : Folder where the encrypted CSV file that contains GUID/Key information from the recorder that is used to decrypt the audio files is written out to

Note: All resources should be created on network shares whenever possible.

Key Extraction Action

Connection string to nice_crypto DB

Create a DB resource that will be used to test the connection string to nice_crypto database

When configuring the key extraction action, a connection string to nice_crypto database is required. Since it is entered directly into the key extraction action, there is no direct feedback on the status of the database connection whether is succeeded or failed..

TIP: Create a DB resource that will be used to test the connection string to nice_crypto database, and copy the string from the DB resource into the key extraction action.

Key Extraction Action

Action - Engage Key Extraction

DeleteClose

EditPropertiesLabelsLogs

NICE crypto database connection string

jdbc:sqlserver://db1:1433;databaseNe

?

Keystore location

D:\DEF FILE SYSTEM\Extraction Key Store\

?

Keystore password

.....

?

Keystore alias

ExtractAlias

?

Logger Identifiers

?

Failure Threshold

0

?

Enter the location of the Keystore that is installed on the Extractor server

Submit

Show Advanced

Fields in bold are required

In the Keystore location field, enter the path to the Keystore\keystore name.ks. This is the location of the Keystore installed on the Extractor server.



Key Extraction Action

Action - Engage Key Extraction

DeleteClose

EditPropertiesLabelsLogs

NICE crypto database connection stringjdbcsqlserver://db1:1433;databaseNc?

Keystore locationD:\DEF FILE SYSTEM\Extraction Key Store?

Keystore password.....eye

Keystore aliasExtractAlias?

Logger Identifiers?

Failure Threshold0?

Submit

Show Advanced

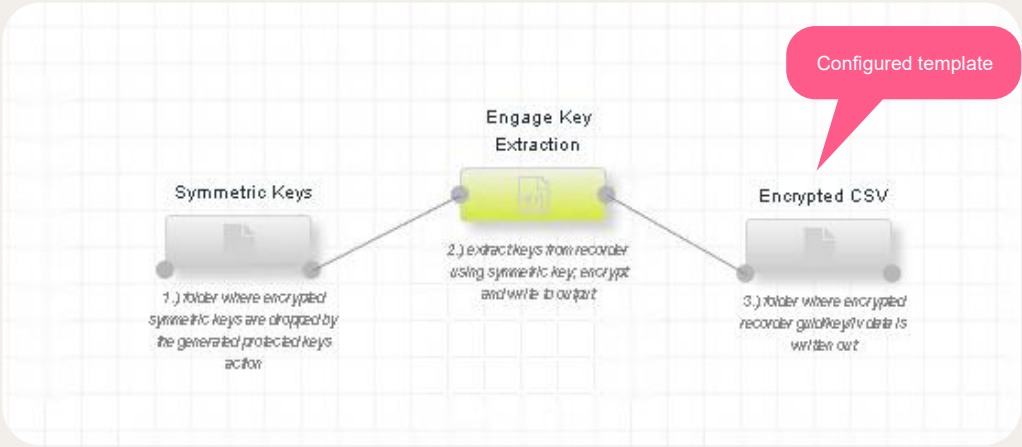
Fields in bold are required

Configured action

NiCE

This is how the Key Extraction action should look like

Key Extraction



scsym
gs.Dmmsm/bpoc.scvshs

ms.Dmmsm/bpoc.scvshs



Do it Yourself

- Refer to Exercises #14



Load Key & NMF
Decryption
(On the Decryption
Server)



Load Key

Add a Process

View Quantum View Schedule

Process Template: NICE Integration Base Template - Load Key

Name: 03-NICE Integration Base Template - Load Key

Description: Not necessary in many cases, as 'Prepare Me'

Start Time: 11/25/2019 00:00

Interval: 15 (min: 1, max: 60)

Interval Type: Minute

Blackout Start Time: 00:00:00

Blackout End Time: 00:00:00

Submit

Rename this process as "03-NICE Integration Base Template - Load Keys files"

We will use the NICE Integration Base Template - Load Keys and add to it the NMF decryption part.

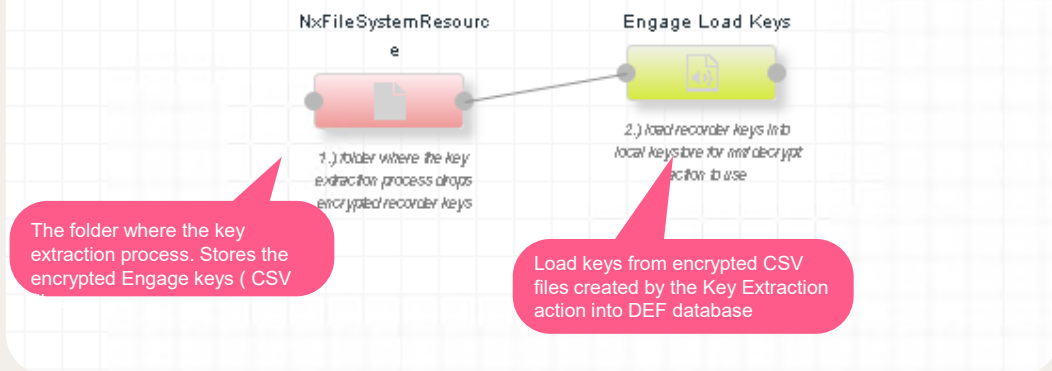
Tip: When creating processes, it is helpful to preface the name with a two-digit number that indicates where this process falls in the DEF flow, rename this process as

03-NICE Integration Base Template - Load Keys

Load Keys and NMF Decryption

TIP: Shift-Click and drag mouse to select multiple components

TIP: Right click to display the floating tool palette



This part of the process decrypted the CSV files created by the Key Extraction action; those files contain the keys that were extracted from the Nice recorder

and load the keys into the DEF database.

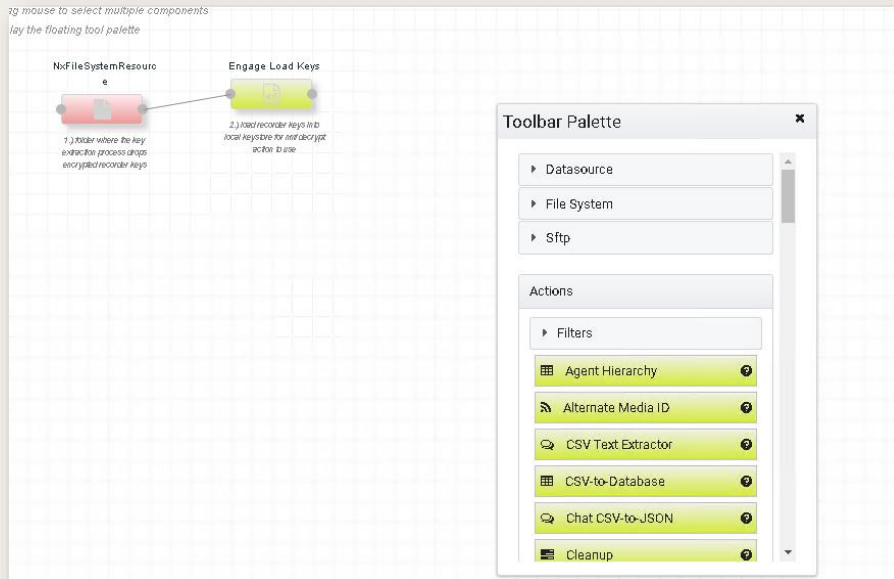
Input NxFileSystemResource - Folder where the Key Extraction process stores the encrypted CSV files that contain the decryption keys for the NMF files. This resource was created when the Key Extraction process was created.

Load Keys - The Load Keys action is responsible for reading the encrypted CSV files generated by the Key Extraction action, decrypting them using the appropriate symmetric key from the keystore in

the GPK action, and importing key and initialization vector information into the DB table nx_nmf_keystore for use by the NMF Decrypt action.



Load Keys and NMF Decryption

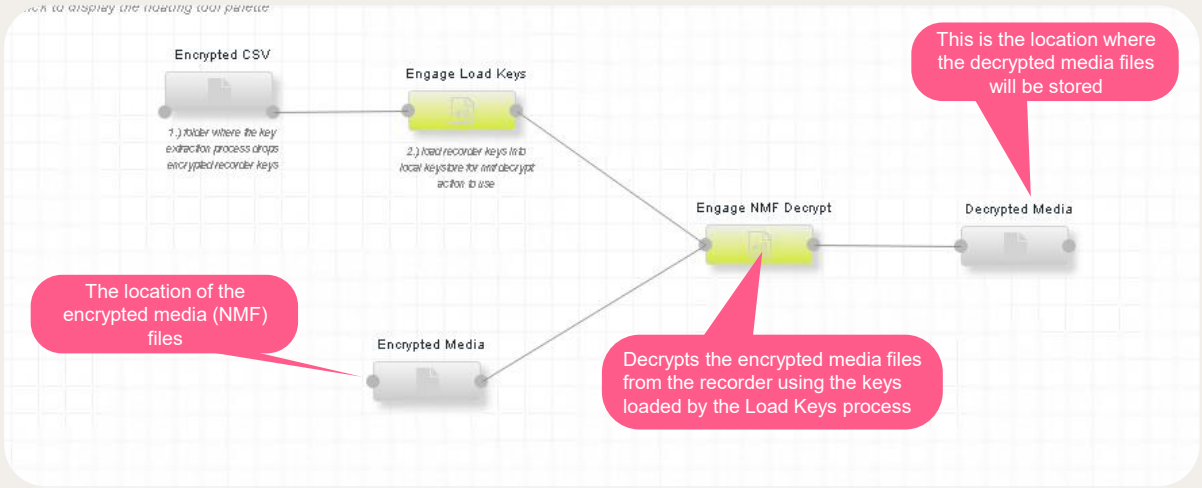


NMF Decrypt - The NMF Decrypt action uses the DB table `nx_nmf_keystore` from the previous action to decrypt incoming NMF files.

We will edit the Load Keys template and add to it the NMF decryption part.

In Design mode right click on the design area and add the 'Engage NMF Decrypt' action and two file system resources to connect as an input and output the 'Engage NMF Decrypt' action

Load Keys and NMF Decryption



The NMF Decrypt process decrypts the encrypted media files from the recorder using the keys loaded by the Load Keys process. The NMF Decrypt process takes one file resource as an input and one as an output.



Load Key Action

Action - Engage Load Keys

DeleteClose

EditPropertiesLabelsLogs

Process files from metadatafalse?

Keystore locationD:\GPK Keystore\GPK_keystore.ks?

Keystore password*****

Delete encrypted CSV files after processingfalse?

Delete keys older than # days0?

Failure Threshold0?

Set the keystore location to the directory that was created on the Decryption server by the GPK action

Show Advanced

Submit

Fields in bold are required

Click on the Engage Load Keys action. Set the keystore location to the directory that was created on the Decryption server by the GPK action, Enter the keystore password. Press Submit.



Engage NMF Decrypt action

Action - Engage NMF Decrypt

DeleteClose

EditPropertiesLabelsLogs

Process files from metadatafalse

File filter*.nmf

Keystore locationD:\GPK Keystore\GPK_keystore.ks

Keystore password*****

Delete encrypted NMF files after processingfalse

Flatten output folder structurefalse

Failure Threshold0

Set the keystore location to the directory that was created on the Decryption server by the GPK action

Show Advanced

Submit

Fields in bold are required

Enter the path to the Keystore\keystore name.ks

NiCE

Do it Yourself

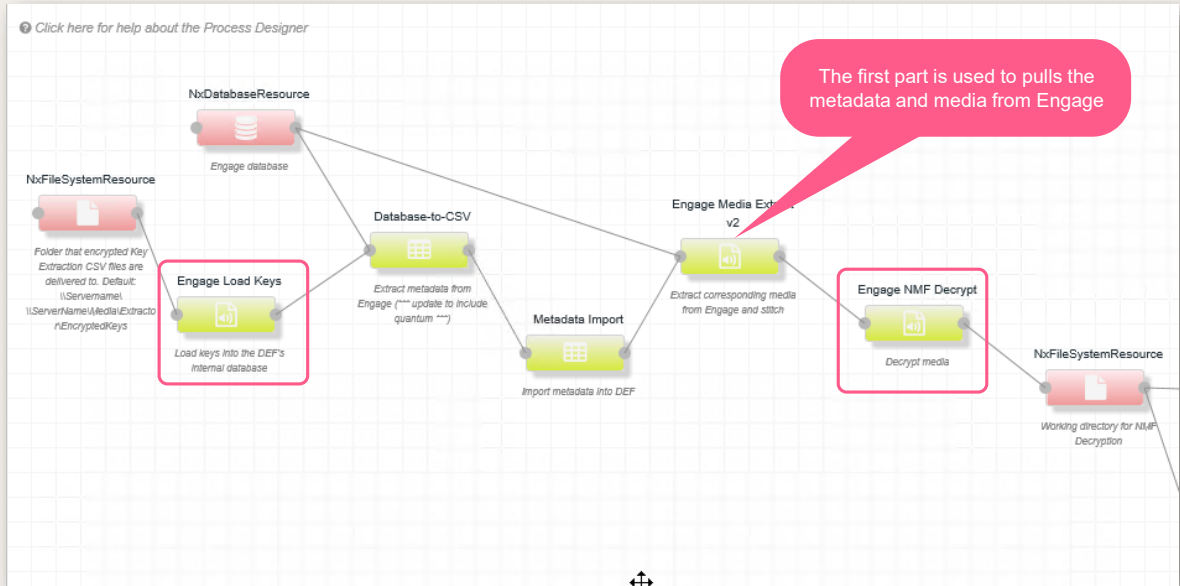
- Refer to Exercises #15



Extract from
Engage



Extract from Engage



From DEF 5.5 and on the Load Keys and NMF Decryption process are included in the Extract from Engage templates

This template is designed to pull encrypted media and metadata from an Engage system

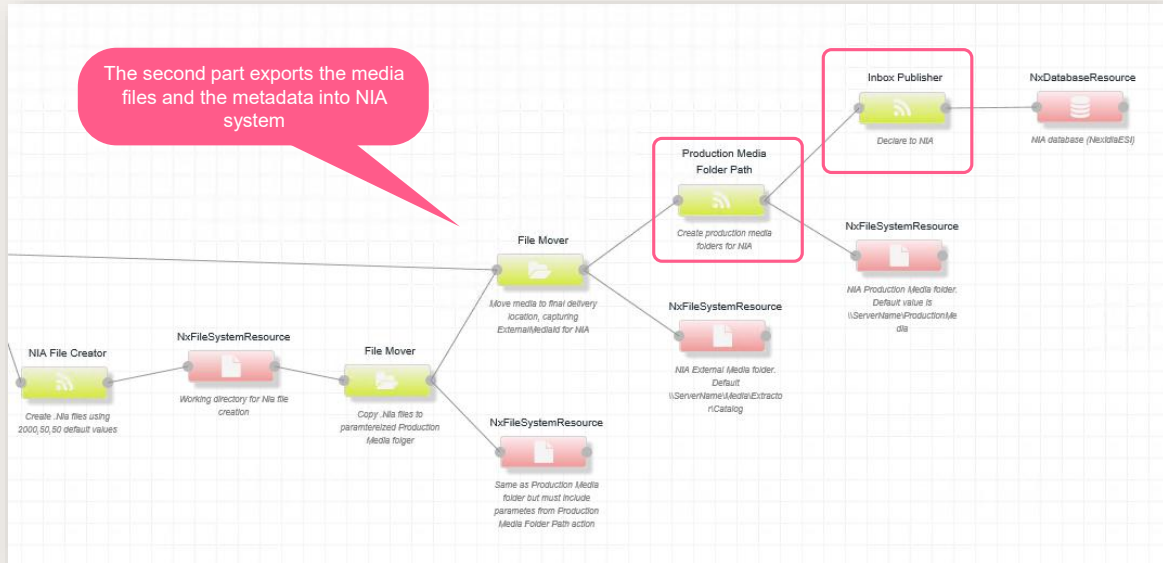
The process created by this template will perform the following:

- Load decryption keys from an Engage database

- Extract media and metadata from file resources

- Decrypt files

Extract from Engage (Continued)



Process continued:

- Create the Production Media Folder path based on audio file metadata
- Push files to the external media storage
- Publish files to NIA

Engage Media Extract

Action - Engage Media Extract v2

[Delete](#) [Close](#)

[Edit](#) [Properties](#) [Labels](#) [Logs](#)

Use CTI data	true	?
Skip Invalid Calls	false	?
Failure Threshold	0	?

[Submit](#)

[Show Advanced](#)

Fields in bold are required

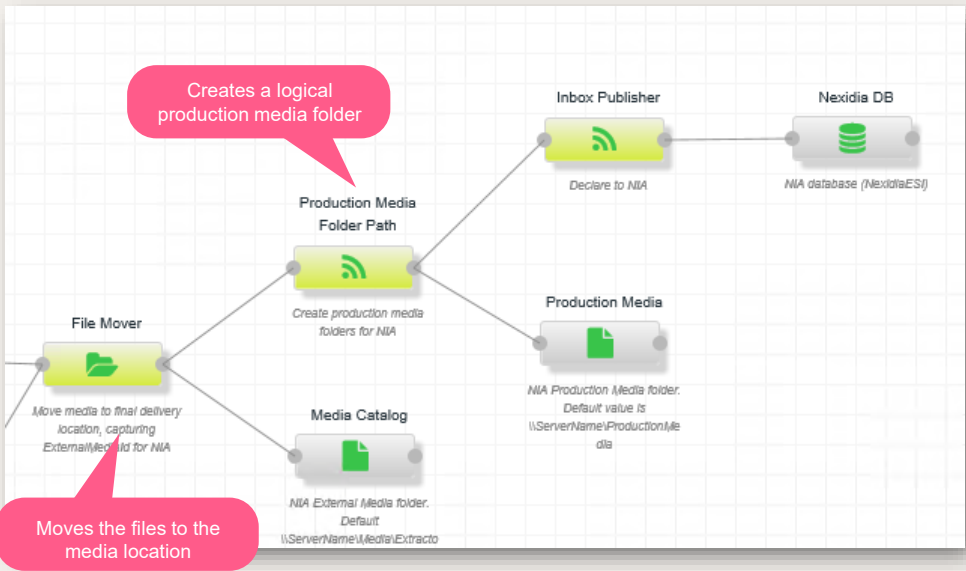
Extracts calls from Engage and stitches the call segments base on the call metadata

Extracts media from Engage recorders based on incoming metadata,

The action queries Engage for additional information required to extract and deliver them to the output resource

Pull the calls from Engage and stitches the call segments base on the call metadata

Audio Files location



The File mover action, moves the media to the media share folder used by NIA, this action must be configured to use capture the 'External Media ID' metadata during the move.



Production Media Folder Path

Action - Production Media Folder Path

[Properties](#)
[Labels](#)
[Logs](#)

Production Media Folder Path

Failure Threshold

[Show Advanced](#)

Fields in bold are required

The media folder structure that will be created in the output resource based on the metadata specified here
Example: media\{metadata1}\{metadata2}

Production Media Folder Path creates the folder structure in output resources without moving the media to that location, but NIA will use this path in the ingest path property.

This path must be visible to NIA when a session is created (create logical production media folder)

Move the media to the media share and create a logical production media folder

Inbox Table

The physical location of the audio file

Results		Messages	
IngestFilePath	MetadataXML	RowId	RowCreateDateUtc
1 \\N\XDB1\Production Media\2019\11\07\6756637315213361153_6756637315213361153.nmf	<?xml version="1.0" encoding="UTF-8"?> <Metada...	6	2019-12-29 08:51:48
2 \\N\XDB1\Production Media\2019\11\07\6756637766184927235_6756637766184927235.nmf	<?xml version="1.0" encoding="UTF-8"?> <Metada...	7	2019-12-29 08:51:48
3 \\N\XDB1\Production Media\2019\11\07\6756639179229167621_6756639179229167621.nmf	<?xml version="1.0" encoding="UTF-8"?> <Metada...	8	2019-12-29 08:51:48
4 \\N\XDB1\Production Media\2019\11\13\6758743094728916993_6758743094728916993.nmf	<?xml version="1.0" encoding="UTF-8"?> <Metada...	9	2019-12-29 08:51:48
5 \\N\XDB1\Production Media\2019\11\13\6758744993104461827_6758744993104461827.nmf	<?xml version="1.0" encoding="UTF-8"?> <Metada...	10	2019-12-29 08:51:48

The logical location of the audio file

The Ingest file path column in the Inbox table contains the logical location of the audio file – this path will be used in the session.

The metate XML holds the physical (actual) location of the audio file.



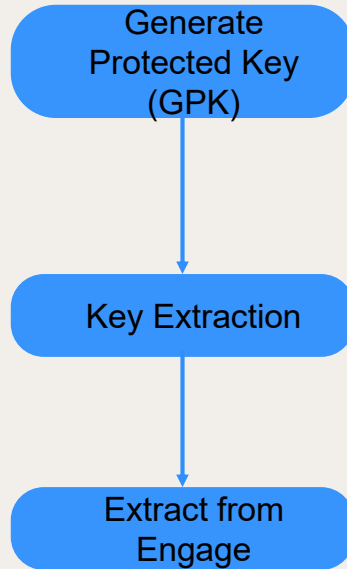
Inbox Table

```
<?xml version="1.0" encoding="UTF-8"?>
<Metadata xmlns="http://www.nexidia.com/2005/schemas/Metadata" xmlns:i="http://www.w3.org/2001/XMLSchema-instance">
  <Property Name="CallDateTime" Value="2019-11-07 20:43:35.94"/>
  <Property Name="CallDirection" Value="Internal"/>
  <Property Name="DNIS" Value="1001"/>
  <Property Name="AgentName" Value="Osika, Saijai"/>
  <Property Name="AgentId" Value="2"/>
  <Property Name="Extension" Value="1001"/>
  <Property Name="UDFvarchar1" Value="6756637315213361153"/>
  <Property Name="UDFvarchar2" Value="6756637315213361153"/>
  <Property Name="UDFvarchar3" Value="2019-11-07 18:43:35.94"/>
  <Property Name="UDFvarchar4" Value="2019-11-07 20:43:35.94"/>
  <Property Name="UDFvarchar5" Value="2019-11-07 20:45:19.457"/>
  <Property Name="UDFvarchar6" Value="2019-11-07 18:43:35.940"/>
  <Property Name="UDFvarchar7" Value="2019-11-07 18:45:19.457"/>
  <Property Name="UDFint1" Value="16795217"/>
  <Property Name="UDFvarchar11" Value="3"/>
  <Property Name="UDFint9" Value="2"/>
  <Property Name="UDFint10" Value="0"/>
  <Property Name="UDFint11" Value="0"/>
  <Property Name="UDFint16" Value="1"/>
  <Property Name="UDFint13" Value="17"/>
  <Property Name="UDFint14" Value="2"/>
  <Property Name="UDFvarchar18" Value="Saijai"/>
  <Property Name="UDFvarchar19" Value="Osika"/>
  <Property Name="UDFdate1" Value="2017-04-12 21:56:03.0"/>
  <Property Name="UDFvarchar22" Value="0.0"/>
  <Property Name="UDFvarchar27" Value="Osika, Saijai"/>
  <Property Name="HasVideo" Value="false"/>
  <Property Name="Channel0" Value="Agent"/>
  <Property Name="ExternalMediaId" Value="\neat1\d$\ Media\6756637315213361153_6756637315213361153.nm"/>
</Metadata>
```

The physical location of the audio file.



Encryption Process Scheduling



NiCE

The GPK process -runs only once for initial key creation, and subsequent key rotation by clicking Launch. This template is a one-time process and does not need a schedule defined for it.

Key extraction and extract from Engage processes are run on schedule bases every 15 minutes

summary

In this lesson we covered:

- How to configure the 3 decryption templates:
 - Generate Protected Key (GPK)
 - Key Extraction
 - Load Key





Thank You



Upgrades

Data Exchange Framework v12.5
Fundamentals

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objectives

By the end of this lesson, you will be able to:

- Perform an Upgrade of an Existing DEF Server
- Perform a Multi-Server Upgrade



Upgrade using
Cumulative Updater



Cumulative Updater

This procedure is only supported for upgrading from DEF version 5.0 or higher



Stop All Processes



Stop the DEF services



Run the CU

The Cumulative Updater tool is used to upgrade the Data Exchange Framework. You can choose to upgrade individual servers or upgrade multiple servers at the same time from one location.

Important! If you are upgrading from a version prior to 5.0, you must install a new DEF instance with a dedicated database (separate from the previous release) , and then recreate the processes there.

Cumulative Updater

⚙️

Process - NICE Decryption - Key Extraction Rev H

👤 Design

📄 Copy

🚀 Launch

▶ Start

■ Stop

⏸ End

🗑 Delete

📊 Manage Tables

➕ Add Process

📈 View Quantum

📅 View Schedule

🔄 Refresh

Preview

Properties

Edit

Recent Activity

Stop all processes

⚙️ Nexidia Data Exchange Framework Admin1	Nexidia Data Exchange Framework Admin	Running	Automatic
⚙️ Nexidia Data Exchange Framework Worker1	Nexidia Data Exchange Framework Worker	Running	Automatic

Stop all DEF Services

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Stop all running process.
Shut down the Data Exchange Framework service, and make sure that the Java processes have terminated in Task Manager.



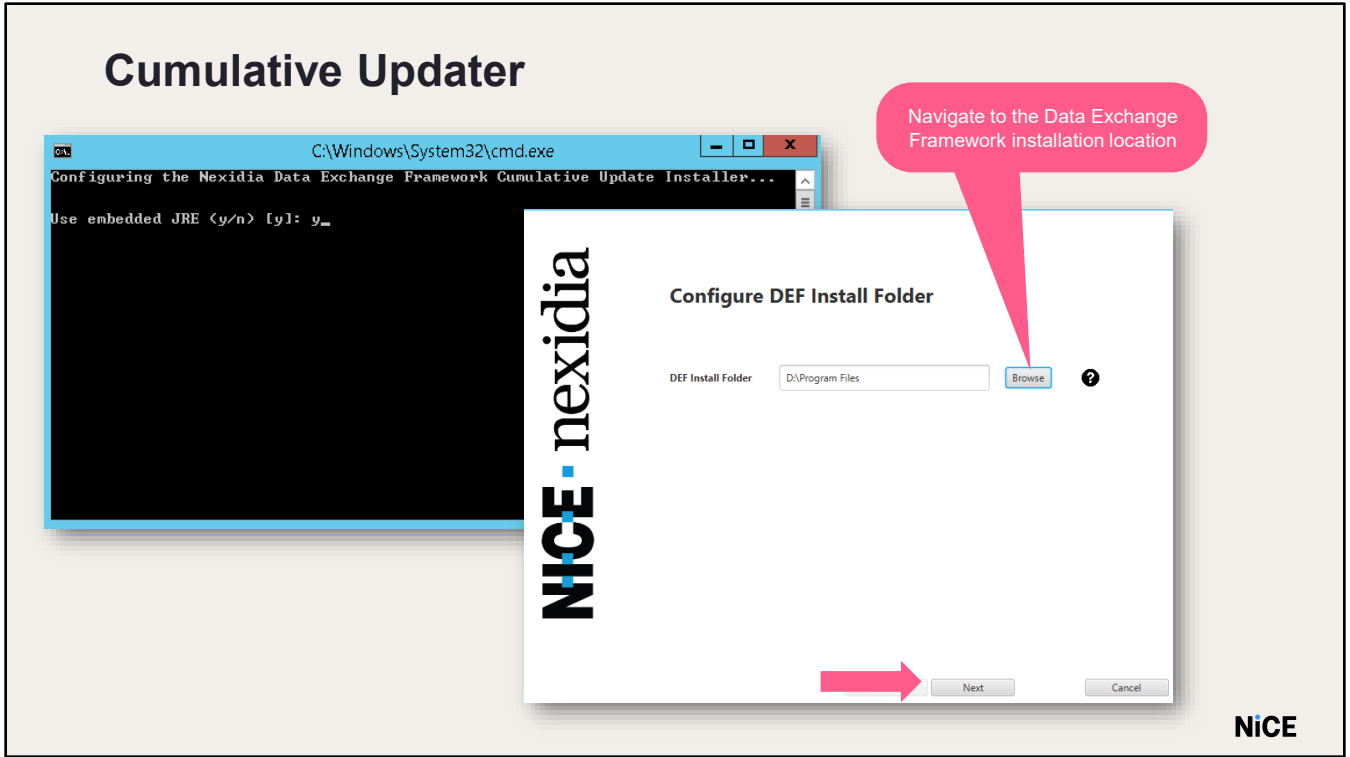
Cumulative Updater

Name	Date modified	Type	Size
bin	10/5/2025 11:18 PM	File folder	
etc	10/5/2025 11:18 PM	File folder	
jre	10/5/2025 11:18 PM	File folder	
lib	10/5/2025 11:18 PM	File folder	
MultiServersCU	10/5/2025 11:18 PM	File folder	
tools	10/5/2025 11:19 PM	File folder	
validateCustomActionInCU	10/5/2025 11:19 PM	File folder	
wb-lib	10/5/2025 11:19 PM	File folder	
CumulativeUpdate.bat	7/24/2025 12:09 PM	Windows Batch File	2 KB

Right-click on the Cumulative Update batch file and select Run as administrator.



Cumulative Updater

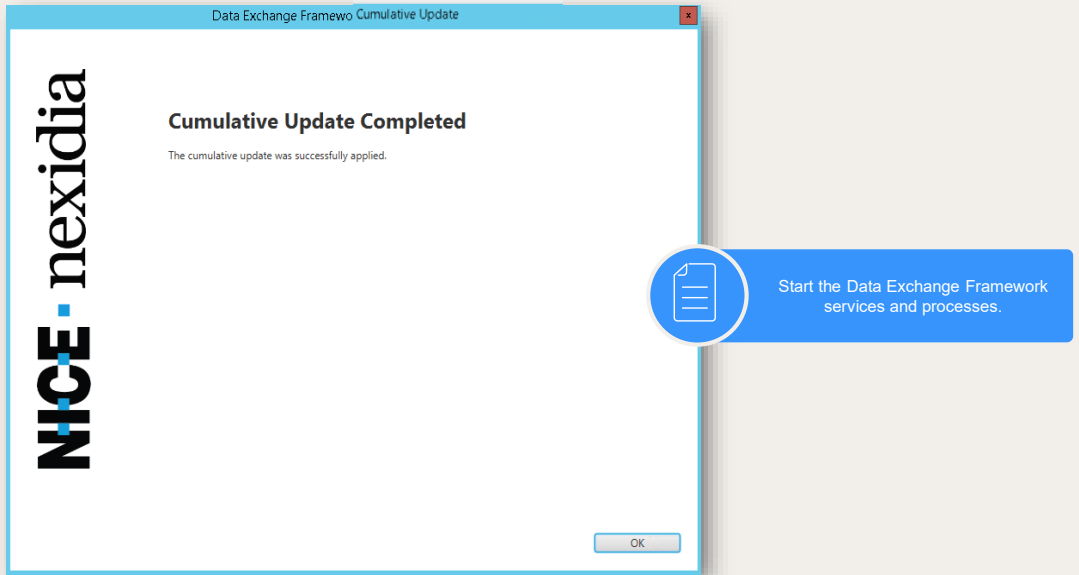


You are prompted to specify if you want to use the embedded JRE or not. Enter “y” or “n”. If you don't want to use the embedded JRE, then you must have JAVA_HOME in your environment. If you have not set up your JAVA_HOME, you are asked to enter the path to the JRE. The DEF Cumulative Updater window opens.

Navigate to the Data Exchange Framework installation location, and select it.

Click Next to confirm the values you selected. The cumulative update status opens.

Cumulative Updater



NiCE

Important! If Interaction analytics Workbench was activated in the previous version, open the process

Upgrading the Application

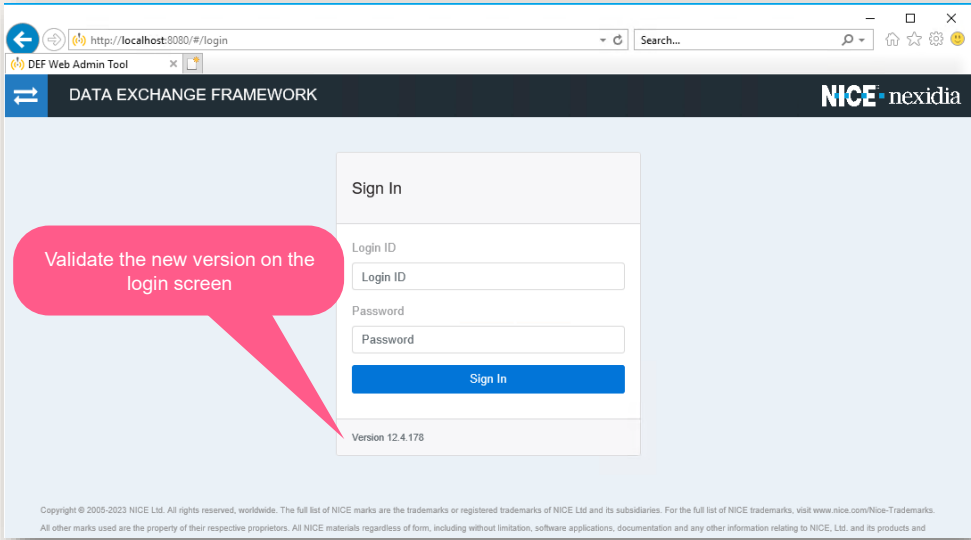
Upgrading an Individual Server

builder and confirm the availability of the Workbench actions. If they aren't there, do not attempt to execute any processes that require Workbench functions. Check the nx_extractor.log file in {DEF_DATA}\logs for the specific error.

If your license has expired, contact your Interaction analytics support representative for a new license file.



Cumulative Updater



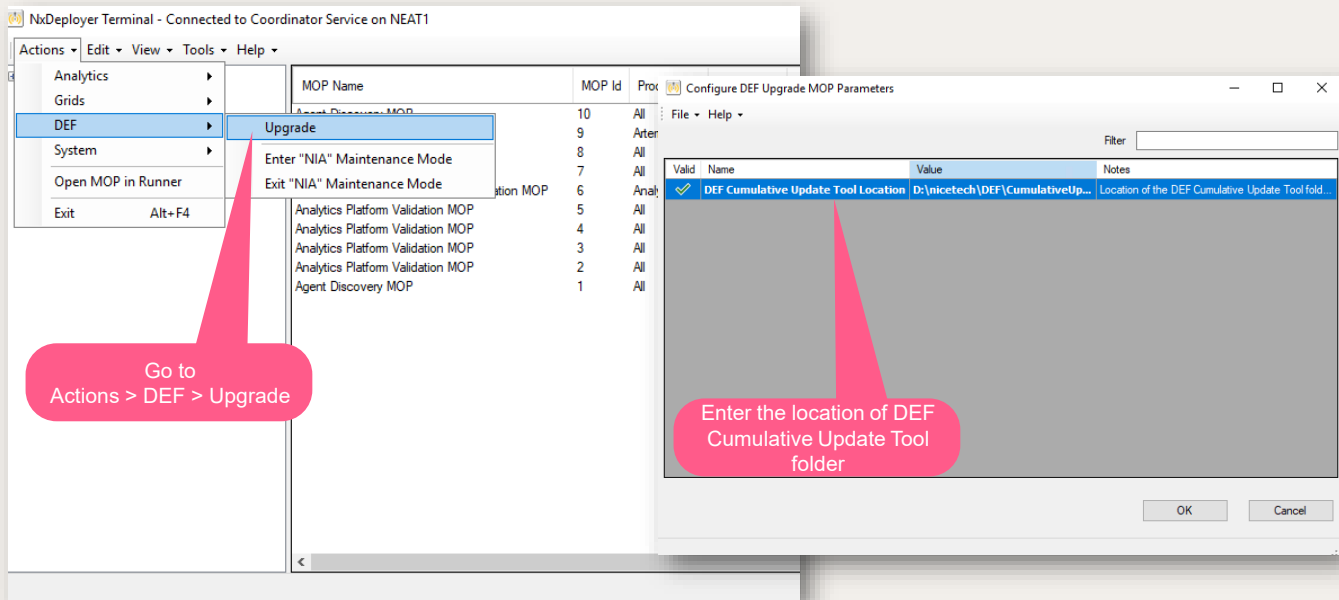
Check the version of the Data Exchange Framework on the login page



Upgrade using
NxDeployer



NxDeployer



You can use NxDeployer to simultaneously upgrade DEF for all agent computers in the Real-Time Grid implementation. To ensure that the scope is limited to a single system, NxDeployer will not a Steps to upgrade DEF: (Here NxDeployer is installed in NEAT1 Server)

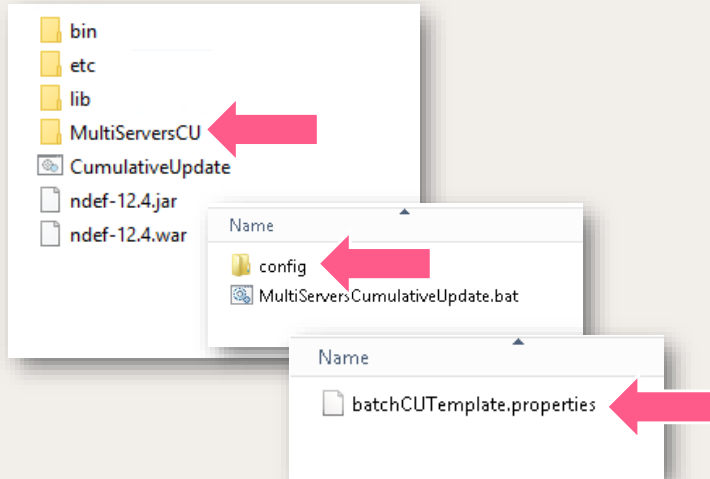
1. In the NxDeployer Terminal window, go to Actions > DEF > Enter "NIA" Maintenance Mode.
2. Then click Actions > DEF > Upgrade
3. The Configure DEF Upgrade MOP Parameters window opens.
4. Enter the location of DEF Cumulative Update Tool folder.
5. The Configure New MOP Run window opens.
6. Optionally enter a name and description of the MOP to identify it in the NXDeployer Terminal, and click OK. The MOP Runner window opens.
7. In the MOP Runner window, click Play. The MOP begins running. Expand the tree to monitor its progress.



Multi Server Upgrade



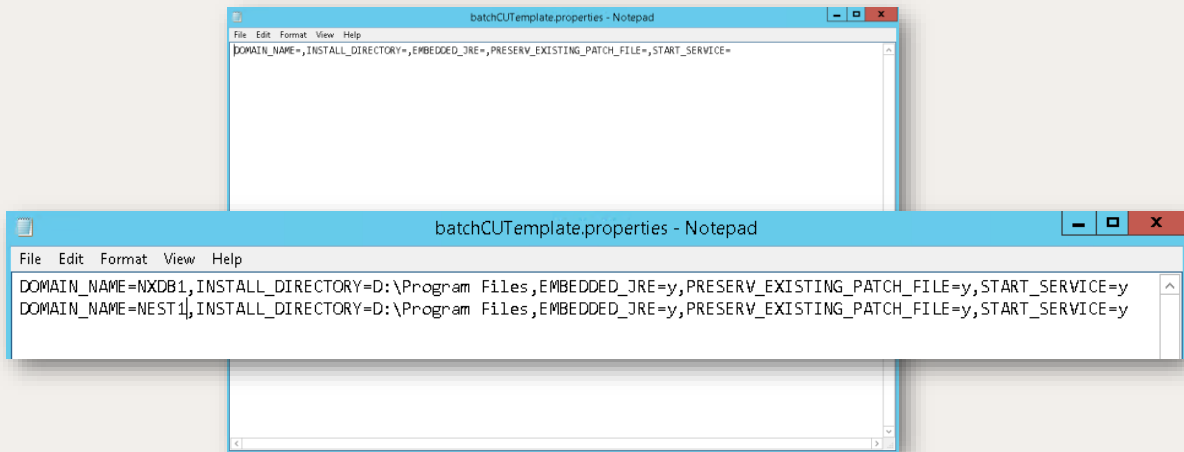
Cumulative Updater



Navigate to the CumulativeUpdateTool zip file and extract it to a local folder.

Navigate to the MultiServersCU folder and open the config folder.

Cumulative Updater



NiCE

Enter values to specify where each instance of DEF to be updated is located. You can add as many lines as you need based on how many servers you have. You must include these values:

DOMAIN_NAME - Enter the server name.

INSTALL_DIRECTORY - The directory where the Data Exchange Framework is installed.

For Example c:\Program Files\Nexidia. Do not include Data Exchange Framework in the path.

EMBEDDED_JRE - Enter y if the ,.. uses JRE or n if it does not.

PRESERVE_EXISTING_PATCH_FILE. If there is an existing patches file in this environment

enter y if you want to preserve it or no if you do not want to preserve it.

START_SERVICE - If you want to start the service after the update leave blank or enter y to start it, or n if you do not want to start it.



summary

In this lesson we covered:

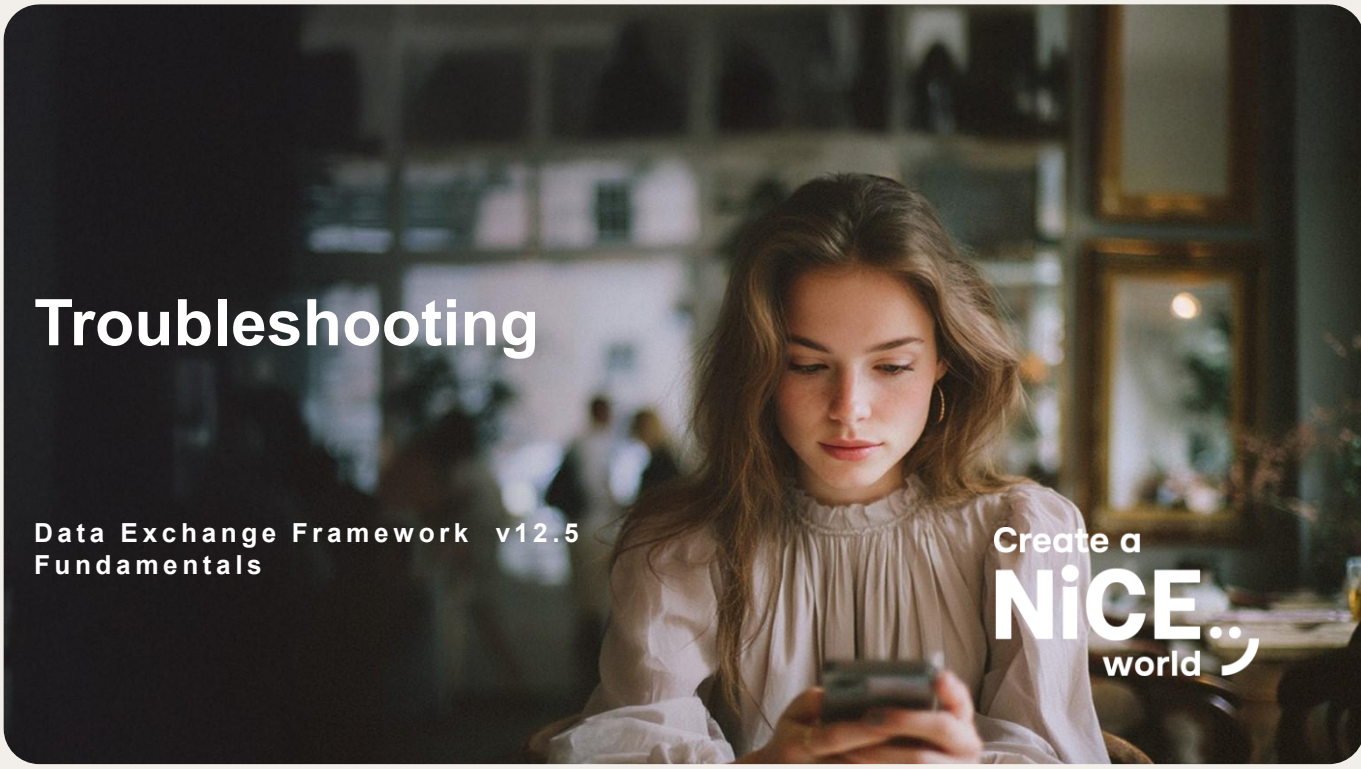
- Upgrade of an Existing DEF Server
- Performing a Multi-Server Upgrade





Thank You





Troubleshooting

Data Exchange Framework v12.5
Fundamentals

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objectives

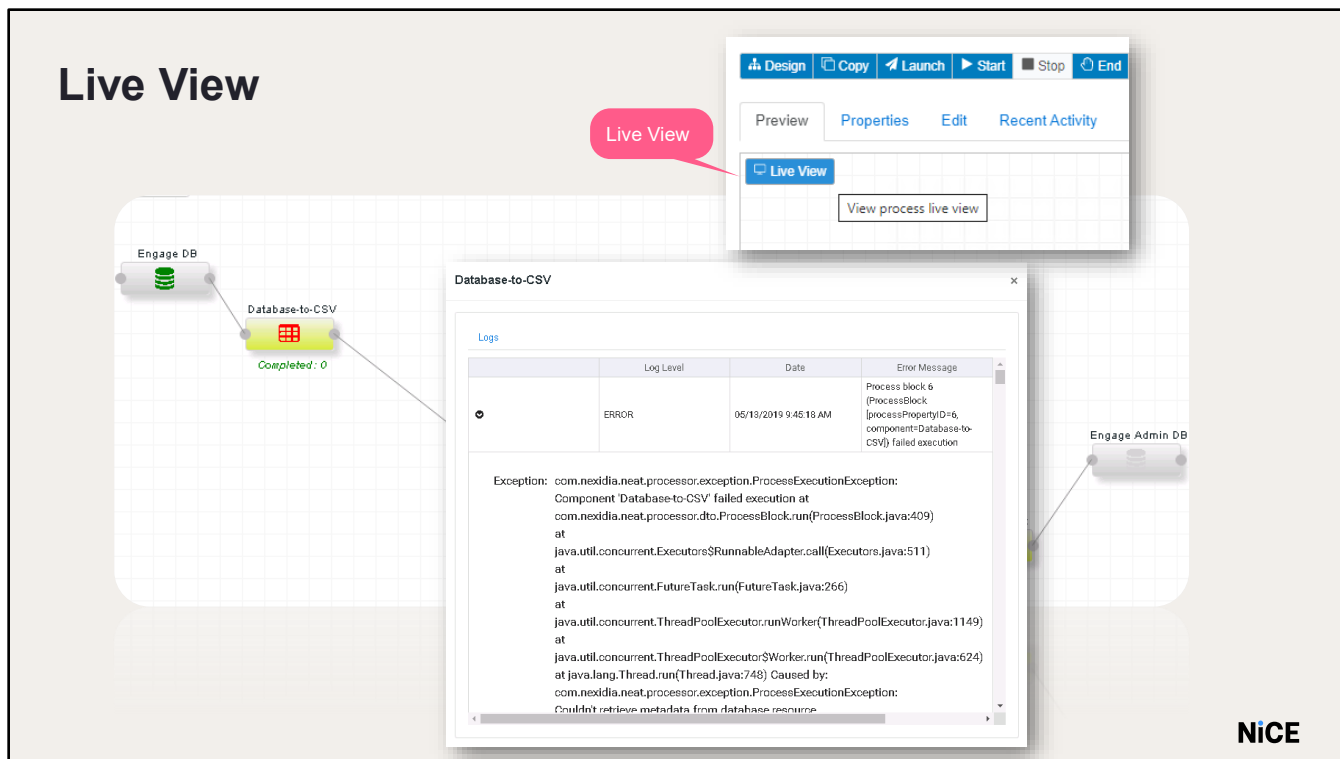
By the end of this lesson, you will be able to:

- Understand the Live View Feature
- Understand the Deployment Reports



Live View





When you select a process from the tree view, a preview of the process as it is currently defined displays. You can optionally select the Live View button to see the following information, which is

refreshed automatically. When live is enabled, LIVE VIEW ON message flashes next to the Live View button.

You can hover over this message to display the date and time stamp for the live view results that are showing.

A live view of the action state(s) based on events.

A working indicator that shows the process work state.

Summing of file counts for each action.

Working Table

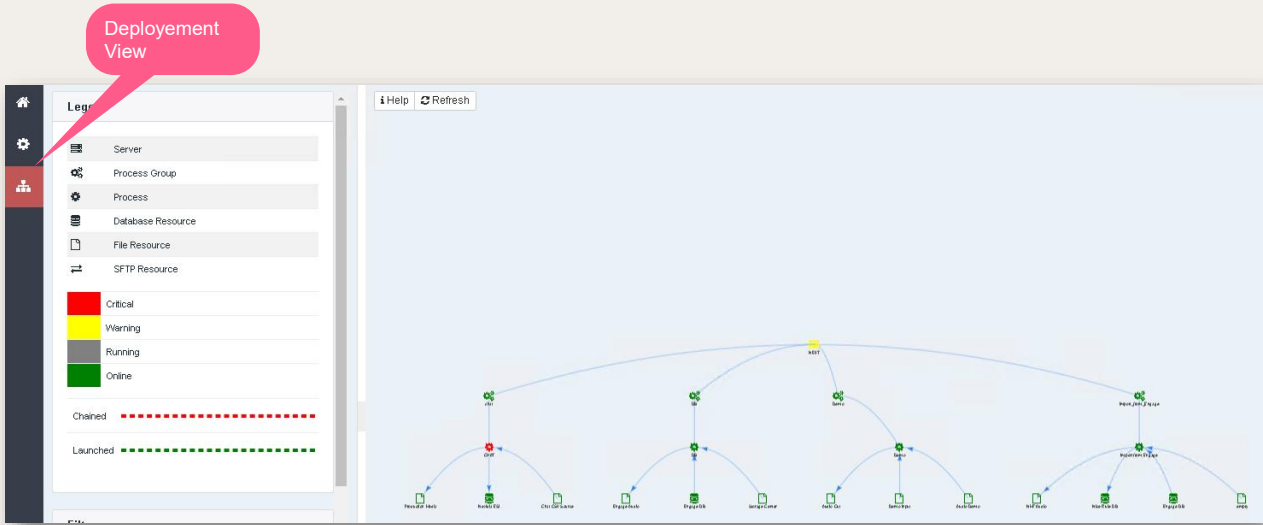
Disposition-FileMoverAction-352	CallId	InteractionId	RecordingSide	NxFailureMessage
NULL	10001	222	1	Message: Channel files don't exist on the file s...
NULL	10001	222	2	Message: Channel files don't exist on the file s...

Action failure
error message

When one of the actions failed, a column "NxFailureMessage" added to the working table with the error message.



Deployment View



The Deployment View provides a graphical representation of all processes for a specific deployment in a single view so that you can easily understand and debug workflows associated with it.



Deployment View

The screenshot displays the NiCE Deployment View. On the left, a 'Legend' panel lists various components: Server, Process Group, Process, Database Resource, File Resource, and SFTP Resource. Below these are status indicators: Critical (red), Warning (yellow), Running (grey), and Online (green). A 'Filters' section is also present. The main workspace shows a process diagram with nodes like 'Engage Action' and 'Engage DB' connected by arrows. A 'Demo' configuration window is open, showing tabs for 'Properties', 'Edit', and 'Recent Activity'. The 'Edit' tab is active, displaying a message: 'This component has not been configured yet. Please use the edit tab to configure this component.' The NiCE logo is visible in the bottom right corner.

Clicking on one of the processes open the process configuration window, the user can edit and launch the process from the deployment view

Do it Yourself

- Refer to Exercise #17



Deployment Reports



Deployment Reports

The screenshot displays the NICE Data Exchange Framework interface. On the left, a navigation tree under 'Companies' shows 'NICE' selected. The main panel shows 'Company - NICE' with a toolbar containing 'Copy', 'Deactivate', 'Delete', 'Deployment Report', and 'Refresh'. A pink callout bubble points to the 'Deployment Report' button with the text 'Deployment Reports'. Below the toolbar, there are tabs for 'Properties', 'Edit', and 'Recent Activity'. The 'Properties' tab is active, showing fields like ID, Name, Status, Site Count, Administrator email, Reporting source code, and Deactivated. A modal window titled 'Deployment Report- NICE' is open, showing 'Deployment report options' with checkboxes for 'Show Quick Start Guide', 'Show Process Components', and 'Show Process Thumbnails'. Below this is a table of 'Available Processes' with columns for ID and Name.

ID	Name
4	EX11
2	EX7
3	EX9A

At the bottom of the modal, there are 'Run Report' and 'Cancel' buttons.

You can generate a detailed report of a customer deployment that includes company, sites, servers, processes, resources ,and admin configuration settings.

Those reports help the Support Team view the process details and diagrams

1. Select a company and click Deployment Report. The Deployment Report window opens.
2. Select the desired configuration options and click Run Report. A confirmation message displays.
3. Click OK. The report is generated and displayed on the screen.

Deployment Reports

Company Deployment Report

Created on Friday, August 18, 2017, 1:04pm

Table of Contents

Admin	Admin Settings	View
Customer	NICE	View
Site	TLV	View
Server	NEAT	View
Process	EX11	View
Resource Component	Engage DB	View
Action Component	Database-to-CSV	View
Resource Component	Engage Stereo Data	View
Resource Component	SC	View
Action Component	Metadata Import	View
Action Component	File Mover	View
Resource Component	Engage Segment	View
Action Component	Media Conversion	View
Resource Component	Converted Audio	View
Action Component	Merge	View
Resource Component	Engage Merged Audio	View
Action Component	NIA File Creator	View
Action Component	File Mover	View
Resource Component	Production Media	View
Resource Component	Production Media	View
Action Component	Inbox Publisher	View
Resource Component	NiDatabaseResource	View
Process	EX7	View
Resource Component	Engage DB	View

List of company, sites, servers, processes, resources

Can drill down to each item



Site and Server Info

← → ↻ localhost:8080/#/deploymentreport/2?showQuickStartGuide=false&showProcessComponents=true&showProcessThumbnail...

Site - TLV	
ID	2
Name	TLV
Status	ONLINE
Type	Site
Administrator email	Not set
Reporting source code	Not set
Deactivated	false
Top	
Server - NEAT	
ID	2
Name	NEAT
State	ONLINE
Status	ONLINE
Threads per action	2
Process debug level	DEBUG
Heartbeat interval (seconds)	60
Server retention (days)	90
Audit log retention (days)	90
Administrator email	Not set
Reporting source code	Not set
Type	Server
Deactivated	false



Deployment Reports

← → ↺

localhost:8080/#/deploymentreport/2?showQuickStartGuide=false&showProcessComponents=true&showProcessThumbnail...

☆

⌵

Notification email on completion	false
Pct variance notification threshold	Not set
Expected volume for variance check	Not set
Maximum days to attempt self-reconciliation	10
Runtime alert threshold (minutes)	120
Terminate process at runtime threshold	false
Auto-tune thread counts	true
Include concurrently-running actions	true
De-dupe rollover table	false
Disable self-cleaning	false
Audit results	true
Compute node availability	Disabled

Top

Process Definition Thumbnail for "EX7"

Process 1.1

Database In-CPU

Copy Data

File Search Report

File Reader

Audio File

View the process details and diagrams



summary

- In this lesson, we covered:
 - Live View Feature
 - Deployment Reports





Thank You



Course Closing

Data Exchange Framework v12.5
Fundamentals

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Course Survey

A screenshot of the "Training satisfaction survey" form. The first question is "How satisfied are you with the overall training experience?" with a 5-point scale from "Very Dissatisfied" (1) to "Very Satisfied" (5). The second question is "On a scale of 1 to 5, how satisfied are you with the following aspects of the training course:". It lists five aspects: "The instructor's knowledge of the subject matter", "The instructor's delivery, attitude and behavior", "The course content and structure (agenda, presentations, documents, etc.)", "The balance's between lectures, demos, and hands-on activities", and "The labs used for demos and exercises". Each aspect has a 5-point scale from "Very Dissatisfied" (1) to "Very Satisfied" (5).

Please leave us comments on how we can improve the learning experience for this course

Scores range from "Very Dissatisfied" (left) to "Very Satisfied" (right)

You should have received a course survey in your email from CSATTeam@nice.com. Please take a few moments to take the course survey before we proceed.

If you did not receive the course survey in your Inbox, check your Junk email folder. If you did not receive the survey, please let your instructor know so that he/she can have it resent to you.



Practical Exam

NiCE

Data Exchange Framework Fundamentals v12.5 – Practical Exam

General Instructions

Students will install and configure a DEF system and build a process according to the requirements described below



Please feel free to refer to any documentation, manuals, or notes to assist in completing the Practical Exam

Duration: 6 hours

You have been assigned install the DEF software and create the necessary processes to import the existing hierarchy from NICE Engage to Interaction Analytics. You will also create a process to import encrypted audio and metadata from Engage to Interaction Analytics.

System info:

- You will need to install the DEF software according to the following site map (The DEF software is located on D:\nicetech)
 - NEAT software installed on the NEAT1 Server. Create the following directory for the NEAT software
 - D:\Program Files\NEAT
 - NEST Software for Extraction installed on the NEAT1 Server. Create the following directory for the NEST software

Read the Practical Exam document carefully

Complete all assigned tasks

Notify the instructor when complete

You will receive the Practical Assessment document from the instructor. Before beginning please read the entire document carefully. Complete all assigned tasks and notify the instructor when everything has been completed.





Thank You



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